Multidimensional Lattice Walk Enumeration Through Coefficient Extraction Operators

by

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Abstract

In this thesis we investigate the application of an operator paradigm to the enumeration of lattice walks in multiple dimensions, in order to automatize some of the procedures involved in finding generating function Q for the walks. In particular, given a lattice walk model in some number of dimensions, with some regional constraints and interactions, and arbitrary step set, we may write a functional equation $(1 - t\Gamma S)Q = q$, where Γ is some coefficient extraction operator. We also use the operator perspective to unify and extend the Algebraic and Obstinate kernel methods. Most of the thesis will deal with lattice walk models with small steps confined to the non-negative orthant and the end of the thesis will cover the generalizations.

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Preface

This work intends to establish a new framework of notation and tools for the enumeration of random walks with or without interactions, and similar problems. The notation makes it much easier to explore the symmetries of the problem and write down summation formulas where it was not possible to do so before (with a simplified notation). The tools make the derivation of functional equations for various problems an almost trivial matter. The real power is in combining both approaches to gain insights into the generating function world of enumerating random walks on constrained regions in multidimensional spaces.

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Glossary

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Chapter 1

Introduction

Why do we study lattice walks? Apart from being an interesting mathematical object that leads to a lot of interesting mathematics, they have some real- world applications. A lot of processes can be modeled via lattice walks: from stock market objects to proteins and polymers. There is a rich theory of continuous and discrete lattice walks. Specifically, in lattice walk enumeration, we attempt to find out things about the numbers of walks of some length and of certain type with certain geometrical constraints. The topic often leads to the use of generating functions as a powerful tool in the analysis. In the end, we gain insight into what makes certain lattice walk models "easy" and others more difficult. In this thesis we look at the enumeration for constrained lattice walks with interactions, in order to gain insight into the deeper workings of lattice walk enumeration problems.

1.1 Recent History

In the most basic case, lattice walks starting at the origin on d dimensions have no restrictions, $\mathcal{D}$ is the whole space, and there are no interactions. We can enumerate these walks easily by writing the full generating function

$$\begin{aligned} Q(x_1, \ldots, x_d; t) &= \frac{1}{1 - tS(x_1, \ldots, x_d)} \\ &= 1 + tS(x_1, \ldots, x_d) + t^2 S(x_1, \ldots, x_d)^2 + \cdots . \end{aligned} \tag{1.1}$$

This works since we can use each individual steps and every walk is uniquely represented by a sequence of steps from $\mathcal{S}$. This works for arbitrary initial conditions too, since we can start the walks at any point and evolve them. The coefficients of each power of t are Laurent polynomials in the variables $x_1, \ldots, x_d$. If we have the generating function for a lattice walk model, we have a lot of information at our hands. Replacing x_k with 0 for each k gives us the generating function for

walks returning to the origin. Similarly, taking higher derivatives and evaluating at zero allows us to find the number of walks ending at specific points. If we set all the $x_k = 1$ inf Q, we obtain the length generating function for the lattice walks in the plane. There are numerous questions we can ask about each of those characteristics of the model.

The next step up is to deal with half-spaces: $\mathcal{D}$ is the region where one of the $x_k \geqslant 0$. These cases are not as simple to solve, but they are doable. Check [3], [4], [5] and [6] for work on the one-dimensional case that can be extended to general d. The generating functions we can obtain are algebraic. Naturally, we are led to consider cases where $\mathcal{D}$ is an intersection of half-spaces. Even for $d = 2$, the problem has generated a rich theory of quadrant (or quarter-plane) walks. We obtain functional equations for the function Q that are not easy to solve in general. Often, the generating functions here are not even D-finite, although there are D-finite examples. The main object of study in these cases has been a group G of bi-rational transformations preserving the polynomial $K = 1 - tS$, called the kernel of the model.

When G is finite, in the 23 non-equivalent cases, we obtain a D-finite generating function (full?), as can be seen in the classifications [7],[8] and [9], [10] for specific cases. The main tool in the study of the finite group cases is the algebraic kernel method, in which we sum different functional equations produced by group actions by elements of G. This is done to cancel the majority of the terms in the functional equation. See [11] for a treatment of models with finite groups and repeated steps.

There are 56 non-equivalent cases of quarter-plane walks with an infinite group. For 51 of them, both the full generating function [12] and the length generating function, for walks returning to the origin, are not D-finite [13]. The remaining 5 cases have been dealt with in [14] and [15]. The main tool used to tackle these 5 cases is the iterated kernel method, in which we use the roots of the kernel K to show the generating function is not D-finite.

Naturally, we are led to invesitigating higher-dimensional variants of this problem. See [16] and [17] for when $\mathcal{D}$ is a Weyl chamber. In three dimensions we have the recent partial classification [18] of lattice walk models with up to six steps, where the authors extend the algebraic kernel method to handle the majority of finite group cases encountered.

Another generalization is to consider walks with interactions and/or self-avoiding walks (or SAWs). These two types of generalizations have important applications in physics (?) because they can be used to model polymers, proteins and other chains. For some research on SAWs with various interactions, see [19],[20], [**?**], [21], [22], [23] and [24]. The topic of SAWs does not lend itself as nicely to using generating functions like the ones considered so far, so, as a simpler first step, we can consider regular lattice walks with interactions. See [25] and [26] for walks with interactions in an octant. See [27], [28], and [29] for walks with interactions in the plane.

A long-standing dream goal in lattice walk enumeration research is the automatization of the process of determining characteristics of the models and similar problems. There has been some progress in this direction. For examples see [30] and [31] for details on symmetric functions, [32] for a holonomic approach to combinatorics, [33] for automatic classification of lattice walks, and finally, [34] for some of the computational challenges involved in automation.

Some automation can be performed in the related subject of the study of lattice walk asymptotics. For examples of the theory, see [35], [36] and [37]. A lot of the works mentioned so far have also discussed asymptotics.

1.2 Thesis Objectives

This thesis grew out of work on regular quarter-plane lattice walks with interactions with the axes. The derivation of the functional equation in the non-interacting case is a simple inclusion-exclusion process, but doing the same for the interacting case is somewhat more involved. It is possible to further simplify the equation via coefficient extraction and back-substitution. The process works out beautifully to give us a much more compact functional equation. However, the process is quite painstaking even for three dimensions, let alone for higher-dimensions, although it is possible.

The mathematicians who study lattice walk enumeration have as a goal to classify lattice walks according to various characteristics like step set, dimension, group size associated to the kernel, and perhaps most importantly: D-finitenes. They also dream of being able to automatize the process whenever possible. In this thesis we aim to further the study of lattice walk models towards this goal of automatization. The optimistic thing would be to suppose that a general algorithm exists that takes an arbitrary lattice walk model, in arbitrarily many dimensions, with arbitrary finite step sets and with arbitrary constraints; and outputs various data about the model: functional equation for the generating function, a solution and information about the model's D-finiteness. It is only logical that, if such an algorithm were to exist, it would not have difficulty with the particulars of the lattice walk model. While this kind of algorithm may be too strong to envision, it may be the case that there are atrocious lattice walk models that we cannot solve, the idea of the existence of such an algorithm has been a strong motivator in our pursuit of generalizations. This thesis succeeds in demonstrating that the first task is achievable. For a vast assortment of lattice walk models, we can almost-automatically write down a functional equation for its generating function counting length, position and interactions. The equation looks like

$$(1 - t\Gamma S)Q = q\,, \tag{1.2}$$

where Γ is an operator on the variables of Q, S is the Laurent step polynomial keeping inventory

of the steps in polynomial form, and q is some Laurent polynomial, serving as an initial condition. The operator idea is powerful and we will use it as much as we can. A secondary achievement is the furthering of the theory of the orbit summation method, which, along with the obstinate kernel method, combines in a nicer theory of operators consisting of group actions. However, time and space have not been sufficient to go far enough in this direction. Near the end we present some full and partial solutions to two-dimensional interacting lattice walk models.

1.3 Layout of Thesis

In the first few chapters we will go through motivating examples that will serve as a springboard into further generalizations. First, we will start off with one-dimensional lattice walks with and without interactions in Chapter 2. Then we will quickly move on to two-dimensional lattice walks, and beyond, in Chapter 3. On the way we will develop some of the basic operator theory involved in the lattice walk models. In Chapter 6 we will tackle some generalization of the problem and show that they all satisfy a similar functional equation.

In Chapter 4 we will take a more practical route to deal with group-theoretic operators relevant tol lattice walk models with small steps, and then deal with extending the Algebraic Kernel method to more dimensions and interactions. In Chapter 5 we will look at the Obstinate Kernel method and solve some highly-symmetrical two-dimensional interacting models.

1.4 Notation and Conventions

1.4.1 Formal Series

We will work in the setting of formal power and Laurent series. Problems with convergence will not be studied in much detail; formal convergence will suffice. If we are given a ring R, we may construct several formal objects with a commutative indeterminate x with coefficients in R.

- $R[x]$: Ring of polynomials with monomials x^k for integers $k \geqslant 0$.
- $R[x, 1/x]$: Ring of Laurent polynomials with monomials x^k for integers k.
- $R[[x]]$: Ring of formal power series with monomials x^k for integers $k \geqslant 0$.
- $R((x))$: Field of formal Laurent series with monomials x^k for integers k.

We can also define their multivariate analogues $R[x_1, \ldots, x_n]$ and $R[x_1, \overline{x_1}, \ldots, x_n, \overline{x}_n]$, and their series counterparts $R[[x_1, \ldots, x_n]]$ and $R((x_1, \ldots, x_n))$. For a thorough treatment of the subject,

please see [1]. When we work with a combination of objects coming from different rings/fields over the indeterminate x, we will consider all objects as part of the most general ring/field at the time. If in doubt, we will always work in the field of formal Laurent series. Given a sequence (w_n), we may consider its generating function $W(x)$ given by $\sum_n w_n x^n$, which lives appropriately in one of the rings over x. The multivariate case for a multi-index sequence $(w_{n_1,\ldots,n_m})$ is defined analogously as

$$W(x_1,\ldots,x_m) = \sum_{n_1,\ldots,n_m} w_{n_1,\ldots,n_m} x_1^{n_1}\cdots x_m^{n_m} \tag{1.3}$$

whenever this is a formal Laurent series. We often need to extract particular coefficients from the formal polynomial/series when we solve enumeration problems via generating functions . In the one-variable case we have the coefficient extraction operators on formal series

$$[x^n]W(x) = w_n\,. \tag{1.4}$$

The multivariate case is handled analogously with the caveat that, whenever the number of variables extracted is less than the number of variables present, we extract by treating those remaining variables as parameters. For a more detailed discussion, please consult [2]. When we deal with lattice walk generating functions, we often work with functional equations. In the broadest sense, in this thesis we will discuss functional equations that may involve the following: a function F, the function evaluated at various arguments, and various coefficient extractions of F (?). This will be done via operators, so in the end, we will have a functional equation involving F and operators. We will write the identity operator as 1.

1.4.2 Lattice Walk Models

In this work we will investigate lattice walk models on the n-dimensional lattice $\mathbb{Z}^n$. A specific model $\mathcal{M}$ consists of several ingredients

- $\mathcal{S}$: a finite set of steps that we can take from a point in $\mathbb{Z}^n$ to another point in $\mathbb{Z}^n$. In general, this step set may differ from point to point, e.g., a honeycomb lattice embedded in $\mathbb{Z}^2$. The steps themselves are written as vectors $s = (s_1,\ldots,s_n)$. For example, $\mathcal{S}$ could be a set of small steps defined by coordinates among $\{-1,\ 0,\ 1\}$. Another example is when we have more than one step of a given kind.

- $\mathcal{D}$: a domain for the lattice walks. This is the region in $\mathbb{Z}^n$ where the lattice walks live. The easiest cases involve the whole space or a half-space. Starting with quarter-plane regions, the

analysis becomes rather complicated. One can further constrain the walks to live on semi-infinite strips.

- $\mathcal{D}_0$: an interaction domain for the lattice walks. This is a sub-region in $\mathcal{D}$ and we want to count how many times lattice walks interact with it by stepping on coordinates that are part of it. We may count different points from $\mathcal{D}_0$ in different ways. This less-studied paradigm may involve marking visits to various places: the origin, the axes in two dimensions, various higher-dimensional hyperplanes, a cone, an infinite strip, the sides of a box and so on.

- $\mathcal{Q}$: a set of lattice walks acting as an initial condition from which all other lattice walks evolve. This could be as trivial as starting with the empty walk ending at the origin to a set of infinitely many walks with some pattern.

Figure 1.1: Illustration of regions in the plane.

It is natural to consider the generating function that keeps track of all the information characterizing the lattice walks in the model. For each lattice walk w, its length $|w|$ is the total number of steps that it has taken and we keep track of this with a parameter t. We keep track of the walk's endpoint coordinates in $\mathbb{Z}^n$ by the indeterminates $x_1, \ldots, x_n$, and its interactions with $\mathcal{D}_0$ via marking parameters a_m, where m is some specific subscript index. The marking is highly dependent on the geometry of $\mathcal{D}_0$ and the way we want to count interactions. If the end coordinates of the lattice walk are $(m_1, \ldots, m_n)$, and if the marking monomial (a product of a_m's) is $a(w)$, we can write the generating function for the lattice walk model as

$$Q(x_1, \ldots, x_n; t) = \sum_{w \in \mathcal{M}} a(w) x_1^{m_1} \cdots x_n^{m_n} t^{|w|} \,. \tag{1.5}$$

Other important objects are the step polynomial, or characteristic polynomial, $S(x_1, \ldots, x_n)$ defined by

$$S(x_1, \ldots, x_n) = \sum_{s \in \mathcal{S}} x_1^{s_1} \cdots x_n^{s_n} \,, \tag{1.6}$$

and the starting polynomial defined by

$$q(x_1, \ldots, x_n; t) = \sum_{w \in \mathcal{Q}} a(w) x_1^{m_1} \cdots x_n^{m_n} t^{|w|} \,. \tag{1.7}$$

Usually, we see S a Laurent polynomial of degree (or valuation) 1 in each variable and we have $q = 1$. Note that in all of these quantities the variables a_m have been supressed. We can include those too, whenever necessary; for example, when there are only finitely many of them. Furthermore, it will be useful to use the quantities $b = a - 1$ and $c = b/a = 1 - \overline{a}$. (?)

Example 1. To illustrate the notation, we shall investigate a lattice walk model in the plane. Consider the problem of lattice walks, which are restricted to the quarter plane, start at the origin, and which only have small diagonal steps. We will count their interactions with the axes. In our terminology we have a lattice walk model $\mathcal{M}$ with $\mathcal{S}$ given by

$$\mathcal{S} = \{(-1,-1),(1,-1),(-1,1),(1,1)\}\,. \tag{1.8}$$

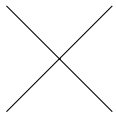


Figure 1.2: The model {NW, NE, SW, SE}.

The region $\mathcal{D}$ consists of all points in $\mathbb{Z}^2$ with non-negative coordinates. Next, $\mathcal{D}_0$ is composed of the two non-negative axes with overlap at the origin; we count visits to the origin as visits to each axis. Finally, $\mathcal{Q}$ consists of the empty walk. In the language of generating functions we have $q = 1$ and

$$S = S(x_1,x_2) = x_1x_2 + \overline{x}_1x_2 + x_1\overline{x}_2 + \overline{x}_1\overline{x}_2 = (x_1+\overline{x}_1)(x_2+\overline{x}_2)\,. \tag{1.9}$$

There is a choice in naming the variables counting the interactions and, as it will prove notationally simpler, we will go with their defining quality: we mark the locations where one coordinate is zero. That is, mark walks ending on $x_k = 0$ by incrementing the power of a_k in the generating function. In this model, the only step we can take at first is $(1,1)$ and this contributes a monomial x_1x_2t. From here on we can build on the single lattice walk of size one by taking a step in one of the four directions. This action leads to the following monomials. ZZPICTUREZZ

- $(1,1)$: monomial $x_1^2x_2^2t^2$ since there are no interactions.
- $(1,$-$1)$: monomial $a_2x_1^2t^2$ since the walk lands on the x_1 axis with coordinate $x_2 = 0$.
- $($-$1,1)$: monomial $a_1x_2^2t^2$ since the walk lands on the x_2 axis with coordinate $x_1 = 0$.

- (-1, -1): monomial $a_1 a_2 t^2$ since the walk lands on the origin, which is on both axes.

Since we have seen all the lattice walks of size at most 2, we see that the generating function $Q = Q(x_1, x_2; t)$ starts as

$$\begin{aligned} Q &= 1 + x_1 x_2 t + (a_1 a_2 + a_1 x_2 + a_2 x_1 + x_1 x_2) t^2 + \cdots \\ &= 1 + x_1 x_2 t + (a_1 + x_1)(a_2 + x_2) t^2 + \cdots . \end{aligned} \tag{1.10}$$

It may be unexpected, that the first few coefficients factor, but this phenomenon is due to the factoring of $S(x_1, x_2)$ itself. We will see a more general result later. (?)

1.4.3 Thesis Notation

In the interest of generalizing the problem-solving strategies involved in lattice walk enumeration, we need powerful notation to handle all the intricacies of the problem. In fact, it is the notation itself that paved the way towards generalization in our originla research because it is suggestive and points towards the deeper connections between the generating functions and quantities involved. With this thesis we aim to make this kind of notation more common in higher-dimensional lattice walk analysis. Typically, in the multivariate setting we see multi-index notation where a bolded, vectorial or even supressed notation denotes the collection of variables involved. We would use this notation as well, but it is insufficient when working with restricted geometries for $\mathcal{D}$ and $\mathcal{D}_0$. Therefore, we have the following notation.

Definition 2. *Let* $\mathbb{0} := (0, \ldots, 0)$ *and* $\pm\mathbb{1} := (\pm 1, \ldots, \pm 1)$ *where we have as many terms in these sequences as needed in the specific context. Also, let* $[n] = \{1, \ldots, n\}$.

Definition 3. *For sequences* $I = (i_1, \ldots, i_m)$ *and* $J = (j_1, \ldots, j_m)$*, and an indeterminate* x*, we have the monomial*

$$x_I^J = x_{i_1}^{j_1} \cdots x_{i_m}^{j_m} , \tag{1.11}$$

with special cases $x_I^{\mathbb{0}} = 1$ *and* $x_I^{\mathbb{1}} = x_I$ *and* $x_I^{-\mathbb{1}} = 1/x_I = \overline{x}_I$.

Definition 4. *For sequences* $I = (i_1, \ldots, i_m)$ *and* $J = (j_1, \ldots, j_m)$*, and an indeterminate* x*, we have the coefficient extraction operator*

$$[x_I^J]F = [x_{i_1}^{j_1} \cdots x_{i_m}^{j_m}]F = F_I^J , \tag{1.12}$$

with special cases:

- $F_I^J = F$, *if* I *is empty.*
- $F_I^J = F_J$, *if* I *is the full set of variables ordered unambiguously.*

We will primarily be using this notation for extractions with respect to the x variables in the generating functions but, sometimes, we will be using such a coefficient extraction with respect to other variables, most commonly t. We will never mix such extractions with this notation.

Definition 5. *(?) For sequences* $I = (i_1, \ldots, i_m)$ *and* $J = (j_1, \ldots, j_m)$*, we define the term extraction operator as*

$$[[x_I^J]] = x_I^J [x_I^J] \,. \tag{1.13}$$

That is, $[[x_I^J]]F = x_I^J F_I^J$.

Example 6. Here is an illustrative example of the notation. If $Q(x_1, x_2; t)$ is given as in the previous example, we have

$$Q_1^0 = [x_1^0]Q(x_1, x_2) = Q(0, x_2) = 1 + a_1(x_2 + a_2)t^2 + \cdots \,, \tag{1.14}$$

$$Q_{1,2}^{0,0} = Q_{0,0} = Q(0, 0) = 1 + a_1 a_2 t^2 + \cdots \,. \tag{1.15}$$

Furthermore, we may write

$$S_1^{-1} = [\overline{x}_1](x_1 + \overline{x}_1)(x_2 + \overline{x}_2) = x_2 + \overline{x}_2 \tag{1.16}$$

and $S_{1,2}^{\pm 1, \pm 1} = 1$.

Chapter 2

One-dimensional Lattice Walk Models

2.1 Derivation of the Functional Equation

Before we tackle the higher-dimensional lattice walk problems, it will be instructive to visit the one-dimensional case which will help illustrate the methods and ideas that we will employ throughout this work. We begin with one-dimensional lattice walk models with smalls steps where the size of each step is at most one unit in each direction. Here the step set is $\mathcal{S} \subseteq \{\text{-}1, 0, 1\}$, which can also be represented via the step polynomial

$$S = S_{\text{-}1}\overline{x} + S_0 + S_1 x\,. \tag{2.1}$$

Figure 2.1: Illustration of regions on the line.

Here S_k is 1 or 0 if k is in $\mathcal{S}$ or not, respectively.

First we will look at a basic way to generate lattice walks and then we will look at how we can encapsulate it into an algebraic method. We start with the empty walk. We can grow it by doing one of three actions: step to the left, step to the right, or stay in one spot. Stepping to the left takes us to the negative axis and we do not want any walks that go there, so we ignore it. Thus we may extend the walk by stepping to the right or staying in one spot. These new walks have length one and they can be extended by each of the three actions. We can easily keep track of the visits to the origin - each time the walks stay or step on the origin, we increment a counter. We may continue this process as long as we wish, always removing any walks that end on negative coordinates.

Ultimately, it is easier to work with polynomials that encapsulate the important features in a smaller package. Also, working with polynomials and power series allows us to use the powerful methods of generating functions to gleam (?) into the inner workings of lattice walk models. To do this, we shall keep track of each walk's length via powers of t, ending position via powers of x, and visits to the origin with powers of a. We can perform the previous analysis in this language. We start with the empty walk of length zero. It corresponds to the monomial $1 = a^0x^0t^0$. Extending the walk corresponds to a multiplication by $\overline{x}$, x or 1, depending on whether we step to the left or to the right, or we stay on the same coordinate. The two resulting walks will have monomials xt and at, respectively. We have removed the walk that would have had a monomial $\overline{x}t$ since it ends on a negative coordinate. The next iteration has walks with monomials at^2, xt^2, x^2t^2, a^2t^2, axt^2. We can continue this procedure as long as we wish.

Of course, it would be difficult to work with individual lattice walks and it would be better to work with the whole generating function enumerating the walks. Suppose that the generating function for walks of length k is Q_kt^k. Then the full generating function is

$$Q = \sum_{k=0}^{\infty} Q_k t^k \,. \tag{2.2}$$

The key idea is to attempt to work with all the walks at the same time and extend all of them simultaneously. Doing this successfully will allow us to encapsulate the process above in an equation for Q and then manipulate it algebraically to find out more about the lattice walk model. We start by writing a functional equation relating Q_{k+1} and Q_k. All the walks of length k are encoded in Q_kt^k. To extend all of them by one step, we multiply the function by tS. By the distributive property, the product of the two sums wil lead to all possible multiples of monomials for lattice walks of length k and monomials for possible steps that we can take. However, some of the resulting monomials may have a negative power of x, corresponding to a walk ending on a negative coordinate. Here is another key idea. To remove the monomials with negative powers of x, we will use some kind of operator Γ, but we will worry about its representation later. For now we have the iteration

$$Q_{k+1}t^{k+1} = \Gamma(tSQ_kt^k) = \Gamma(SQ_k)t^{k+1} \,. \tag{2.3}$$

In other words, $Q_{k+1} = \Gamma(SQ_k)$ for all $k \geqslant 0$. We may sum both sides of the equation for $k \geqslant 0$ to find

$$\sum_{k=0}^{\infty} Q_{k+1}t^{k+1} = \sum_{k=0}^{\infty} \Gamma(SQ_k)t^k \tag{2.4}$$

We are working with formal series, after all, and we are looking for the easiest approach right now. It will be safe to assume that Γ does not interact with t and that we may factor Γ outside of the summation. After all, we are looking for an operator that will be able to work with the whole generating function. With these assumptions in hand, we may manipulate the equation as follows.

$$Q - Q_0 = t\Gamma S \sum_{k=0}^{\infty} Q_k t^k \tag{2.5}$$

$$Q = 1 + t\Gamma(SQ)\,. \tag{2.6}$$

We may rewrite the functional equation as

$$(1 - t\Gamma S)Q = 1\,, \tag{2.7}$$

so as to highlight the operator nature of the equation.

We just need to find a suitable representation for the operator Γ and will have completed the analysis. The operator needs remove all monomials with negative powers of x. However, the lowest order of x we may attain in SQ_k or SQ is -1. That is, all valid walks have power of x that is greater or equal to zero, and extending them by taking one step can only go as far as -1 on the axis. It is enough to take out the $\overline{x}$ term from the generating function. This can be achieved via coefficient extractions. Applying the operator $\overline{x}[\overline{x}]$ on a function extracts the $\overline{x}$ term from the function. The operator that eliminates such a term from a function is just

$$E = 1 - \overline{x}[\overline{x}]\,, \tag{2.8}$$

where 1 is the identity operator. We can also find an operator that marks walks ending on the origin. It is enough to multiply their monomials, which have no x, with the label a. This can be represented by the operator

$$L = 1 + (a-1)[x^0] = 1 + b[x^0] \tag{2.9}$$

where $b = a - 1$. This operator takes a function and add to it $(a-1)$ times the function's x^0 term. The final function has a x^0 term with coefficient $1 + (a-1) = a$.

Finally, to find Γ it is enough to multiply out the operators L and E as

$$\begin{aligned}\Gamma = LE &= (1-\overline{x}[\overline{x}])(1+b[x^0]) \\ &= 1 + b[x^0] - \overline{x}[\overline{x}] - \overline{x}[\overline{x}]b[x^0] \\ &= 1 + b[x^0] - \overline{x}[\overline{x}]\,.\end{aligned} \tag{2.10}$$

We would have obtained the same result if expaned the product EL instead. The last term vanished because once we have made the first extraction using $[x^0]$, that term no longer contains a variable x, and the $[\overline{x}]$ extraction picks out nothing. Here is an example that illustrates how the operators work in an actual problem.

Example 7. Suppose that $S = x + \overline{x}$. Starting with $Q_0 = 1$ we have

$$Q_1 = LES(1) = (1 + b[x^0] - \overline{x}[\overline{x}])(x + \overline{x}) = x + \overline{x} - \overline{x} = x \tag{2.11}$$

$$Q_2 = LES(Q_1) = (1 + b[x^0] - \overline{x}[\overline{x}])(x^2 + 1) = x^2 + 1 + b = a + x^2 \tag{2.12}$$

$$Q_3 = LES(Q_2) = (1 + b[x^0] - \overline{x}[\overline{x}])(a\overline{x} + ax + x + x^3)(?) \tag{2.13}$$

$$= a\overline{x} + ax + x + x^3 - a\overline{x} = (1+a)x + x^3\,. \tag{2.14}$$

Therefore,

$$\begin{aligned}Q &= Q_0 + Q_1 t + Q_2 t^2 + Q_3 t^3 + \cdots \\ &= 1 + xt + (a + x^2)t^2 + (ax + x + x^3)t^3 + \cdots\,.\end{aligned} \tag{2.15}$$

Note that if we apply Γ to tSQ, we obtain Q again but with the first term, $Q_0 = 1$, missing.

The theory so far is consistent with the present methods for finding functional equations for the generating functions for lattice walk models because in the end we reach the same functional equations. However, the perspective is different and this allows us to go further than before. To show the consistency with the theory so far, we will expand the operator form and write down a functional equation that does not involve operators. For the rest of the chapter we make a change of notation to reduce clutter. Let $Q_k = [x^k]Q$. We start with

$$\begin{aligned}[x^0]SQ &= [x^0]((S_{-1}\overline{x} + S_0 + S_1 x)Q) \\ &= (S_{-1}x[x] + S_0\overline{x}[\overline{x}] + S_1\overline{x}^2[\overline{x}^2])Q \\ &= S_{-1}Q_1 + S_0 Q_0\,,\end{aligned} \tag{2.16}$$

where last line follows since $Q_{-1} = 0$ since that would include walks ending on $\overline{x}$. A similar

computation shows that $[\overline{x}]SQ = S_{\text{-}1}Q_0$ (?) and this gives the functional equation

$$\begin{aligned} 1 &= Q - tLE(SQ) = Q - t(SQ + b[x^0]SQ - \overline{x}[\overline{x}]SQ \\ &= Q - t(SQ + b(S_{\text{-}1}Q_1 + S_0Q_0) - \overline{x}(S_{\text{-}1}Q_0)) \,. \end{aligned} \tag{2.17}$$

We can rewrite it in the following form

$$(1 - tS)Q - t(bS_0 - \overline{x}S_{\text{-}1})Q_0 - btS_{\text{-}1}Q_1 = 1 \,. \tag{2.18}$$

Here $Q = Q(x)$, $Q_0 = Q(0)$ and $Q_1 = Q'(0)$. It turns out that the Q_1 term is not needed. To see this, we first eliminate the Q_1 term by applying $[x^0]$ to both sides of the equation.

$$\begin{aligned} 1 &= Q_1^0 - t(S_{\text{-}1}Q_1 + S_0Q_0) - tbS_0Q_0 - btS_{\text{-}1}Q_1 \\ &= (1 - atS_0)Q_0 - atS_{\text{-}1}Q_1 \,. \end{aligned} \tag{2.19}$$

We can solve this equation for Q_1, that is,

$$tS_{\text{-}1}Q_1 = \overline{a}((1 - atS_0)Q_0 - 1) \,, \tag{2.20}$$

and substitute the result into the first functional equation. For now we do not worry about the possibility of $S_{\text{-}1} = 0$, because it leads to the same conclusion anyway. Then

$$\begin{aligned} 1 &= (1 - tS)Q - t(bS_0 - \overline{x}S_{\text{-}1})Q_0 - c((1 - atS_0)Q_0 - 1) \\ &= (1 - tS)Q - (c - t\overline{x}S_{\text{-}1})Q_0 + c \end{aligned} \tag{2.21}$$

and we can rewrite the equation as

$$(1 - tS)Q - (c - t\overline{x}S_{\text{-}1})Q_0 = \overline{a} \,, \tag{2.22}$$

since $1 - c = 1 - (1 - \overline{a}) = \overline{a}$. We will call this the master functional equation.

Example 8. When $S = x + \overline{x}$ we have the functional equation

$$(1 - t(x + \overline{x}))Q - (c - t\overline{x})Q_0 = \overline{a} \,. \tag{2.23}$$

It turns out that the back-substitution is unnecesary. This is a good thing because in higher dimensions we would have to perform an exponential number of back-substitutions to obtain such an equation: 2^d substitutions for a d-dimensional problem. Previous approaches in [29] and [**?**]

have only dealt with two-dimensional problems that only have a few back-substitutions. It is the use of operators that allows us to circumvent the problem altogether. To see why, consider the action of the marking operator L. To mark a monomial, we just multiply it by a. This means that, to undo the marking, we only need to multiply the monomial by $1/a$. That is, L has an inverse and we can write it down immediately. We have

$$L^{-1} = 1 + \left(\frac{1}{a} - 1\right)[x^0] = 1 - c[x^0]\,. \tag{2.24}$$

If we apply this to the functional equation, the right-hand side is simply $L^{-1}(1) = 1 - c = \overline{a}$ and the left-hand side is

$$\begin{aligned} L^{-1}(1 - tLES)Q &= L^{-1}Q - tESQ && (2.25)\\ &= (Q - cQ_0) - t(SQ - \overline{x}S_{-1}Q_0) && (2.26)\\ &= (1 - tS)Q - (c - t\overline{x}S_{-1})Q_0\,. && (2.27) \end{aligned}$$

This is a great shortcut indeed.

Proposition 9. *Let $S = S_{-1}\overline{x} + S_0 + S_1 x$ be the Laurent step polynomial for a one-dimensional lattice walk model restricted to the non-negative axis. Then the generating function Q satisfies the functional equations*

$$\begin{aligned} (L^{-1} - tES)Q &= \overline{a}\,, && (2.28)\\ (1 - tS)Q - (c - t\overline{x}S_{-1})Q_0 &= \overline{a}\,. && (2.29) \end{aligned}$$

The first one we call an operator equation, and the second, master functional equation.

The operator notation/perspective is powerful. It will allow us to derive functional equations for many lattice walk models in arbitrary dimensions with ease. Usually, derivations have employed inclusion-exclusion and careful book-keeping ([8], [15]) to write master functional equations, or their precursors, in a direct manner. Unfortunately, that approach is difficult to generalize to arbitrary dimensions and more dificult restrictions/marking.

We can also deal with coefficient extraction in a neater way. Instead of extracting $[x^k]$ from the master functional equation, it is better to first go through the operator equation, since $[x^k]$ interacts

with L and E in a nice way. We have

$$[x^0]L^{-1} = [x^0] - [x^0]c[x^0] = [x^0] - c[x^0] = \overline{a}[x^0] \tag{2.30}$$

$$[x^0]E = [x^0] - [x^0]\overline{x}[\overline{x}] = [x^0]\,. \tag{2.31}$$

When we extract the x^0 term from the functional equation, we have

$$\begin{aligned}[x^0]\overline{a} = \overline{a} &= [x^0](L^{-1} - tES)Q \\ &= ([x^0]L^{-1} - t[x^0]ES)Q \\ &= (\overline{a}[x^0] - t[x^0]S)Q \\ &= [x^0](\overline{a} - tS)Q\,.\end{aligned} \tag{2.32}$$

Therefore, we have a functional equation for the first coefficient

$$[x^0](1 - atS)Q = 1\,. \tag{2.33}$$

We can perform the extraction immediately and it simplifies to

$$(1 - atS_0)Q_0 - atS_{-1}Q_1 = 1\,. \tag{2.34}$$

Extracting higher coefficients is not difficult either. If $k \geqslant 1$, we have

$$[x^k]L^{-1} = [x^k] - [x^k]c[x^0] = [x^k]\,, \tag{2.35}$$

$$[x^k]E = [x^k] - [x^k]\overline{x}[\overline{x}] = [x^k]\,, \tag{2.36}$$

and

$$\begin{aligned}[x^k]\overline{a} = 0 &= [x^k](L^{-1} - tES)Q \\ &= ([x^k] - t[x^k]S)Q \\ &= [x^k](1 - tS)Q\,.\end{aligned} \tag{2.37}$$

Expanding this yields

$$\begin{aligned}Q_k - t(S_{-1}Q_{k+1} + S_0Q_k + S_1Q_{k-1}) &= 0 \\ (1 - S_0t)Q_k - t(S_{-1}Q_{k+1} + S_1Q_{k-1}) &= 0\,.\end{aligned} \tag{2.38}$$

Proposition 10. *Let $S = S_{-1}\overline{x} + S_0 + S_1x$ be the Laurent step polynomial for a one-dimensional*

lattice walk model. Let $Q_k = [x^k]Q$ be the generating functions for the walks ending at coordinate k. These functions satisfy the equations

$$(1 - atS_0)Q_0 - atS_{-1}Q_1 = 1\,, \tag{2.39}$$

$$(1 - S_0t)Q_k - t(S_{-1}Q_{k+1} + S_1Q_{k-1}) = 0\,, \tag{2.40}$$

where $k \geqslant 1$.

2.2 Solving Functional Equations For Various Step Sets

If a model has steps which can never be taken, we might as well remove them, and obtain a simpler model. The generating function will be the same in each case. This allows us to identify three major cases: $S = \delta$ or $S = \delta + x$ or $S = \delta + \overline{x} + x$. We will handle the first two cases simultaneously, and the third one, later.

2.2.1 The $S = \delta$ or $S = \delta + x$ case

This condition allows us to find Q_0 directly from the $[x^0]$ equation.

$$Q_0 = \frac{1}{1 - a\delta t}\,. \tag{2.41}$$

We can substitute this into the master functional equation.

Proposition 11. *Let $S = \delta$ or let $S = \delta + x$. Then the full generating function is given by*

$$Q(x, a; t) = \frac{\overline{a}}{1 - tS}\left(1 + \frac{b}{1 - a\delta t}\right) = \frac{1 - \delta t}{(1 - a\delta t)(1 - tS)}\,. \tag{2.42}$$

If we let $a = 1$ we find that

$$\widehat{Q} = \frac{1}{(1 - tS)}\,, \tag{2.43}$$

and therefore we have the linear relation

$$(1 - a\delta t)Q - (1 - \delta t)\widehat{Q} = 0\,, \tag{2.44}$$

We can write it as $B((1-a\delta t)Q) = 0$ using the operator $B = 1-[b^0]$. There is a similar relationship

for Q_0. Extracting the x^0 term from the last equation leads to

$$B((1-a\delta t)Q_0) = 0 \tag{2.45}$$

or in other words,

$$Q_0 = \frac{1-a\delta t}{1-\delta t}\widehat{Q}_0\,. \tag{2.46}$$

Proposition 12. *Let $B = 1 - [b^0]$ and let Q be the generating function for a lattice walk model with $S = \delta$ or $S = \delta + x$. Then Q and Q_0 satisfy the functional equations*

$$B((1-a\delta t)Q) = 0 \quad \textit{and} \quad B((1-a\delta t)Q_0) = 0\,. \tag{2.47}$$

In other words,

$$Q = \frac{1-a\delta t}{1-\delta t}\widehat{Q} \quad \textit{and} \quad Q_0 = \frac{1-a\delta t}{1-\delta t}\widehat{Q}_0\,. \tag{2.48}$$

Even such small examples already show connections between Q and $\widehat{Q}$. It will be interesting to find such relationships for higher-dimensional lattice walk models.

2.2.2 The $S = \delta + \overline{x} + x$ case

In this case we will apply the Obstinate Kernel method, as is done in the two-dimensional cases in [29], page 11 of [**?** , p=11]. ZZCITEZZ It requires finding an x value that makes the kernel $K = 1 - tS$ vanish in the functional equation

$$(1-tS)Q - (c - t\overline{x})Q_0 = \overline{a}\,. \tag{2.49}$$

The equation $1 - tS = tx^2 - (1-\delta t)x + t = 0$ is quadratic with two roots that are functions of t. We have to pick the one that has no singularity at $t = 0$. The roots are

$$\sigma = \frac{1-\delta t - \sqrt{(1-\delta t)^2 - 4t^2}}{2t} \quad \text{and} \quad \sigma^* = \frac{1-\delta t + \sqrt{(1-\delta t)^2 - 4t^2}}{2t}\,, \tag{2.50}$$

and they multiply to 1, so $\overline{\sigma} = \sigma^*$.

Next, we substitute $x = \sigma$ in the equation. For now we have to assume that the whole term KQ

vanishes and we obtain

$$-(c - t\overline{\sigma})Q_0 = \overline{a}\,. \tag{2.51}$$

Now we can solve for Q explicitly. We calculate

$$Q = \frac{\overline{a}}{1 - tS}\left(1 - \frac{t\overline{x} - c}{t\overline{\sigma} - c}\right) = \frac{t}{1 - tS}\left(\frac{\overline{x} - \overline{\sigma}}{b - a\overline{\sigma}t}\right)\,. \tag{2.52}$$

Proposition 13. *Let $S = \delta + \overline{x} + x$. Then the generating functions for this model are*

$$Q_0 = \frac{1}{a\overline{\sigma}t - b} \quad \textit{and} \quad Q = \frac{t(\overline{\sigma} - \overline{x})}{(1 - tS)(a\overline{\sigma}t - b)}\,, \tag{2.53}$$

where $\overline{\sigma} = (1 - \delta t + \sqrt{(1 - \delta t)^2 - 4t^2})/(2t)$.

As before, we can find a linear relationship between $Q(x, a; t)$ and $Q(x, 1; t)$. Note that $(a\overline{\sigma}t - b)Q_0$ and $(a\overline{\sigma}t - b)Q$ are functions that do not depend on a. Therefore, they are annihilated by the operator B.

Proposition 14. *Let $S = \delta + \overline{x} + x$ and $f = a\overline{\sigma}t - b$. Then $B(fQ) = B(fQ_0) = 0$.*

We can also look at the problem from a different angle - that of the Algebraic Kernel method used extensively in the quarter-plane model classification ZZCITEZZ. We find a polynomial $f(a, t)$, for which $B(fQ_0) = c$. This will serve to illustrate the group approach together with the kernel method. Note that S, and therefore K, is invariant under the map $* : x \mapsto \overline{x}$, and this allows us to write a second functional equation to complement the first. That is,

$$(1 - tS)Q - (c - t\overline{x})Q_0 = \overline{a}\,, \tag{2.54}$$

$$(1 - tS)Q^* - (c - tx)Q_0 = \overline{a}\,. \tag{2.55}$$

To eliminate the Q_0 term ,we multiply the first equation by $c - tx$, the second by $c - t\overline{x}$, and subtract one equation from the other. We divide everything by $1 - tS$ and the equation that remains is

$$(c - tx)Q(x) - (c - t\overline{x})Q(\overline{x}) = \overline{a}F\,, \tag{2.56}$$

where

$$F = \frac{t(\overline{x} - x)}{1 - tS} \tag{2.57}$$

and crucially, F does not depent on a. We extract the x^1 coefficient from the equation to acquire an equation relating Q_1 and Q_0

$$cQ_1 - tQ_0 = \overline{a}F_1\,. \tag{2.58}$$

We can set $a = 1$ here to find that

$$F_1 = -t\widehat{Q}_0\,, \tag{2.59}$$

and we can back-substitute this quantity into the original equation.

$$\begin{aligned} cQ_1 &= tQ_0 - \overline{a}t\widehat{Q}_0 \\ bQ_1 &= t(aQ_0 - \widehat{Q}_0)\,. \end{aligned} \tag{2.60}$$

Earlier we had the equation

$$(1 - a\delta t)Q_0 = 1 + atQ_1 = 1 + atQ_1\,, \tag{2.61}$$

which we can solve for Q_1 this time and insert the solution into the above equation

$$b((1 - a\delta t)Q_0 - 1) = t^2a(aQ_0 - \widehat{Q}_0)\,. \tag{2.62}$$

Collecting coefficients and simplifying leads us to the equation

$$(c(1 - a\delta t) - at^2)Q_0 - (-t^2)\widehat{Q}_0 = c\,, \tag{2.63}$$

and therefore, $f(a,t) = c(1 - a\delta t) - at^2$.

Proposition 15. *Let $S = \delta + \overline{x} + x$ and $f = c(1 - a\delta t) - at^2$. Then $B(fQ_0) = c$.*

With two different approaches we obtained slightly different relationships upon applying the operator B.

2.2.3 Conclusion

In the one-dimensional case, we were able to find explicit solutions for Q_0 and Q various relationships for Q and $\widehat{Q}$, together with the operator B. In higher dimensions this analysis will be more difficult, but we will find some B-operator relationships for the functions enumerating lattice walks returning to the origin. Here is a table showing the resuls for the one-dimensional lattice walk models with small steps and interaction with the origin.

S	Q_0	Q	$f(a,t)$
δ	$\dfrac{1}{1-a\delta t}$	$\dfrac{1}{1-a\delta t}$	$1-a\delta t$
$\delta+x$	$\dfrac{1}{1-a\delta t}$	$\dfrac{1-\delta t}{(1-t(\delta+x))(1-a\delta t)}$	$1-a\delta t$
$\delta+\overline{x}+x$	$\dfrac{1}{a\overline{\sigma}t-b}$	$\dfrac{t(\overline{\sigma}-\overline{x})}{(1-t(\delta+\overline{x}+x))(a\overline{\sigma}t-b)}$	$a\overline{\sigma}t-b$

Table 2.1: Generating functions for walks with small steps on the non-negative half-line, where $B(fQ_0) = B(fQ) = 0$.

Chapter 3

Higher-dimensional Lattice Walk Models

We can easily generalize the one-dimensional results to two or more dimensions. The following diagram illustrates the two-dimensional case.

Figure 3.1: Two-dimensional lattice walk model constraints.

In the past we have performed inclusion-exclusion to arrive at functional equations for the two-dimensional problems, but when it comes to generalizing to higher dimensions, the process is difficult and prone to error. For a view of difficulties involved in not using the operator notation, see [29] where the authors had to wrestle with case-by-case analysis. Instead, we use operators to derive the functional equations cleanly and without pain. First, we will illustrate the inclusion-exclusion with the non-interacting two-dimensional problem. This is the usual procedure carried out in problems so far [8]. Similarly to the one-dimensional problem, we start with the generating function Q for the lattice walks on the quarter-plane. Then we can extend the walks by one step via a multiplication by tS and remove those that step outside of the quarter-plane. To do this we can subtract the walks that end on coordinates $(\text{-}1, k)$ or $(k, \text{-}1)$ for integers k. However, we have removed too many walks. We need to re-add those walks that end on $(\text{-}1, \text{-}1)$.

Figure 3.2: Inclusion-exclusion of regions.

Definition 16. *The ZZCLASSICZZ d-dimensional lattice walk model is the one with small steps in the positive orthant in $\mathbb{Z}^d$. The interacting ZZCLASSICZZ model is the ZZCLASSICZZ d-dimensional model with interactions with the hyperplanes $x_k = 0$ bounding the positive orthant.*

The walks that step on $(\text{-}1, k)$ are those coming from the boundary and stepping with steps in the negative x_1 direction. To compute the generating function term, we start with Q_1^0 and multiply by $t\overline{x}_1 S_1^{-1}$. The situation is similar for the $(k, \text{-}1)$ coordinates. Finally, we re-add those walks that step on $(\text{-}1, \text{-}1)$. Those are walks coming from the origin in a diagonal step, so we multiply $Q_{1,2}^{0,0}$ by $t\overline{x}_1\overline{x}_2 S_{1,2}^{-1,-1}$. We have obtained Q again but with the t^0 term missing.

This leads to the functional equation for Q

$$Q = 1 + tSQ - t\overline{x}_1 S_1^{-1} Q_1^0 - t\overline{x}_2 S_2^{-2} Q_2^0 + t\overline{x}_1\overline{x}_2 S_{1,2}^{-1,-1} Q_{1,2}^{0,0}\,, \tag{3.1}$$

which is rather large even for two dimensions. Contrast this notation to the one in [8] (page 10). The problem quickly runs into notational difficulties for d-dimensional problems, but that can be dealt with by writing the equation as a sum over subsets of $\{1, \ldots, d\}$, like it is done for the non-interacting case in [15]. This method runs into trouble when we start considering interactions. We take the functional equation for the non-interacting case and modify it to add interactions. This has been done before in ZZCITEZZ by adding in terms like $(a_1 - 1)T$ where T is some expression involving extractions of Q and S, but we will not go into the messy details. Instead, we will use the operator notation to handle the problem in a much more efficient manner.

3.1 Using Operators to Derive the Functional Equation

If we use operators we can build the generalizations to higher dimensions in a straight-forward manner. As we have done before, we multiply Q by tS, so we can extend the lattice walks by one step. Then we apply an operator E, to be determined later, to remove all the walks venturing outside of the positive orthant (nonzero coordinates x_k). Finally, we apply a marking operator to this term that marks the walks on the boundary hyperplanes $x_k = 0$. Since we have extended all walks by one step, we no longer have walks of size 0. Adding that term to the expression leads us back to Q itself. That is,

$$Q = 1 + LE(tSQ) \tag{3.2}$$

and the only task remaining is to find L and E. Like in the one-dimensional problem, we start removing monomials with coordinates outside of the positive orthant. The important idea here is that, if we have operators that just remove monomials, then it does not matter in what order we

remove them. Let $E_k = 1 - \overline{x}_k[\overline{x}_k]$ be the operator removing monomials with x_k coordinate equal to -1. Then we may write $E = E_1 \cdots E_d$. For example,

$$E_1E_2 = (1 - \overline{x}_1[\overline{x}_1])(1 - \overline{x}_2[\overline{x}_2]) = 1 - \overline{x}_1[\overline{x}_1] - \overline{x}_2[\overline{x}_2] + \overline{x}_1\overline{x}_2[\overline{x}_1\overline{x}_2]\,, \tag{3.3}$$

and this shows that the inclusion-exclusion is a natural consequence of how the operator is set up. Similarly, if we let $L_k = 1 + (a_k - 1)[x_k^0]$, then we may write $L = L_1 \cdots L_d$. Here order does not matter since we are marking monomials with variables a_k. It does not matter if we first multiply by a_1 and then by a_2, or vice-versa. As in the one-dimensional case, the operators L_k are invertible with inverses $L_k^{-1} = 1 - c_k[x_k^0]$, and thus, the operator L is also invertible. We can apply the operator L^{-1} to the functional equation

$$L^{-1}(1 - tLES)Q = (L^{-1} - tES)Q = L^{-1}(1) = \sum_{I \subseteq [d]} c_I = \overline{a}_{[d]}\,. \tag{3.4}$$

This allows us to write down the following result.

Proposition 17. *The ZZCLASSICZZ interacting d-dimensional lattice walk model has a full generating function Q satisfying the functional equations*

$$(1 - tLES)Q = 1\,, \tag{3.5}$$

$$(L^{-1} - tES)Q = \overline{a}_{[d]}\,, \tag{3.6}$$

where L, E, L^{-1} are of the form $\Gamma = \Gamma_1 \cdots \Gamma_d$.

Previously, we have had to deal with coefficient extractions and back-substitution to find the expanded form of the second functional equation (page 9 of [29] and page 12 of [**?**]). Using the operator notation allows us to skip this process. In general, the second functional equation has many less terms. To see this, we take a look at the operator L_kE_k. We have

$$L_kE_k = 1 + b_k[x_k^0] - \overline{x}_k[\overline{x}_k] \tag{3.7}$$

and if we apply the operator to an equation, we increases the number of terms in the equation by a factor of 3. We have 3^d terms relating to SQ by the end of applying LE. On the other hand, once we apply the operator L^{-1}, we introduce 2^d terms relating to Q, but since we only have E applied on SQ, we would only have at most 2^d terms from this. For the case of walks of step size at most 1 and labelling the hyperplane interactions, we have exactly 2^d terms $Q_I^{\emptyset}$, one for each subset $I \subseteq [d]$.

In this case we can write out the functional equation explicitly. Firstly,

$$\begin{aligned} L^{-1}Q &= (1 - c_1[x_1^0]) \cdots (1 - c_d[x_d^0])Q \\ &= \sum_{I \subseteq [d]} (-1)^{|I|} c_I Q_I^{\mathbb{0}} \end{aligned} \tag{3.8}$$

and

$$\begin{aligned} ESQ &= (1 - \overline{x}_1[\overline{x}_1]) \cdots (1 - \overline{x}_d[\overline{x}_d])(SQ) \\ &= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I [\overline{x}_I^{\mathbb{1}}](SQ) \\ &= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{1}} Q_I^{\mathbb{0}} \,, \end{aligned} \tag{3.9}$$

where the last line follows from the fact that there are no negative powers of x_k in Q for any k. Therefore, the functional equation is

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x}_I S_I^{-\mathbb{1}}) Q_I^{\mathbb{0}} = \overline{a}_{[d]} \,. \tag{3.10}$$

As in the two-dimensional case, we let $K_I = c_I - t\overline{x}_I S_I^{-\mathbb{1}}$ where, if $I = \varnothing$, we have the empty products and empty extractions, leaving us with $K_\varnothing = K$. Finally, we can write the master equation in d dimensions.

Theorem 18. *The ZZCLASSICZZ interacting d-dimensional lattice walk model, has a full generating function Q satisfying the functional equation*

$$\sum_{I \subseteq [d]} (-1)^{|I|} K_I Q_I^{\mathbb{0}} = \overline{a}_{[d]} \tag{3.11}$$

where $K_I = (c_I - t\overline{x}_I S_I^{-\mathbb{1}})$.

This is the interacting version of the functional equation found in [15]. In our notation this is

$$KQ - \sum_{\substack{I \subseteq [d] \\ I \neq \varnothing}} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{1}} Q_I^{\mathbb{0}} = 1 \,. \tag{3.12}$$

3.2 Coefficient Extractions

3.2.1 Using Operator Notation For Fast Coefficient Extraction

As in the two-dimensional case, we can now perform coefficient extractions with $[x_I^{\mathbb{0}}]$ to attain further functional equations relating $Q_I^{\mathbb{0}}$ to other Q terms of the same variables or fewer variables. The operator $[x_i^k]$ commutes with all operators M_j and E_j with $j \neq i$, so we only need to consider the action on the operators $M_i = L_i^{-1}$ and E_i. We have already done so in the one-dimensional case. The zeroth power extraction yields

$$[x_i^0]M_i = [x_i^0] - [x_i^0]c_i[x_i^0] = [x_i^0] - c_i[x_i^0] = \overline{a}_i[x_i^0]\,, \tag{3.13}$$

$$[x_i^0]E_i = [x_i^0] - [x_i^0]\overline{x}_i[\overline{x}_i] = [x_i^0]\,, \tag{3.14}$$

and the k-th power extraction for $k \geqslant 1$ yields

$$[x_i^k]M_i = [x_i^k] - [x_i^k]c_i[x_i^0] = [x_i^k]\,, \tag{3.15}$$

$$[x_i^k]E_i = [x_i^k] - [x_i^k]\overline{x}_i[\overline{x}_i] = [x_i^k]\,. \tag{3.16}$$

Therefore, for $k \geqslant 0$ the operator $[x_i^k]$ eliminates both M_i and E_i from the picture, while for $k = 0$ it removes E_i but leaves a residual $\overline{a}_i$ behind for M_i. Therefore, if we extract $[x_I^{\mathbb{0}}]$ from the functional equation, we have

$$\begin{aligned}[x_I^{\mathbb{0}}]a_{[d]} = \overline{a}_{[d]} &= [x_I^{\mathbb{0}}](M_{[d]} - tE_{[d]}S)Q \\ &= \left(\overline{a}_I M_{[d]\setminus I}[x_I^{\mathbb{0}}] - tE_{[d]\setminus I}[x_I^{\mathbb{0}}]S\right)Q \\ &= [x_I^{\mathbb{0}}](\overline{a}_I M_{[d]\setminus I} - tE_{[d]\setminus I}S)Q\,,\end{aligned} \tag{3.17}$$

so that

$$\overline{a}_{[d]\setminus I} = [x_I^{\mathbb{0}}](M_{[d]\setminus I} - a_I tE_{[d]\setminus I}S)Q\,. \tag{3.18}$$

If we rewrite this, we have

$$\overline{a}_I = [x_{[d]\setminus I}^{\mathbb{0}}](M_I - a_{[d]\setminus I}tE_I S)Q \tag{3.19}$$

and we can use whichever notation is more useful at the time. The latter notation underscores the idea that, when we extract coefficients for walks that end at some hyperplane intersection, we are having a reduced version of the functional equation at the cost of an added a term. When we extract

a general term, we can use the operator $[x_I^{\mathbb{0}} x_J^P]$, where P is a set of powers greater than 0 and $I \cap J = \varnothing$, $|P| \geqslant 1$ and $I \cup J = K$. We find that

$$\begin{aligned} [x_I^{\mathbb{0}} x_J^P] a_{[d]} = 0 &= [x_I^{\mathbb{0}} x_J^P](M_{[d]} - tE_{[d]}S)Q \\ &= [x_I^{\mathbb{0}} x_J^P](\overline{a}_I M_{[d]\setminus K} - tE_{[d]\setminus K}S)Q\,, \end{aligned} \tag{3.20}$$

which we can write as

$$[x_I^{\mathbb{0}} x_J^P](M_{[d]\setminus K} - a_I tE_{[d]\setminus K}S)Q = 0\,. \tag{3.21}$$

If I is empty, then

$$[x_J^P](M_{[d]\setminus J} - tE_{[d]\setminus J}S)Q = 0\,. \tag{3.22}$$

The main role of these calculations is to provide a shortcut for manual calculations. After all, we need these to obtain functional equations for $Q_I^{\mathbb{0}}$ for $I \subseteq [d]$.

3.2.2 Further Functional Equations

We need to go further in order to find equations similar to the master functional equation. We know that

$$[x_I^{\mathbb{0}}]SQ = S_I^{\mathbb{0}} Q_I^{\mathbb{0}} + R_I\,, \tag{3.23}$$

where R_I is a collection of $S_I^{-P} Q_I^P$ terms that involves sets P whose terms are not all zero. We can use this inside the functional equation

$$\begin{aligned} \overline{a}_{[d]\setminus I} &= M_{[d]\setminus I}[x_I^{\mathbb{0}}]Q - a_I tE_{[d]\setminus I}[x_I^{\mathbb{0}}](SQ) \\ &= M_{[d]\setminus I} Q_I^{\mathbb{0}} - a_I tE_{[d]\setminus I}(S_I^{\mathbb{0}} Q_I^{\mathbb{0}} + R_I) \\ &= (M_{[d]\setminus I} - a_I tE_{[d]\setminus I} S_I^{\mathbb{0}}) Q_I^{\mathbb{0}} - a_I tE_{[d]\setminus I} R_I \end{aligned} \tag{3.24}$$

so we can rewrite it as

$$(M_{[d]\setminus I} - a_I tE_{[d]\setminus I} S_I^{\mathbb{0}}) Q_I^{\mathbb{0}} = a_I(\overline{a}_{[d]} + tR_I')\,, \tag{3.25}$$

where $R'_I = E_{[d]\setminus I} R_I$. We compute the following terms

$$M_{[d]\setminus I} Q_I^{\mathbb{0}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} c_J Q_{I,J}^{\mathbb{0},\mathbb{0}}\,, \tag{3.26}$$

$$a_I t E_{[d]\setminus I} S_I^{\mathbb{0}} Q_I^{\mathbb{0}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} a_I t \overline{x}_J S_{I,J}^{\mathbb{0},\text{-}\mathbb{1}} Q_{I,J}^{\mathbb{0},\mathbb{0}}\,, \tag{3.27}$$

and the functional equation can be written as

$$\sum_{J \subseteq [d]\setminus I} (-1)^{|J|} (c_J - a_I t \overline{x}_J S_{I,J}^{\mathbb{0},\text{-}\mathbb{1}}) Q_{I,J}^{\mathbb{0},\mathbb{0}} = a_I (\overline{a}_{[d]} + t R'_I)\,. \tag{3.28}$$

This suggests the following notation.

Definition 19. *For sets I and J that do not share any elements, we write*

$$K_J(I) = c_J - a_I t \overline{x}_J S_{I,J}^{\mathbb{0},\text{-}\mathbb{1}} \tag{3.29}$$

where if $I = \varnothing$, then $K_J(I) = K_J$.

This notation allows us to write the 2^d functional equations for $Q_I^{\mathbb{0}}$ and $I \subseteq [d]$ in simplified form.

Corollary 20. *Let I be a subset of $\{1, \ldots, d\}$. Then the generating function $Q_I^{\mathbb{0}}$ for the d-dimensional interacting ZZCLASSICZZ lattice walk model satisfies the functional equation*

$$\sum_{J \subseteq [d]\setminus I} (-1)^{|J|} K_J(I) Q_{I,J}^{\mathbb{0},\mathbb{0}} = \overline{a}_{[d]\setminus I} + a_I t R'_I \tag{3.30}$$

where the R'_I term only involves higher-order coefficient extractions from Q; that is, where not all the powers of variables x are zero.

This is a clean way to write down all the equations for $Q_I^{\mathbb{0}}$ in a unifying way. To compute the R'_I terms we first determine what R_I is. It is a sum of terms $S_I^{-P} Q_I^P$ for various non-negative P with size $|I|$ whose elements are not all 0. To calculate R'_I we apply the operator $E_{[d]\setminus I}$ on R_I. We can try calculating some of these quantities. A key observation is that the terms Q_I^P are power series in

the remaining variables, so they do not have non-negative coefficients. Let $sq = S_I^{-P} Q_I^P$. Then

$$\begin{aligned} E_j sq &= sq - \overline{x}_j (sq)_j^{-1} \\ &= sq - \overline{x}_j (s_j^{-1} q_j^0 + s_j^0 q_j^{-1} + s_j^1 q_j^{-2}) \\ &= sq - \overline{x}_j s_j^{-1} q_j^0 \,. \end{aligned} \tag{3.31}$$

Hence

$$E_K sq = \sum_{J \subseteq K} (-1)^{|J|} \overline{x}_J s_J^{-\mathbb{1}} q_J^{\mathbb{0}} \,. \tag{3.32}$$

Finally, we can write

$$\begin{aligned} R'_I &= E_{[d]\setminus I} R_I \\ &= \sum_{P > \mathbb{0}} E_{[d]\setminus I} S_I^{-P} Q_I^P \\ &= \sum_{P > \mathbb{0}} \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} S_{I,J}^{-P,-\mathbb{1}} Q_{I,J}^{P,\mathbb{0}} \\ &= \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} \sum_{P > \mathbb{0}} S_{I,J}^{-P,-\mathbb{1}} Q_{I,J}^{P,\mathbb{0}} \,. \end{aligned} \tag{3.33}$$

Proposition 21. *The functions R'_I are given by*

$$R'_I = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} \sum_{P > \mathbb{0}} S_{I,J}^{-P,-\mathbb{1}} Q_{I,J}^{P,\mathbb{0}} \,. \tag{3.34}$$

Of course, when $d =$ or $d = 2$ we do not need the full generality of this formula, but for higher dimensions, it would be quite difficult to write the equations by hand without it.

3.2.3 Kernel Forms

We can write out the functional equations in a way that highlights the kernel K in the equation. We start out as follows

$$\begin{aligned} (1 - tLES)Q &= 1 \\ (1 - t(LE - 1 + 1)S)Q &= 1 \\ KQ + t(1 - LE)SQ &= 1 \,. \end{aligned} \tag{3.35}$$

This is the form we see in operator functional equation. Here is another way we could have expanded.

$$\begin{aligned}(1 - tLES)Q &= 1\\ (1 + LE(1 - tS - 1))Q &= 1\\ LEKQ + (1 - LE)Q &= 1\\ LEKQ + (1 - L)Q &= 1\,,\end{aligned} \tag{3.36}$$

where $EQ = Q$ since Q has no terms that need removing. Something interesting happens when we set all the a_k's to 1 in the last equation This is equivalent to removing the marking operator. The equation becomes.

$$E(KQ) = 1\,. \tag{3.37}$$

The form of this equation suggests that we use the following substitution

$$Q = \frac{W}{x_1 \cdots x_d K}\,. \tag{3.38}$$

Then we multiply both sides of the functional equation by $x_1 \cdots x_d$ and obtain

$$x_1 \cdots x_d = x_1 \cdots x_d E(KQ) = x_1 \cdots x_d E(\overline{x}_1 \cdots \overline{x}_d W) = E'W\,, \tag{3.39}$$

where $E' = E'_1 \cdots E'_d$ and

$$E'_k = x_k E_k \overline{x}_k = x_k(1 - \overline{x}_k[\overline{x}_k])\overline{x}_k = 1 - [x_k]^0\,. \tag{3.40}$$

Proposition 22. *Let $W = x_1 \cdots x_d KQ$, where Q is the generating function for walks without interactions in the positive orthant. Then*

$$E'W = \sum_{I \subseteq [d]} (-1)^{|I|} W_I^{\mathbb{0}} = x_1 \cdots x_d\,. \tag{3.41}$$

For example, in two dimensions we have

$$W(x_1, x_2) - W_1^0(x_2) - W_2^0(x_1) + W_{1,2}^{0,0} = x_1 x_2\,. \tag{3.42}$$

Thus, we can write the solution to the problem as

$$Q = \frac{1}{x_1 \cdots x_d K} \left(x_1 \cdots x_d - \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} W_I^{\mathbb{0}} \right). \tag{3.43}$$

In two dimensions this becomes

$$Q(x_1, x_2) = \frac{x_1 x_2 + W_1^0(x_2) + W_2^0(x_1) - W_{1,2}^{0,0}}{x_1 x_2 (1 - tS)}. \tag{3.44}$$

Example 23. Consider the one-dimensional model on the non-negative line with $S = \overline{x} + x$. Here the solution for the interacting version of the model is given by

$$Q = \frac{t(\overline{\sigma} - \overline{x})}{(1 - tS)(a\overline{\sigma}t - b)}, \tag{3.45}$$

so when we set $a = 1$ and simplify the expression, we find that the generating function for the non-interacting model is given by

$$\widehat{Q} = \frac{x - \sigma}{x(1 - tS)}, \tag{3.46}$$

as predicted by the analysis of this section. Here $W_0 = \sigma$.

Of course, each $W_I^{\mathbb{0}}$ depends on $Q_I^{\mathbb{0}}$, so we could have seen this relationship from the master functional equation for Q too. Interestingly enough, this calculation points to a differential equation. Clearly, W satisfies the differential equation

$$\partial_1 \partial_2 W = 1 \tag{3.47}$$

and a similar relationship exists for the d-dimensional models.

Proposition 24. *The full generating function Q for d-dimensional non-interacting lattice walks in the positive orthant with small steps, satisfies the inhomogeneous PDE*

$$\partial_1 \cdots \partial_d W = \partial_1 \cdots \partial_d (x_1 \cdots x_d K Q) = 1. \tag{3.48}$$

For further generalization, see Chapter 6.

3.3 First Solutions

3.3.1 Formal Neumann Series

So far we have obtained the equation

$$(1 - tLES)Q = 1\,. \tag{3.49}$$

Let $\Gamma = LE$. Then we can think of the formal inverse of this operator given as Neumann series as

$$(1 - t\Gamma S)^{-1} = \frac{1}{1 - t\Gamma S} = \sum_{k=0}^{\infty} t^k (\Gamma S)^k\,. \tag{3.50}$$

We can write the series solution

$$Q = \sum_{k=0}^{\infty} t^k (\Gamma S)^k(1) \tag{3.51}$$

where $(\Gamma S)^k(1) = \Gamma S(\cdots(\Gamma S(1))\cdots)$. This allows us to compute the series expansion for Q directly using an iteration. The next coefficient is the previous one with ΓS applied to it. It can be implemented easily in symbolic calculation software like Maple or Mathematica. The iteration is a formalized way of finding the first few terms of the series in the naive way: by taking a set of walks of length n, extending them by 1 step, removing the ones that end outside of the quarter-plane, and finally by marking the ones that interact with the hyperplanes $x_i = 0$ at their endpoint. We have not been able to find closed forms for the repeated operator applications in general. Here is an example that illustrates the use of the operators.

Example 25. Let $S = x_1 + \overline{x}_1 + x_2 + \overline{x}_2 = S_1(x_1) + S_2(x_2)$. We start with $\Gamma S^0(1) = 1$. Then

$$\begin{aligned}
\Gamma S^1(1) &= LE(S_1 + S_2) = L(x_1 + x_2) = a_2x_1 + a_1x_2\,,\\
\Gamma S^2(1) &= LE(S_1 + S_2)(a_2x_1 + a_1x_2)\\
&= LE(a_2S_1x_1 + a_1x_2S_2 + a_2x_1S_2 + a_1x_2S_1)\\
&= L(a_2S_1x_1 + a_1x_2S_2 + (a_1 + a_2)x_1x_2)\\
&= a_1a_2^2 + a_2^2x_1^2 + a_1a_2^2 + a_1^2x_2^2 + (a_1 + a_2)x_1x_2
\end{aligned}$$

and this allows us to write

$$Q = 1 + (a_2x_1 + a_1x_2)t + \left(a_1a_2(a_1 + a_2) + (a_2^2x_1^2 + a_1^2x_2^2) + (a_1 + a_2)x_1x_2\right)t^2 + \cdots . \tag{3.52}$$

3.3.2 Hadamard Products

We take a moment to see a result that falls out of the operator notation and which was covered in more generality in [18], where the authors consider a generalization of the situation $S(x_1, x_2) = S_0(x_1) + S_1(x_1)S_2(x_2)$ but for the non-interacting case. We consider the simpler case $S = S_1(x_1)S_2(x_2)$ and its generalizations to arbitrarily-many dimensions, for the interacting case.

Example 26. When $d = 2$ we have the special case $S = S_1(x_1)S_2(x_2)$. The operator ΓS factors as

$$\Gamma S = \Gamma_1\Gamma_2 S_1 S_2 = (\Gamma_1 S_1)(\Gamma_2 S_2) \tag{3.53}$$

because Γ_2 does not interact with S_1 and the operators commute. Then

$$(\Gamma S)^k = (\Gamma_1\Gamma_2 S_1 S_2)^k = (\Gamma_1 S_1)^k(\Gamma_2 S_2)^k \tag{3.54}$$

and we use the Neumann series to find

$$Q = \sum_{k=0}^{\infty} t^k(\Gamma S)^k = \sum_{k=0}^{\infty} t^k(\Gamma_1 S_1)^k(\Gamma_2 S_2)^k = \left(\sum_{k=0}^{\infty} t^k(\Gamma_1 S_1)^k\right) \odot_t \left(\sum_{k=0}^{\infty} t^k(\Gamma_2 S_2)^k\right), \tag{3.55}$$

where $\odot_t$ denotes the Hadamard product of two power series with respect to variable t. From now on we will drop the subscript t unless it is needed. Here we may write

$$Q_i(a_i, x_i; t) = \sum_{k=0}^{\infty} t^k(\Gamma_i S_i)^k \tag{3.56}$$

and by reversing the iteration, we reach the functional equations

$$(1 - t\Gamma_i S_i)Q_i = 1 \tag{3.57}$$

for each i. The functions Q_i are solutions to one-dimensional problems with step sets S_i and Q is

expressed as

$$Q(a_1, a_2, x_1, x_2; t) = Q_1(a_1, x_1; t) \odot_t Q_2(a_2, x_2; t) \tag{3.58}$$

Example 27. Once we know the form of the solution, we can prove that it works by a method that does not use the series solution. Suppose that Q_i is a solution to the functional equation

$$(L_i^{-1} - tE_iS_i)Q_i = \overline{a}_i \tag{3.59}$$

for $i = 1$ and $i = 2$. Then we claim that $Q = Q_1 \odot Q_2$ is a solution to the functional equation

$$(L^{-1} - tES)Q = \overline{a}_1\overline{a}_2\,. \tag{3.60}$$

To this end, we compute

$$\begin{aligned}(L^{-1} - tES)Q &= L^{-1}(Q_1 \odot Q_2) - tES(Q_1 \odot Q_2) \\ &= (L_1^{-1}Q_1 \odot L_2^{-1}Q_2) - t((E_1S_1Q_1) \odot (E_2S_2Q_2))\,.\end{aligned} \tag{3.61}$$

Then we substitute $L_i^{-1}Q_i = \overline{a}_i + tE_iS_iQ_i$ into the first term and expand it to find

$$\begin{aligned}(L_1^{-1}Q_1 \odot L_2^{-1}Q_2) &= (\overline{a}_1 + tE_1S_1Q_1) \odot (\overline{a}_2 + tE_2S_2Q_2) \\ &= \overline{a}_1 \odot \overline{a}_2 + \overline{a}_1 \odot (tE_2S_2Q_2) + \overline{a}_2 \odot (tE_1S_1Q_1) \\ &\quad + (tE_1S_1Q_1) \odot (tE_2S_2Q_2))\,.\end{aligned} \tag{3.62}$$

Since we are working with power series,

$$\overline{a}_i \odot (tE_jS_jQ_j) = \overline{a}_i[t^0]tE_jS_jQ_j = 0\,, \tag{3.63}$$

and thus,

$$(L_1^{-1}Q_1 \odot L_2^{-1}Q_2) = \overline{a}_1\overline{a}_2 + t((E_1S_1Q_1) \odot (E_2S_2Q_2))\,, \tag{3.64}$$

establishing that Q is indeed a solution to the equation.

Lemma 28. *Let Q be the generating function for a ZZCLASSICZZ two-dimensional interacting lattice walk model, with $S = S_1(x_1)S_2(x_2)$. Let Q_i be the generating function for a ZZCLASSICZZ*

one-dimensional lattice walk model with step polynomial S_i. Then

$$Q(x_1, x_2, a_1, a_2; t) = Q_1(x_1, a_1; t) \odot Q_2(x_2, a_2; t)\,. \tag{3.65}$$

This allows us to prove the following result via induction.

Theorem 29. *Consider a ZZCLASSICZZ interacting d-dimensional lattice walk model. Suppose that $S(x_1, \ldots, x_d) = S_1 \cdots S_r$ where S_i and S_j do not share variables when $i \neq j$. Then the full generating function Q is given by*

$$Q = Q_1 \odot \cdots \odot Q_r \tag{3.66}$$

where Q_k is the full generating function for the ZZCLASSICZZ interacting sub-d-dimensional problem for the variables present in S_k.

Example 30. Consider the two-dimensional model with diagonal steps, that is, $S = (x_1 + \overline{x}_1)(x_2 + \overline{x}_2)$. We will look at this case in more detail in Chapter 5. The one-dimensional problems are the same and have solutions

$$\widetilde{Q}_0 = \frac{\sigma}{at - b\sigma} \quad \text{and} \quad \widetilde{Q} = \frac{t(x - \sigma)}{x(1 - t(x + \overline{x})(at - b\sigma)}\,, \tag{3.67}$$

where

$$\sigma = \frac{1 - \sqrt{1 - 4t^2}}{2t}\,. \tag{3.68}$$

We can expand and simplify $\widetilde{Q}_0$ and $\widetilde{Q}$ to find

$$\widetilde{Q}_0 = \frac{2 - a - a\sqrt{1 - 4t^2}}{2(1 - a + a^2t^2)}\,, \tag{3.69}$$

$$\widetilde{Q} = \frac{((a-2)t + (1-a)x + 2axt^2) + (at + (1-a)x)\sqrt{1 - 4t^2}}{2(1 - a + a^2t^2)(x - t - tx^2)}\,. \tag{3.70}$$

We have the expansion

$$\widetilde{Q} = 1 + xt + (a + x^2)t^2 + x(1 + a + x^2)t^3 + \cdots\,. \tag{3.71}$$

Finally, the solution for the two-dimensional model can be expressed as

$$\begin{aligned} Q(x_1, x_2, a_1, a_2; t) =& \widetilde{Q}(x_1, a_1; t) \odot \widetilde{Q}(x_1, a_2; t) \\ =& 1 + x_1 x_2 t + (a_1 + x_1^2)(a_2 + x_2^2) t^2 \\ &+ x_1 x_2 (1 + a_1 + x_1^2)(1 + a_2 + x_2^2) t^3 + \cdots . \end{aligned} \tag{3.72}$$

However, obtaining an explicit function via the Hadamard product seems to be rather messy. The function is not algebraic since the unmarked variant is not algebraic. However, the function Q is D-finite since both functions in the product are.

3.3.3 Reducible Models and Groups For Walks With Small Steps

Here is another observation that will allow us to find some solutions to models in an easier way. If the model has some steps that cannot be used in the construction of lattice walks, then we can safely remove them from the step set, and we would still have the same generating function enumerating the lattice walks. This works for any step sets and any regional constraints.

Definition 31. *Given a model $\mathcal{M}$ with step set $\mathcal{S}$, we define the minimal step set $\mathcal{S}'$ for the model to be the set of steps that can eventually be used by lattice walks in the model. We call a model reducible if $\mathcal{S}'$ is a strict subset of S.*

Lemma 32. *The full generating function for a lattice walk model with any step set and constraints is the same whether we use the a step set or its minimal version.*

It is easier to find the generating function for a model if we have more functional equations that will help us cancel out some of the terms. One way to find more equations is to apply a transformation on tuple $(x_1, \ldots, x_d)$ that leaves intact as many parts of the functional equation as possible but still adds some information in the form of a non-equivalent equation. We could map each $x_1, \ldots, x_d$ to some functions separately and then examine the group generated by such maps. We often eliminate the kernel in front of Q in order to find equations for the extracted forms of Q, so it is useful to consider transformations that preserve the kernel. This is a fruitful approach in the non-interacting case, and it has been covered in ZZCITEZZ for two-dimensional models. We will call this group $G(K)$, although it is not the full group of rational transformations preserving the kernel. For example, a lot of the time $(x_1, x_2) \mapsto (x_2, x_1)$ is not in the group, but it is a valid transformation preserving the kernel. If a transformation on $(x_1, \ldots, x_d)$ preserves the kernel $K = 1 - tS$, then it must also preserve the step polynomial S. We may write S as

$$S = S_k^{-1} \overline{x} + S_k^0 + S_k^1 x \tag{3.73}$$

for some k from 1 to d. We consider sending $x_k \mapsto y_k$ and solving $S(x_k) = S(y_k)$ for y_k to determine the possible transformations we can perform. After some simplification we have the equation

$$(S_k^{-1} - S_k^1 x_k y_k)(y_k - x_k) = 0\,. \tag{3.74}$$

We have three cases depending on whether S_k^{-1} and S_k^1 are zero.

1. If both are zero, the equation is trivially-true and all transformations preserve the kernel.

2. If only one of them is zero, then $y_k = x_k$, so only the identity transformation preserves the kernel.

3. If neither one is zero, then we have the solutions $y_k = x_k$ and

$$y_k = \overline{x}_k \frac{S_k^{-1}}{S_k^1}\,. \tag{3.75}$$

   Note that the transformation $g_k(x_k) = y_k$, in the latter case, is an involution since

$$g_k^2(x_k) = g_k\left(S_k^{-1} x_k S_k^1\right) = x_k\,. \tag{3.76}$$

The first two cases are somewhat trivial and we may say something about the generating function Q right away.

1. If $S_k^{-1} = 0$ and $S_k^1 = 0$, then $S = S_k^0$ and the model takes all of its steps inside the $x_k = 0$ hyperplane. Therefore, the model is equivalent to a lower-dimensional model with no x_k coordinate. Suppose that the generating function in the latter case is $q(t)$. Then the generating function in the former case is given by $Q(x_k, a_k; t) = q(a_k t)$, where the rest of the variables have been suppressed from the notation. We can also see this from the functional equation directly. Let $M = M'M_k$ and $E = E'E_k$

$$\begin{aligned}
[x_k^0](M - tES)Q &= \overline{a}_{[d]} \\
(a_k M'[x_k^0] - tE'[x_k^0]S)Q &= \overline{a}_{[d]} \\
(\overline{a}_k M' - tE'S_k^0)Q_k^0 &= \overline{a}_{[d]} \\
(M' - a_k tE'S_k^0)Q_k^0 &= \overline{a}_{[d]\setminus\{k\}} \\
(M' - a_k tE'S)Q &= \overline{a}_{[d]\setminus\{k\}}\,,
\end{aligned} \tag{3.77}$$

where the last line follows since Q and S do not depend on x_k. This is just an equation for a $(d-1)$-dimensional model with an extra scaling of t by a_k.

2. If $S_k^{-1} \neq 0$ and $S_k^1 = 0$, then the model is reducible to one with $S = S_k^0$, that is, the model reduces to case 1.

3. If $S_k^{-1} = 0$, $S_k^0 = 0$ and $S_k^1 \neq 0$, then the walks in model leave the $x_k = 0$ hyperplane and never return. Here we an essentially lower-dimensional problem as well, since we have $S = S_k^1 x_k$. If the generating function for the lower-dimensional problem is $q(t)$, then the generating function for this model is $Q(x_k, a_k; t) = q(x_k t)$, where the rest of the variables have been suppressed from the notation. Similarly to case 1, we may derive this result from the functional equation directly.

All the other cases are more complicated and require more careful analysis.

We can define a group G_S of the nontrivial transformations.

Definition 33. *For a specific step polynomial S, let $G = G_S = G(K)$ be the group generated by the nontrivial transformations g_k.*

It is easy to see that there are infinitely many such groups, just consider $S = \prod_{k=1}^{d}(x_k + \overline{x}_k)$ with $g_k(x_k) = \overline{x}_k$. The group here has order 2^d. The orders of the finite groups set in 2 dimensions are 1, 2, 4, 6 and 8, from Theorem 3 in [8]. If we were to consider walks with multiple copies of some steps, we may obtain even larger group orders as seen in [11]. The d-dimensional groups defined above are Coxeter groups — a generalization of the dihedral groups encountered in dimension 2 lattice walk models. Coxeter groups are defined as follows.

Definition 34. *The group*

$$G = \langle r_1, \ldots, r_d | (r_i r_j)^{m_{i,j}} = 1 \rangle \tag{3.78}$$

is known as a Coxeter group. Here $m_{i,i} = 1$ and $m_{i,j} \geqslant 2$ for all $i \neq j$. Also $m_{i,j}$ could be infinite.

We call these r's reflections and $m_{i,j} = 2$, they commute. Also, $m_{i,j} = m_{j,i}$. The *rank* of G is the number of generators in this specific representation of G. Often G has more than one representation with different numbers of generators. We can draw a graph for each Coxeter group by the following procedure.

- Label vertices with the generators of G.
- Connect the vertices g_i and g_j if $m_{i,j} \geqslant 3$.

- Label the edges with $m_{i,j}$ if $m_{i,j} \geqslant 4$.

If the graph has different components, the Coxeter group factors as a direct product of Coxeter groups represented by the individual components. We can see this in terms of the step polynomial S.

Proposition 35. *If the step polynomia S factors into two disjoint step polynomials T and R as*

$$S = T(x_1, \ldots, x_k)R(x_{k+1}, \ldots, x_d)\,, \tag{3.79}$$

then the group G_S factors as

$$G_S = G_T \times G_R\,, \tag{3.80}$$

where G_S, G_T and G_R are the groups for the step polynomials S, T and R, respectively.

Proof. For a fixed integer $1 \leqslant i \leqslant k$,

$$S_i^j = [x_i^j]S = R[x_i^j]T = RT_i^j \tag{3.81}$$

and therefore,

$$g_i(x_i) = \overline{x}_i \frac{T_i^{-1}R}{T_i^1 R} = \overline{x}_i \frac{T_i^{-1}}{T_i^1}\,. \tag{3.82}$$

Similarly, if $k+1 \leqslant j \leqslant d$, then

$$g_j(x_j) = \overline{x}_j \frac{R_i^{-1}}{R_i^1}\,. \tag{3.83}$$

Furthermore, g_i and g_j commute since they do not affect each other's variables. Since i and j were arbitrary, this means that the generators from different sets commute and the Coxeter graph of the group factors into two pieces, that is, the group itself factors. We can write

$$G_S = G_T \times G_R\,. \tag{3.84}$$

□

Perhaps a similar result can be proven in the other direction.

3.4 Specializing to Two Dimensions

Until the end of the chapter we will focus on the two-dimensional models with interactions. We can set $d = 2$ in Equation 3.11 and Equation 3.30 to find the two-dimensional master functional equations.

Corollary 36. *The full generating function function Q for the ZZCLASSICZZ interacting two-dimensional models satisfies the master functional equation*

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \overline{a}_1\overline{a}_2\,. \tag{3.85}$$

Corollary 37. *The full generating function Q for the ZZCLASSICZZ interacting two-dimensional models satisfies the additional functional equations*

$$\begin{aligned}
(1 - ta_1S_1^0)Q_1^0 - (c_2 - ta_1\overline{x}_2S_{1,2}^{0,-1})Q_{1,2}^{0,0} - a_1t(S_1^{-1}Q_1^1 - \overline{x}_2S_{1,2}^{-1,-1}Q_{1,2}^{1,0}) &= \overline{a}_2\,,\\
(1 - ta_2S_2^0)Q_2^0 - (c_1 - ta_2\overline{x}_1S_{1,2}^{-1,0})Q_{1,2}^{0,0} - a_2t(S_2^{-1}Q_2^1 - \overline{x}_1S_{1,2}^{-1,-1}Q_{1,2}^{0,1}) &= \overline{a}_1\,,\\
(1 - a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1a_2t(S_{1,2}^{-1,-1}Q_{1,2}^{1,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{0,-1}Q_{1,2}^{0,1}) &= 1\,.
\end{aligned} \tag{3.86}$$

3.4.1 First Examples in the Plane

The ZZCLASSICZZ two-dimensional lattice walk models have been investigated thoroughly but the interacting version has not. We want to extend the ZZCLASSICZZ model methods to the interacting setting or create new ones. Since the generating functions for the interacting models need to reduce to those for the non-interacting models, we do not expect to find solutions for every model. In fact, some of the ZZCLASSICZZ models lead to generating functions that are not even D-finite, so their interacting counterparts will not be D-finite either. For the rest of the problems we hope to find the generating functions Q explicitly, or at least the functions $Q_{1,2}^{0,0}$. This appears to be difficult in most nontrivial cases. Recent research ZZCITEZZ has shown that there are models where we have an explicit relationship between Q or $Q_{1,2}^{0,0}$ and unmarked variants. As we saw in the previous chapter, the one-dimensional cases admit a linear relationship with polynomial coefficients. There are other two-dimensional models that exhibit a rational relationship. The way things have turned with regards to the ZZCLASSICZZ models, it would not be surprising if there are two-dimensional interacting models for which neither a linear nor a rational relationship would exist between Q, $Q_{1,2}^{0,0}$ and various unmarked variants. In this section we will cover the first few examples of quarter-plane models with interactions.

Our first step is to find all the reducible models and we can do this by looking at which of the steps $\{$N, NE, E$\}$ they have in their step set.

1. None of the three steps are present. We have models where walks cannot leave the origin and they are equivalent to ones with $\mathcal{S} = \varnothing$. There are $2^5 = 32$ such models and they are all reducible.



Figure 3.3: Set of steps allowed in models that do not leave the origin. The models are equivalent to a model without steps.

2. There is a N step or an E step. The models are equivalent to one-dimensional ones. There are $2 \cdot 2^4 = 32$ such models.

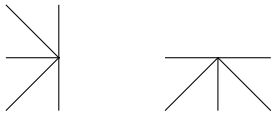


Figure 3.4: Sets of steps allowed in models with an N step or an E step. The models are essentially one-dimensional.

3. There are two or more steps from {N, NE, E} or NE $\subseteq \mathcal{S}$. In this case the models feature walks that leave both axes and once that happens, such walks may use any step from the step set. Therefore, these models are never reducible.

Cases 1 and 2 contain all of the quarter plane models reducible to one-dimensional models. We will look at the case of a model equivalent to a one-dimensional model along the x_2 axis. From the analysis in the previous section, since both $S_1^{-1} = 0$ and $S_1^1 = 0$, we have the first situation. If the solution to the one-dimensional problem is $q(x, a; t) = q(x_2, a_2; t)$, then the solution to the models in cases 1 and 2 is $Q(x_1, x_2, a_1, a_2; t) = q(x_2, a_2; a_1 t)$. To see this in another way, consider the polynomial S itself in the reduced models. Since S factors, the function Q is a Hadamard product between two one-dimensional models.

Since the models are essentially one-dimensional, we also have a linear relationship using the operators B_1 and B_2. If $f(a, t)$ is the function that goes together with $q(x, a, t)$, then the function

$F(a_1, a_2, t)$ that goes together with $Q(x_1, x_2, a_1, a_2, t)$ is given by

$$F(a_1, a_2, t) = f(a_2, a_1 t)\,. \tag{3.87}$$

Then we have $B_2(FQ) = 0$ and consequently, $B_1 B_2(FQ) = 0$. We wrap this section up with the following proposition.

Proposition 38. *Each of the 64 two-dimensional reducible models is equibalent to a one-dimensional model confined to either the x_1 or the x_2 axis. Their solutions are $Q = q(x_1, a_1; a_2 t)$ or $Q = q(x_2, a_2; a_1 t)$, respectively, for a generating function $q(x, a; t)$ for a one-dimensional model. Furthermore, they all satisfy a linear relation $B(FQ) = 0$ for operator $B = B_1 B_2$ and function F, corresponding to q. That is, $F = f(a_2, a_1 t)$ or $f(a_1, a_2 t)$, respectively.*

3.4.2 Kernel Group Considerations

We can partition the two-dimensional lattice walk models in another way — with respect to the nature of the group operations in G_S. We will combine cases for analysis in the form "case $i.j$": we have case i for x_1 and case j for x_2, from the section on groups. Of course, case $i.j$ is equivalent to case $j.i$. From the analysis in the previous sections, cases 1.1, 1.2 and 1.3 only feature reducible models, so we may skip them. We are left with the following cases

Case 2.2: In this case both x_1 and x_2 can only be transformed via the identity transformation. Therefore, exactly one of S_1^{-1} and S_1^1 is zero and exactly one of S_2^{-1} and S_2^1 is zero. The models where $S_1^1 = 0$ or $S_2^1 = 0$ already have solutions since they are reducible. The only remaining unsolved case is when $S_1^{-1} = 0$ and $S_2^{-1} = 0$. The possible models are the ones with steps in {N, E, NE} except the ones going either only in the N direction or only in the E direction. Here are the models in the positive quadrant which we have not solved yet.

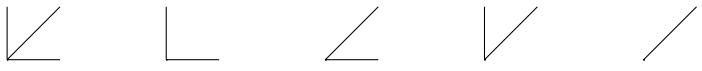


Figure 3.5: Models from case 2.2.

Case 2.3: In this case there is only one transformation for x_1 and two for x_2. This means $S_1^{-1} = 0$ (positive case) or $S_1^1 = 0$ (negative case) but not both. The negative case is reducible, so we

only seek solutions for the positive one. Furthermore, $S_2^{-1} \neq 0$ and $S_2^1 \neq 0$, so there must be at least one step going in the positive x_2 direction and one in the negative x_2 direction. If both NE and SE are in $\mathcal{S}$, then we have 8 cases. If one of NE and SE is in $\mathcal{S}$ but not both, then we have another 8 cases. Finally, if neither is in $\mathcal{S}$, we must have the other three steps present, in order to satisfy the conditions. We have 17 cases in total, the group size is two, and none of the models are reducible. Here are all of these models.

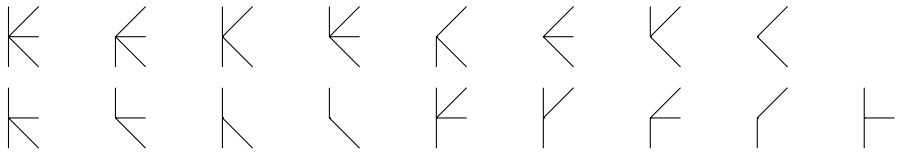


Figure 3.6: Models from case 2.3.

Case 3.3: This is the most interesting case with two transformations for x_1 and x_2 each. This means that there must be at least one step going in a positive and one going in a negative direction, for each of x_1 and x_2. If we look at which diagonal steps are present, similarly to case 2.2, we find that there are 145 such models. Previous results show that the group $G(K)$ has orders 4, 6, 8 and infinity, depending on the step set in question. None of these cases are reducible.

3.5 Solving Basic Two-dimensional Models

3.5.1 Case 2.2 – Group of Order One

The only models that are not reducible are those where $S_1^{-1} = 0$ and $S_2^{-1} = 0$. Here

$$S = \delta + \alpha x_1 + \beta x_2 + \gamma x_1 x_2 \,. \tag{3.88}$$

The functional equation and its zeroth term extractions are given by

$$\begin{aligned} (1 - tS)Q - c_1 Q_1^0 - c_2 Q_2^0 + c_1 c_2 Q_{1,2}^{0,0} &= \overline{a}_1 \overline{a}_2 \,, \\ (1 - t a_1(\delta + \beta x_2))Q_1^0 - c_2 Q_{1,2}^{0,0} &= \overline{a}_2 \,, \\ (1 - t a_2(\delta + \alpha x_1))Q_2^0 - c_1 Q_{1,2}^{0,0} &= \overline{a}_1 \,, \\ (1 - a_1 a_2 t\delta)Q_{1,2}^{0,0} &= 1 \,. \end{aligned} \tag{3.89}$$

This is just a linear system of four equations for the four unknowns, $Q_{1,2}^{0,0}$, Q_1^0, Q_2^0 and Q, in echelon form, and solve it by back-substitution.

$$Q_{1,2}^{0,0} = \frac{1}{1-a_1a_2\delta t}\,, \tag{3.90}$$

$$Q_1^0 = \frac{\overline{a}_2}{1-ta_2S_1^0}\left(1+\frac{b_2}{1-a_1a_2\delta t}\right) = \frac{1-a_1\delta t}{(1-ta_1S_1^0)(1-a_1a_2\delta t)}\,, \tag{3.91}$$

$$Q_2^0 = \frac{\overline{a}_1}{1-ta_1S_2^0}\left(1+\frac{b_1}{1-a_1a_2\delta t}\right) = \frac{1-a_2\delta t}{(1-ta_2S_2^0)(1-a_1a_2\delta t)}\,, \tag{3.92}$$

$$Q = \frac{1}{1-tS}(\overline{a}_1\overline{a}_2 + c_1Q_1^0 + c_2Q_2^0 - c_1c_2Q_{1,2}^{0,0}) \tag{3.93}$$

$$= \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1-ta_1a_2\delta}\left(c_1\frac{1-a_1\delta t}{1-ta_1S_1^0} + c_2\frac{1-a_2\delta t}{1-ta_2S_2^0} - c_1c_2\right)\right) \tag{3.94}$$

$$= \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1-a_1a_2\delta t}(c_1f_1(a_1,t) + c_2f_2(a_2,t) - c_1c_2)\right)\,. \tag{3.95}$$

Note how Q has a dependency on functions f_1 and f_2 that do not depend on one of a_1 and a_2. We shall see that the linear relation $B(FQ) = 0$ is satisfied by the function

$$F(a_1,a_2,t) = 1 - a_1a_2\delta t\,. \tag{3.96}$$

First,

$$\begin{aligned} B((1-a_1a_2\delta t)\overline{a}_1\overline{a}_2) &= B(\overline{a}_1\overline{a}_2 - t\delta) = B(\overline{a}_1\overline{a}_2) - B(t\delta) \\ &= \overline{a}_1\overline{a}_2 - \overline{a}_2 - \overline{a}_1 + 1 = c_1c_2\,. \end{aligned} \tag{3.97}$$

Then,

$$B(c_1f_1 + c_2f_2 - c_1c_2) = c_1f_1 + c_2f_2 - c_1c_2 - c_2f_2 - c_1f_1 = -c_2c_2\,. \tag{3.98}$$

Putting the two quantities together, along with dividing by $1 - tS$, shows that $B(FQ) = 0$. Of course, this works for $G = Q_{1,2}^{0,0}$ too.

Example 39. Consider the model with $S = x_1 + x_2 + x_1x_2$. Then

$$Q_{1,2}^{0,0} = 1, \quad Q_1^0 = \frac{1}{1-ta_1x_2}, \quad Q_2^0 = \frac{1}{1-ta_2x_1}\,, \tag{3.99}$$

and finally,

$$Q = \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{c_1}{1-ta_1x_2} + \frac{c_2}{1-ta_2x_1}\right). \tag{3.100}$$

3.5.2 Case 2.3 – Group of Order Two

In this case the functional equation and its extracted forms read as

$$\begin{aligned}
(1-tS)Q - c_1Q_1^0 - (c_2 - t\overline{x}_2S_1^{-1})Q_2^0 + c_1c_2Q_{1,2}^{0,0} &= \overline{a}_1\overline{a}_2\\
(1-ta_1S_1^0)Q_1^0 - (c_2 - ta_1\overline{x}_2S_{1,2}^{0,-1})Q_{1,2}^{0,0} &= \overline{a}_2\\
(1-ta_2S_2^0)Q_2^0 - c_1Q_{1,2}^{0,0} - a_2tS_2^{-1}Q_2^1 &= \overline{a}_1\\
(1-a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1a_2S_{1,2}^{0,-1}Q_{1,2}^{0,1} &= 1\,.
\end{aligned} \tag{3.101}$$

We can identify a subproblem that is essentially one-dimensional and therefore, already solved. To see this, take the second and fourth equations, and make the substitutions $a_i = a$, $c_i = c$, $x_2 = x$, $a_1t = \tau$, $S_1^0 = T$ and $Q_1^0(x_2) = q(x)$. The equations become

$$(1-\tau T)q - (c - \tau\overline{x}T_{-1})q_0 = \overline{a}\,, \tag{3.102}$$

$$(1-a\tau T_0)q_0 - a\tau T_{-1}q_1 = 1\,, \tag{3.103}$$

where $q_0 = q(a,\tau)$. We can see that this is a one-dimensional problem again. The solution here is $q(x,a,\tau)$, obtained by the methods in the previous chapter, so the original problem has solutions

$$Q_1^0(x_2) = q(x_2, a_2, a_1t) \quad \text{and} \quad Q_{1,2}^{0,0} = q_0(a_2, a_1t)\,. \tag{3.104}$$

The reason the method worked in this case is that we have steps that keep a lattice walk staying on the x_2 axis, steps that take it away from the x_2, but there are no steps that let it return. This is similar to the case where a lattice walk leaves the x_2 axis, never to return. Essentially, the problem splits into a one-dimensional one with regards to the x_2 axis, and a further two-dimensional one the walks step off the axis. We rewrite the master equation as

$$(1-tS)Q - (c_2 - t\overline{x}_2S_2^{-1})Q_2^0 = \overline{a}_1(\overline{a}_2 + b_1q - b_1c_2q_0) = \overline{a}_1\phi(x_2)\,. \tag{3.105}$$

Since $S_1^{-1} \neq 0$ and $S_2^1 \neq 0$, we have a root $x_2 = \sigma_2$ that allows us to eliminate the $(1 - tS)Q$ term. That is, we are finding a root for x_2 again, since $x = x_2$ in the previous equations. We find that

$$Q_2^0 = -\frac{\overline{a}_1 \phi(\sigma_2)}{c_2 - t\overline{\sigma}_2 S_2^{-1}} \tag{3.106}$$

and we write Q as

$$Q = \frac{1}{1 - tS}\left(\overline{a}_1 \phi(x_2) - \frac{c_2 - t\overline{x}_2 S_2^{-1}}{c_2 - t\overline{\sigma}_2 S_2^{-1}} \overline{a}_1 \phi(\sigma_2)\right). \tag{3.107}$$

Of course, this can be expanded and simplified based on what q and q_0 are from the one-dimensional setting, but the important thing is that we can build the solution in this case via one-dimensional model equations.

3.5.3 Miscellaneous Case

There is one case that is easy to solve, is not reducible, but has an infinite group. Consider the case of $S = x_1 x_2 + \overline{x}_1 \overline{x}_2$. The group operations are $g_1(x_1) = \overline{x}_1 \overline{x}_2^2$ and $g_2(x_2) = \overline{x}_2 \overline{x}_1^2$. If we apply these transformations on (x_1, x_2) in an alternating sequence, each time the absolute powers of x_1 and x_2 will increase in one of the coordinates. This shows that the group $G(K)$ is infinite. However, this case is easy to solve, because the model is essentially one-dimensional. It is equivalent to the model on the non-negative line with $S = x + \overline{x}$ and marking visits to the origin with a, where $x = x_1 x_2$ and $a = a_1 a_2$. The solution is

$$Q(x_1, x_2, a_1, a_2; t) = q(x_1 x_2, a_1 a_2; t). \tag{3.108}$$

3.5.4 Counting Classes of Models

We can tally all the classes of lattice walk models on the quarter-plane. First, we have 64 reducible models. Then we have 5 new cases with group of order one, and 34 new cases with group of order two. Also, there are 29 models with group of order four, 6 with group of order six, and 4 with group of order eight. We also have the easy-to-solve miscellaneous case above. So far we have 143 cases of models we can solve for easily or models with a finite group, some of which w ecan solve easily. Next, by inspection of the table in ZZCITEZZ (walks w/ small steps), we see that there are 120 models with group of infinite order: 12 with three steps, 40 with four steps, 44 with five steps, and 24 with six steps. Adding all of those together yields 263 cases, so we must have over-counted. There are 7 reducible models with an infinite group. This brings the total count to 256, and we are

done.

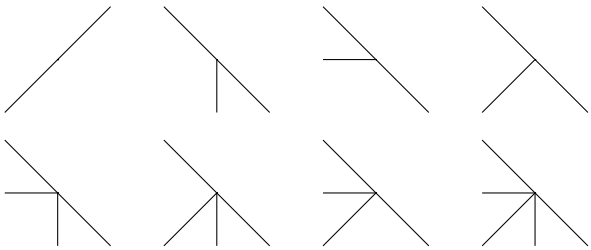


Figure 3.7: Reducible models with an infinite group.

So far we have solved the 64 reducible models, the 39 other models with group of order one or two, and the single new case of easy-to-solve model with an infinite group, for a grand total of 104 trivial or easy models. Out of the remaining 152 models, 113 have an infinite group, and their analysis will not be a part of this thesis. This leaves the 39 cases with a group of size four, six or eight. The rest of the thesis will deal with the analysis of those cases.

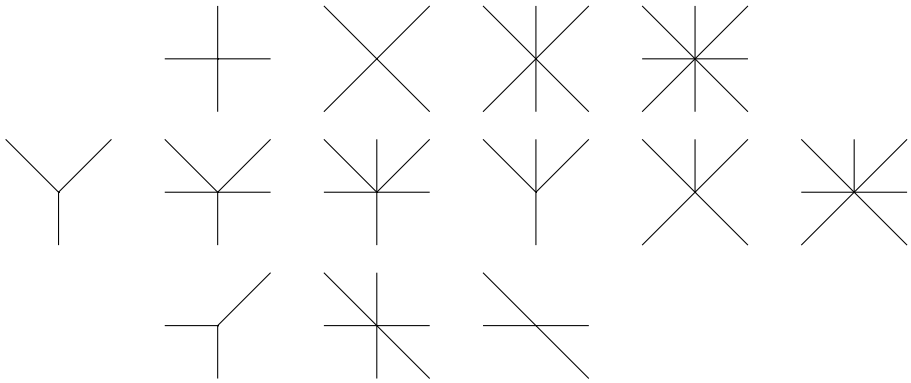


Figure 3.8: Models with a finite group.

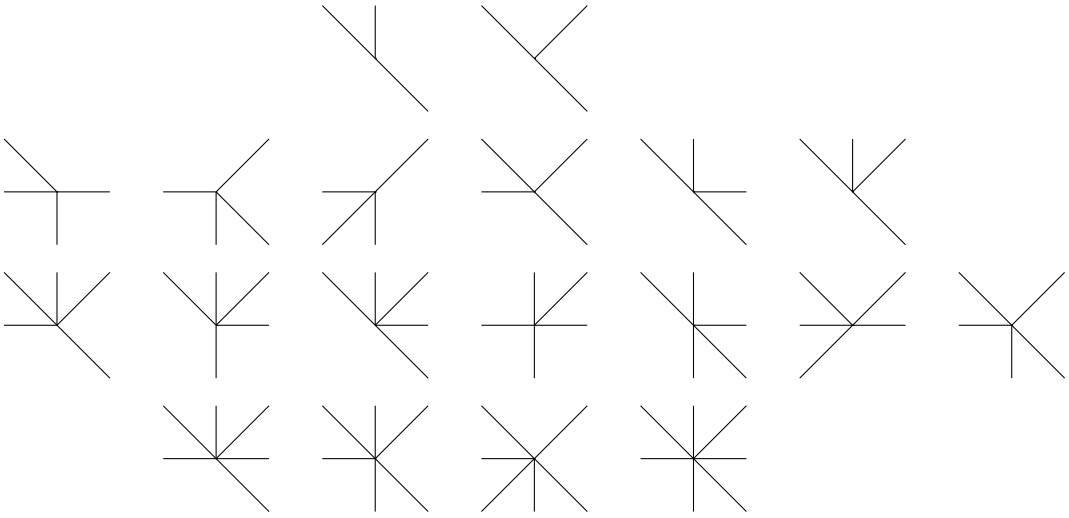


Figure 3.9: Models with an infinite group. To get the rest of them, apply one of the symmetries of the square.

Chapter 4

The Algebraic Kernel Method

We will explore the algebraic kernel method as seen in [18] with extensions to interacting lattice walk models. We start with the non-interacting version of the problem.

4.1 Models Without Interactions

At the core of the Algebraic Kernel method we have the idea that we can apply transformations from a group G onto the master functional equation and acquire different equations for each $g \in G$. After we sum these equations in a nice way, we manage to eliminate the functions Q_1^0, Q_2^0, $Q_{1,2}^{0,0}$ and their orbits, only leaving the orbit of the kernel term KQ and a term on the right-hand side. What makes this possible is pre-multiplying the whole equation by $x_1 x_2$. This also works in higher dimensions, as we will show later. Although the generalization has been implemented before (for example, page 11 of [8] for the two-dimensional case and page 10 of [18] for the three-dimensional one), we will present it in a compact and easy-to-generalize way. Instead of summing equations as in [8], we will use an operator D_G that wil do the same thing. We will define D_G in a way that is analogous to the way the Algebraic kernel method is currently applied.

Definition 40. *For a lattice walk model with small steps, let G be the group generated by the transformations $g_k : x_k \mapsto \overline{x}_k S_k^{-1}/S_k^1$ preserving the kernel. Then let D_G be the operator defined as*

$$D_G = \sum_{g \in G} (-1)^{|g|} g \,, \tag{4.1}$$

where $|g|$ is the sign of g.

In two dimensions we can find the whole group by iterating over the two generators. That is, the

group is $\{1, g_i, g_j g_i, g_i g_j g_i\}$. It is cleaner to write g_{iji} instead of $g_i g_j g_i$, so we have the following notation.

Definition 41. *For a sequence $L = (l_1, l_2, \ldots)$ of positive integers, define*

$$g_L = g_{l_1 l_2 \ldots} = g_{l_1} g_{l_2} \cdots , \tag{4.2}$$

when the list is finite or when there is convergence, in the case of an infinite list.

In the case of a finite group in two dimensions we have $(g_1 g_2)^n = 1$ for some minimal n. This allows us to rewrite D_G in a useful way. The following lemma is just a restatement of the orbit sum method on page 11 of [8], but it is recast in operator notation.

Lemma 42. *For a two-dimensional model with small steps and a group of order $2n$ with generators g_1 and g_2, we can write the operator D_G as*

$$D_G = (1 + g_{ij} + \cdots + (g_{ij})^{n-1})(1 - g_i) . \tag{4.3}$$

Proof. We can just check the identity by expanding it as

$$\begin{aligned} D_G &= (1 + g_{ij} + \cdots + (g_{ij})^{n-1})(1 - g_i) \\ &= 1 - g_i + g_{ij} - g_{iji} + \cdots - (g_{ij})^{n-1} g_i , \end{aligned} \tag{4.4}$$

giving us one of writing the orbit. □

Here are some examples of small groups.

Example 43. When $|G| = 4$ the elements commute, so the orbit is $\{1, g_1, g_{12} = g_{21}, g_2\}$, and the operator factors as

$$D_G = 1 - g_1 - g_2 + g_{12} = (1 - g_1)(1 - g_2) . \tag{4.5}$$

When $|G| = 6$ the orbit is $\{1, g_1, g_{21}, g_{121} = g_{212}, g_{12}, g_2\}$. We can write the operator as

$$\begin{aligned} D_G &= (1 + g_{12} + g_{1212})(1 - g_1) \\ &= 1 + g_{12} + g_{1212} - g_1 - g_{121} - g_{12121} \\ &= 1 - g_1 - g_2 + g_{12} + g_{21} - g_{121} . \end{aligned} \tag{4.6}$$

We can generalize the lemma for a multidimensional group G.

Lemma 44. *Consider a d-dimensional lattice walk model with small steps and group G. For a nontrivial generator g_k of G, we have $D_G = D_H(1 - g_k)$, where H is a subgroup of G.*

Proof. Fix a representation of G. Each element either ends in g_k or it does not. We have a bijection between the two classes — just multiply by the involution g_k from the right. The two classes must have the opposite parity, so we can split G into two parts H and Hg_k. We need to make sure that H is a group. Then we can split the summation for D_G into two parts. We will keep track of the sign changes with $e_g = (-1)^{|g|}$. We have

$$\begin{aligned} D_G &= \sum_{g \in H} e_g g + \sum_{g \in Hg_k} e_g g \\ &= \sum_{g \in H} e_g g + \sum_{g \in H} (-1) e_g g g_k \\ &= \left(\sum_{g \in H} e_g g \right) (1 - g_k) \\ &= D_H(1 - g_k)\,. \end{aligned} \tag{4.7}$$

(Simpler proof(?)): Take the subgroup H isomorphic to $G/\langle g_k \rangle$. Since g_k is of order two, $|H|$ has $|G|/2$ elements. Consider the elements of G that commute with g_k, that is, $h \in G$ where $g_k h = h g_k$. Multiplying by $g_k^{-1} = g_k$ we see that $h = g_k h g_k$. Let H be the subgroup of G generated by those elements. There are two cosets H and Hg_k. □

Lemma 45. *If I is a nonempty subset of $\{1, \ldots, d\}$, then the operator D_G annihilates the term $tS_I^{-\mathbb{1}} Q_I^{\mathbb{0}}$.*

Proof. By the previous lemma, for some $i \in I$,

$$D_G(tS_I^{-\mathbb{1}} Q_I^{\mathbb{0}}) = D_H((1 - g_i) tS_I^{-\mathbb{1}} Q_I^{\mathbb{0}})) = D_H(0) = 0\,, \tag{4.8}$$

since the variable x_i is not present in that term. □

We obtain the d-dimensional generalization to Proposition 5 of [8].

Theorem 46. *For a d-dimensional lattice walk model with small steps confined to the positive orthant, we have*

$$D_G(x_1 \cdots x_d K Q) = D_G(x_1 \cdots x_d)\,, \tag{4.9}$$

where the right-hand side may sometimes equal zero.

Proof. When we multiply the master functional equation by $x_1 \cdots x_d$, we transform almost all of the terms of the equation to ones of the form $tS_I^{-\mathbb{1}}Q_I^{\mathbb{0}}$. When we apply D_G, the only term on the left-hand side that survives is $x_1 \cdots x_d KQ$, and the equality follows. □

This is a generalization of the orbit summation found in [15]. The authors in [8] and [18] have found that sometimes the algebraic kernel method fails; that is, $D_G(x_1 \cdots x_d) = 0$ and the equation does not give us new information. The operator notation allows us to a more general instance of when this situation occurs.

Corollary 47. *If there exists an element h of G that permutes the variables $x_1, \ldots, x_d$ and has order two, then the regular Algebraic kernel method fails. That is,*

$$D_G(x_1 \cdots x_d KQ) = 0\,, \tag{4.10}$$

and we would have to find information about Q in another way.

Proof. Since h is an involution, the same method applies as before and we can factor $D_G = D_H(1-h)$ for some $H \subseteq G$. The action $1 - h$ eliminates the monomial $x_1 \cdots x_d$. □

Example 48. The two-dimensional models for which the algebraic kernel method fails are precisely thosefor which we have a permutation as in the corollary. That is, $h(x_1, x_2) = h(x_2, x_1)$ and h has order two. Proposition 6 in [8] follows.

Example 49. In three dimensions and models with at most 6 steps, we see that the Algebraic kernel method fails in 62 three-dimensional models (page 13 of [18]). For example, there is a model with group size 12 for which the Algebraic kernel method fails. For that model, one of the group element is a permutation of two of the variables — in that case x_2 and x_3. Here G contain the group seen in the previous example for the variables x_2 and x_3. This means that every higher-dimensional model that has group G, containing the group above, will fail in the regular Algebraic Kernel method.

A natural question is can we take D_H instead of $D_G = D_H(1-h)$, and check if we can perfom a half-orbit summation, and to what success. The answer is yes. The operator D_H eliminates some of the Q variables and introduces others, but the result is a function symmetric under permutation by h. One may hope to perform a method analogous to the half-orbit summation method to prove algebraicity or D-finiteness.

Example 50. There is another way we can prove that D_G acts as advertised. As in Chapter 3, we start with the kernel form $E(KQ) = 1$ where we make the substitution $Q = W/(x_1 \cdots x_d K)$, and

rewrite the equation as

$$E'W = x_1 \cdots x_d \,, \tag{4.11}$$

where $E' = E'_1 \cdots E'_d$ and $E'_k = 1 - [x_k^0]$. Therefore,

$$W = x_1 \cdots x_d - (1 - E')W = x_1 \cdots x_d - \sum_{\varnothing \neq I \subseteq [d]} W_I^{\mathbb{0}} \,. \tag{4.12}$$

We immediately see that

$$\begin{aligned} D_G(x_1 \cdots x_d KQ) = D_G(W) &= D_G\left(x_1 \cdots x_d - \sum_{\varnothing \neq I \subseteq [d]} W_I^{\mathbb{0}} \right) \\ &= D_G(x_1 \cdots x_d) \,, \end{aligned} \tag{4.13}$$

since the $W_I^{\mathbb{0}}$ terms have missing variables. Let us think about this for a moment. The form of W is the same for all lattice walk models with small steps so, if there are two models whose operators both annihilate expressions of the form $W_I^{\mathbb{0}}$ for all nonempty $I \subseteq [d]$, then both models have equations of the above type with each other's operators. In particular, we may take the group G of the highly-symmetric models and apply it via the operator D_G to another model, and we retrieve an equation of the type aboe. We will look into this further.

Definition 51. *Consider the group G for a highly-symmetric model, where $g_k(x_k) = \overline{x}_k$. Define the operator $\Delta_k = 1 - g_k$. Let $\Delta = \Delta_1 \cdots \Delta_d$.*

Theorem 52. *Let G be the group generated by $g_k(x_k) = \overline{x}_k$ for $1 \leqslant k \leqslant d$. Then*

$$\Delta(x_1 \cdots x_d KQ) = \Delta(x_1 \cdots x_d) = (x_1 - \overline{x}_1) \cdots (x_d - \overline{x}_d) \tag{4.14}$$

for all d-dimensional non-interacting lattice walk models with small steps in the positive orthant.

Proof. It is enough to see that Δ annihilates terms of the form $W_I^{\mathbb{0}}$ for $I \neq \varnothing$. Since the operators Δ_k commute, we only need to apply one Δ_i for some i first, and find that

$$\Delta_i(W_I^{\mathbb{0}}) = W_I^{\mathbb{0}} - g_i(W_I^{\mathbb{0}}) = 0 \,, \tag{4.15}$$

since the variable x_i is not present in this term. The rest follows by linearity. □

This will not help to show a specific model has a D-finite solution because, in general, we cannot

factor out the K term from the left-hand side. However, the idea is important. Next, we will look into generalizing the ideas involved in the algebraic kernel method.

Definition 53. *Let G be a group acting on $x_1, \ldots, x_d$. We define a weighted orbit summation operator to be an operator of the form*

$$D = \sum_{g \in G} R_g g \,, \tag{4.16}$$

where R_g are functions in the variables $x_1, \ldots, x_d$.

This is a generalization of the orbit summation operator that can be written as a weighted operator with $R_g = (-1)^{|g|}$. A first glimpse of this kind of operator can be seen on page 11 of [**?**], where the authors use an orbit summation with weights. We can capture various possible strategies in this notation.

Example 54. Since the Algebraic Kernel method fails to give a nice result when G contains permutations, we may try to apply the operator D_G on a slightly different equation — one where we multiply the whole equation by a function $T \neq x_1 \cdots x_d$. For a function f we have

$$D_G(Tf) = \sum_{g \in G} e_g g(Tf) = \sum_{g \in G} e_g(g(T))g(f) = \sum_{g \in G} R_g g(f) \,, \tag{4.17}$$

where $R_g = (-1)^{|g|} g(T)$. therefore, this strategy is the same as applying a weighted orbit summation operator.

Example 55. Another strategy is to consider applying a weighted operator several times in a row in the hopes of attaining an easier equation to work with. However, this is no different than looking at a weighted operator in the first place. This is because G is a group and the composition D_G^2 leads to another weighted operator, as we shall prove shortly. Also, since some of the operators D_G have nontrivial kernels, they are not invertible, so the set of operators is not a group. However, we can show that the set is a monoid under composition.

Proposition 56. *Given a group G, the set of weighted orbit summation operators forms a monoid under composition.*

Proof. Since G is a group, we have an identity element 1, which can also be written as a weighted orbit summation operator with $R_g = 0$ for all $g \neq 1$ and $R_1 = 1$. Thus, the set of operators has an identity element. Since composition of operators is associative, we only need to check that the set

of operators is closed under composition. We have

$$D_1 D_2 = \sum_{g\in G}\sum_{h\in H} R_g g T_h h = \sum_{g\in G}\sum_{h\in H} R_g T_{gh} gh = \sum_{g\in G}\left(\sum_{h\in H} R_{gh_{-1}} T_g\right) g\,, \tag{4.18}$$

which is also an operator of this form once we collect terms corresponding to the same group elements. □

The main question is whether the weighted summation operators will help us in analyzing the models where the Algebraic kernel method fails. We will answer this question in the negative. In general, there is no other weighted orbital operator that eliminates all the unwanted terms of the master functional equation, except the operator with weights $R_g = 1$, that does not also eliminate $x_1 \cdots x_d$. However, this setback does not diminish the usefulness of weighted operators, because their strength is in the analysis of interacting models.

Here is the strategy we will employ. If we apply the operator D on the master functional equation, we need to eliminate all terms except KQ. That is, D needs to annihilate all the Q terms that are missing variables. It is enough to work on the fundamental functional equation for $W = x_1 \cdots x_d KQ$. To go back to the master functional equation case we pre-divide by $x_1 \cdots x_d$. The resulting operator is also weighted.

Lemma 57. *A weighted operator D exists such that $D(KQ) = D(1)$ if and only if a weighted operator D' exists such that $D'(f_I^{\mathbb{O}})$ for generic functions $f_I^{\mathbb{O}}$ which are missing the variables indexed by I, for all I.*

Proof. The discussion above provides the direction from left to right for $f = W$. To prove it for generic functions f, it suffices to note that $f_I^{\mathbb{O}}$ and $f_J^{\mathbb{O}}$ are independent from each other if $I \neq J$. That is, $D(f_I^{\mathbb{O}})$ and $D(f_J^{\mathbb{O}})$ do not share terms. The converse direction is proved by setting $f = W$ and setting $D = D'\overline{x}_1 \cdots \overline{x}_d$, where $(D'\overline{x}_1 \cdots \overline{x}_d)(f) = D'(\overline{x}_1 \cdots \overline{x}_d f)$. □

This essentially turns the question into a linear algebra problem. We just need to apply D on generic functions $W_I^{\mathbb{O}}$, which are missing the variables indexed by I, collect the different results and make their coefficients zero. That is, for every $f = W_I^{\mathbb{O}}$ we look at the orbit

$$\mathrm{Orbit}_G(f) = \{g(f) | g \in G\} \tag{4.19}$$

and collect coefficients in

$$0 = D(f) = \sum_{g\in G} R_g g(f) = \sum_{f'\in Orbit(f)} R'_g f'\,, \tag{4.20}$$

where we set

$$R'_g = \sum_{\substack{g(f)=f' \\ g\in G}} R_g = 0\,. \tag{4.21}$$

Thus, we reach a linear system of $|\mathrm{Orbit}_G(f)|$ equations in $|G|$ unknowns. Of course, we can do this for all $2^d - 1$ terms $f = W_I^{\mathbb{0}}$ and hopefully find a system with a nontrivial solution. Here is a simple example

Example 58. Consider a two-dimensional model with group of order four given by

$$G = \{1, g_1, g_2, g_{12} = g_{21}\}\,. \tag{4.22}$$

In the two-dimensional problem we have the generic functions $W(0,0) = c$, a constant, and $W(0,x_2)$ and $W(x_1,0)$. We can write them as $1, f(x_1), f(x_2)$. A weighted operator for this group can be written as

$$D = R - R_1 g_1 - R_2 g_2 + R_{12} g_{12}\,. \tag{4.23}$$

Consider the different orbits. The orbit of 1 under G is trivial since the function does not change under the group actions. Next we have

$$\mathrm{Orbit}_G(f(x_1)) = \{f(x_1), f(\overline{x}_1), f(\overline{x}_1), f(\overline{x}_1)\} = \{f(x_1), f(\overline{x}_1)\} \tag{4.24}$$

and similarly,

$$\mathrm{Orbit}_G(f(x_2)) = \{f(x_2), f(\overline{x}_2)\}\,. \tag{4.25}$$

Therefore, the equations for the weights are

$$R - R_1 + R_2 - R_{12} = 0\,, \tag{4.26}$$

$$R = R_2, \quad R_1 = R_{12}\,, \tag{4.27}$$

$$R = R_1, \quad R_2 = R_{12}\,. \tag{4.28}$$

When we solve this system of equations, we find $R = R_1 = R_2 = R_{12}$ and the operator is essentially trivial. The simplest choice is to set $R = 1$ and we find that every weighted orbit summation operator for this group G must have identical weights if it is to annihilate generic functions with missing variables.

Definition 59. *A trivial, or unweighted, operator is a weighted operator with weights that are all the same.*

Lemma 60. *Consider a d-dimensional non-interacting lattice walk model with small steps in the positive orthant. If its group is finite and has d generators, then all weighted orbital operators are trivial.*

Proof. (?) It is enough to show that $D(W_k^0) = 0$ for all k implies that the operator is trivial. To this end suppose that $D(f) = 0$ where $f = W_k^0$. Then we can partition the group G as before into sets H and Hg_k. We see that $hg_k(f) = h(f)$, so the orbit of f contains at most $|G|/2$ elements. However, these elements must all be distinct otherwise they would come from the same operations, but the group elements in H are all distinct. Therefore, $|\mathrm{Orbit}(f) = |G|/2$. This means that we have the equations $R_h = R_{hg_k}$ for every $h \in H$. Once we go through every k we have overlaps $R = R_1 = \cdots = R_d$ and $R_1 = R_{12} = R_{13}$, and so on. This means that all the R's are the same and the operator is trivial. □

This shows that in general, there is no clever weighted summation operator that also satisfies the condition $D(x_1 \cdots x_d) \neq 0$. However, the idea will be fruitful in that we may find nontrivial partial solution in the interacting case.

4.2 Models with interactions

It is natural to consider extending the algebraic kernel method to one working for interacting models, but we quickly see that the regular orbit summation operator is not sufficient. This is where the power of the weighted orbital summation operators starts to show. The analysis for two-dimensional models is particularly nice because we end up having the same number of equations as unknowns, and we are able to find the required weights. The linear systems can be written as matrix equations where the determinant of the matrix tells us when we have a useful orbit summation operator. In the end, the method does not extend fully because we do not eliminate the $Q_{1,2}^{0,0}$ term. The D-finiteness of the full generating function Q depends on the D-finiteness of the function $Q_{1,2}^{0,0}$. We will only consider the models with finite groups that have two nontrivial generators. Therefore, the group G has order 4, 6 or 8, and the functional equation in two dimensions is

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \alpha\,. \tag{4.29}$$

4.2.1 Groups of Order Four

The orbital summation operator for the non-interacting case is

$$D_G = 1 - g_1 - g_2 + g_1 g_2 = (1 - g_1)(1 - g_2) = \Delta_1 \Delta_2 \tag{4.30}$$

but this operator does not eliminate the unknown functions in the interacting case. Instead of using this operator, we will use a weighted one and we will determine its weights. We will find a linear system of equations by applying the group operations on the original functional equation. The unknowns will be functions Q_1^0, Q_2^0 and their orbits under G. If the associated matrix has a nonzero determinant, the system has a unique solution and unfortunately, we will not be able to eliminate variables. However, there are some cases where the determinant is actually zero, and this allows us to eliminate some of the unknowns.

Since the transformations commute, we have $g_i Q_i^0 = Q_i^0$ and $g_i g_j Q_i^0 = g_j Q_i^0$. The variables and their orbits are given by the next table.

Group action	Effect on Q_1^0	Effect on Q_2^0
1	Q_1^0	Q_2^0
g_1	Q_1^0	$g_1 Q_2^0$
g_2	$g_2 Q_1^0$	Q_2^0
$g_{12} = g_{21}$	$g_2 Q_1^0$	$g_1 Q_2^0$

Table 4.1: Effects of group actions $g \in G$ on Q_1^0 and Q_2^0.

Therefore, the only variables that we need to eliminate are in the set

$$E = \{Q_1^0,\, Q_2^0,\, g_2 Q_1^0,\, g_1 Q_2^0\}\,. \tag{4.31}$$

This is looking promising since we also have four equations. We rewrite the functional equation in a way that highlights those variables.

$$K_1 Q_1^0 + K_2 Q_2^0 = KQ + K_{1,2} Q_{1,2}^{0,0} - \alpha = \beta\,. \tag{4.32}$$

Then we apply the transformations to produce the system of equations

$$K_1 Q_1^0 + K_2 Q_2^0 = \beta\,, \tag{4.33}$$

$$g_1 K_1 Q_1^0 + g_1 K_2 g_1 Q_2^0 = g_1\beta = \beta_1\,, \tag{4.34}$$

$$g_2 K_1 g_2 Q_2^0 + g_2 K_2 Q_2^0 = g_2\beta = \beta_2\,, \tag{4.35}$$

$$g_{12} K_1 g_2 Q_1^0 + g_{12} K_2 g_1 Q_2^0 = g_{12}\beta = \beta_{12}\,. \tag{4.36}$$

Using the order of variables in E we can write the associated matrix M of this linear system as

$$M_4 = \begin{bmatrix} K_1 & K_2 & 0 & 0 \\ g_1 K_1 & 0 & 0 & g_1 K_2 \\ 0 & g_2 K_2 & g_2 K_1 & 0 \\ 0 & 0 & g_{12} K_1 & g_{12} K_2 \end{bmatrix}. \tag{4.37}$$

Then

$$\begin{aligned} \det M_4 &= K_1 g_{12} K_1 g_1 K_2 g_2 K_2 - g_1 K_1 g_2 K_1 K_2 g_{12} K_2 \\ &= K_1 g_{12} K_1 g_1 K_2 g_2 K_2 - g_1 (g_1 K_1 g_{12} K_1 g_1 K_2 g_2 K_2) \\ &= \Delta_1 (K_1 g_{12} K_1 g_1 K_2 g_2 K_2) \\ &= \Delta_2 (g_1 K_1 g_2 K_1 K_2 g_{12} K_2)\,, \end{aligned} \tag{4.38}$$

where the last equation is obtained by using g_2 instead of g_1 in the factoring..

In order to be able to eliminate some variables, the determinant must equal zero, and that only happens when the expression in Δ_1 or Δ_2 is in the kernel of the respective operator. Suppose that f is a Laurent polynomial in the variables x_i and t and that f is in the kernel of Δ_k for one of the k's. We have

$$\Delta_k f = \sum_n f_n (1 - g_k) x_k^n = \sum_n (f_n - f_{-n} \mu_k^{-n}) x_i^n \tag{4.39}$$

where $\mu_k = S_k^{-1}/S_k^1$. This implies that

$$f_n = f_{-n} \mu_k^{-n} \tag{4.40}$$

$$f_n (S_k^1)^n = f_{-n} (S_k^{-1})^n \tag{4.41}$$

for all $n \in \mathbb{Z}$. The models with groups of order four exhibit at least one symmetry across a line — either vertical or horizontal symmetry, or both. Without loss of generality, let the model have

symmetry across the x_2 axis. That is,

$$g_1(x_1) = x_1, \quad g_1(S_2^i) = S_2^i \quad \text{and} \quad g_1(S_1^i) = S_1^{-i}\,. \tag{4.42}$$

This allows us to compute some things. For example,

$$g_1 K_2 = c_2 - t\overline{x}_2 g_1 S_2^{-1} = c_2 - t\overline{x}_2 S_2^{-1} = K_2\,, \tag{4.43}$$

$$g_1 g_2 K_2 = g_2 g_1 K_2\,. \tag{4.44}$$

Therefore, Δ_1 fixes both K_2 and $g_2 K_2$. We see that the determinant is

$$\det M_4 = \Delta_1(K_1 K_2 g_{12} K_1 g_2 K_2) = K_2 g_2 K_2 \Delta_1(K_1 g_{12} K_1)\,. \tag{4.45}$$

This is zero if and only if $\Delta_1(K_1 g_{12} K_1) = 0$, so we have

$$\begin{aligned} 0 &= \Delta_1(K_1 g_{12} K_1) \\ &= K_1 g_{12} K_1 - g_1 K_1 g_2 K_2 \\ &= (c_1 - t\overline{x}_1 S_1^{-1})(c_1 - t x_1 g_2 S_1^{-1}) - (c_1 - t x_1 S_1^{-1})(c_1 - t\overline{x}_1 g_2 S_1^{-1}) \\ &= c_1 t\overline{x}_1(g_2 S_1^{-1} - S_1^{-1}) - c_1 t x_1(g_2 S_1^{-1} - S_1^{-1}) \\ &= c_1 t(\overline{x}_1 - x_1)(g_2 S_1^{-1} - S_1^{-1}) \\ &= -c_1 t(\overline{x}_1 - x_1)\Delta_2(S_1^{-1})\,. \end{aligned} \tag{4.46}$$

This is true if and only if

$$g_2 S_1^{-1} = S_1^{-1}\,. \tag{4.47}$$

We have

$$g_2(S_{1,2}^{-1,-1}\overline{x}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1} x_2) = S_{1,2}^{-1,-1} x_2 \overline{\mu}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1}\overline{x}_2 \mu_2 \tag{4.48}$$

$$S_{1,2}^{-1,-1} x_2 \overline{\mu}_2 + S_{1,2}^{-1,1}\overline{x}_2 \mu_2 = S_{1,2}^{-1,-1}\overline{x}_2 + S_{1,2}^{-1,1} x_2\,, \tag{4.49}$$

and we can write this succinctly as

$$S_{1,2}^{-1,-1} = \mu_2 S_{1,2}^{-1,1}\,. \tag{4.50}$$

If one of $S_{1,2}^{-1,-1}$ and $S_{1,2}^{-1,1}$ is zero, then so is the other, and since the step set has vertical symmetry,

$S_{1,2}^{1,1}$ and $S_{1,2}^{1,-1}$ are also zero. That is, the model has no diagonal steps. However, since the group has two nontrivial generators, we must have both steps with monomials $\overline{x}_2$ and x_2, and the model is highly-symmetric. If both of the S values are equal to one, then this implies $S_2^{-1} = S_2^1$ and again, the model is highly-symmetric.

Conversely, if we started with a highly-symmetric model, we would have obtained $\det M_4 = 0$. This leads to the following theorem.

Theorem 61. *The determinant* $\det M_4 = 0$ *if and only if the model is highly-symmetric.*

Corollary 62. *The Algebraic kernel method extends to the non-interacting case for* $|G| = 4$ *if and only if the model is highly-symmetric.*

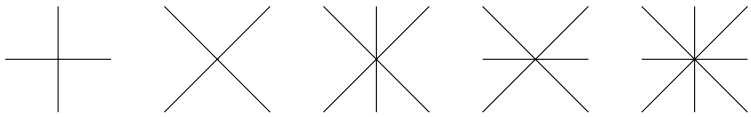


Figure 4.1: The highly-symmetric models on the plane with two nontrivial generators for the group G.

We are finally ready to use the orbit summation method. There are 8 highly-symmetric two-dimensional models, but only five of them have two nontrivial generators for the group G. Furthermore, two of the models are reflections of each other, so we are really left with four distinct cases. From left to right in the figure above, they are the Cartesian model, the Diagonal Cartesian model, the model with two horizontal steps missing, and the full model. If we are able to eliminate all the variables from the set E in the beginning of this section, then we will have a linear relation like

$$\gamma\beta - \gamma_1\beta_1 - \gamma_2\beta_2 + \gamma_{12}\beta_{12} = 0\,. \tag{4.51}$$

Here the choice of signs is inspired by the orbit summation operators. Let $\bullet$ and $*$ stand for applications of g_1 and g_2, respectively. Note that we have $K_2^\bullet = K_2$ and $K_1^* = K_1$. Next, we let

$$x = Q_1^0\,, \quad y = Q_2^0\,, \quad u = Q_1^{0,*}\,, \quad v = Q_2^{0,\bullet}\,, \tag{4.52}$$

and

$$a = K_1\,, \quad b = K_1^\bullet\,, \quad c = K_2\,, \quad d = K_2^*\,. \tag{4.53}$$

The original equations attain a simplified format given by

$$\beta = ax + cy\,, \quad \beta_1 = bx + cv\,, \tag{4.54}$$

$$\beta_2 = dy + au\,, \quad \beta_{12} = bu + dv\,, \tag{4.55}$$

Pluggin these into the linear relation above yields

$$\begin{aligned} 0 &= (ax + cy)\gamma - (bx + cv)\gamma_1 - (dy + au)\gamma_2 + (bu + dv)\gamma_{12} \\ &= (a\gamma - b\gamma_1)x + (c\gamma - d\gamma_2)y + (b\gamma_{12} - a\gamma_2)u + (d\gamma_{12} - c\gamma_1)v\,, \end{aligned} \tag{4.56}$$

and in order to eliminate the variables x, y, u and v, we need all their coefficients to vanish. We obtain the equations

$$a\gamma = b\gamma_1\,, \quad c\gamma = d\gamma_2\,, \quad b\gamma_{12} = a\gamma_2,\,, \quad d\gamma_{12} = c\gamma_1\,. \tag{4.57}$$

If we set $\gamma = 1$, then $\gamma_1 = a/b$ and $\gamma_2 = c/d$ and $\gamma_{12} = (ac)/(bd)$. So we can multiply them out by bd to find the relation

$$bd\beta - ad\beta_1 - cb\beta_2 + ac\beta_{12} = 0\,, \tag{4.58}$$

where we have successfully eliminated Q_1^0, Q_2^0, $Q_1^{0,*}$ and $Q_2^{0,\bullet}$. In terms of the K's we have

$$K_1^\bullet K_2^* \beta - K_1 K_2^* \beta^\bullet - K_1^\bullet K_2 \beta^* + K_1 K_2 \beta^{\bullet *} = 0\,. \tag{4.59}$$

We can rewrite this equation in the following nicer way.

Theorem 63. *For a two-dimensional highly-symmetric model with a group with two nontrivial generators, we have the functional equation*

$$\Delta(K_1^\bullet K_2^* \beta) = 0\,, \tag{4.60}$$

where $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$.

This is a generalization of the result in [**?**] where the authors use a weighted orbit sum to eliminate boundary terms. We can expand the functional equation to produce a different form. Let

$K_0 = K_1^\bullet K_2^*$. We find that

$$\begin{aligned}\Delta(K_0\beta) &= \Delta(K_0(KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha)) \\ &= \Delta(K_0KQ) + \Delta(K_0K_{1,2}Q_{1,2}^{0,0}) - \Delta(K_0\alpha) \\ &= K\Delta(K_0Q) + \Delta(K_0K_{1,2})Q_{1,2}^{0,0} - \Delta(K_0)\alpha\,,\end{aligned} \tag{4.61}$$

where the last equality follows since Δ commutes with K and functions with some variables x_i missing. All that remains is to expand the various quantities involved.

$$\Delta g_i K_i = -\Delta K_i = -(c_i - tx_iS_i^{-1} - c_i + t\overline{x}_iS_i^{-1}) = -t(\overline{x}_i - x_i)S_i^{-1} \tag{4.62}$$

and

$$\Delta K_0 = \Delta_1(K_1^\bullet)\Delta_2(K_2^*) = t^2(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)S_1^{-1}S_2^{-1}\,. \tag{4.63}$$

Next,

$$\Delta(K_0K_{1,2}) = \Delta_1K_1^\bullet\Delta_2K_2^*K_{1,2} = \Delta_1K_1^\bullet(K_2^*K_{1,2} - K_2K_{1,2}^*)\,. \tag{4.64}$$

where the quantity in parentheses is

$$\begin{aligned}&K_2^*K_{1,2} - K_2K_{1,2}^* = \\ =&(c_2 - tx_2S_2^{-1})(c_1c_2 - t\overline{x}_1\overline{x}_2S_{1,2}^{-1,-1}) - (c_2 - t\overline{x}_2S_2^{-1})(c_1c_2 - t\overline{x}_1x_2S_{1,2}^{-1,-1}) \\ =&c_1c_2tS_2^{-1}(\overline{x}_2 - x_2) - c_2t\overline{x}_2S_{1,2}^{-1,-1}(\overline{x}_2 - x_2) \\ =&c_2t(\overline{x}_2 - x_2)(c_1S_2^{-1} - \overline{x}_1S_{1,2}^{-1,-1})\,.\end{aligned} \tag{4.65}$$

Finally,

$$\begin{aligned}\Delta(K_0K_{1,2}) &= c_2t(\overline{x}_2 - x_2)\Delta_1K_1^\bullet(c_1S_2^{-1} - \overline{x}_1S_{1,2}^{-1,-1}) \\ &= c_2t(\overline{x}_2 - x_2)\Delta_1K_1^\bullet(c_1S_2^{-1} - \overline{x}_1S_{1,2}^{-1,-1})\,.(?)\end{aligned} \tag{4.66}$$

To simplify this quantity note that

$$\Delta_1K_1^\bullet c_1S_2^{-1} = c_1S_2^{-1}\Delta_1K_1^\bullet = -c_1tS_2^{-1}(\overline{x}_1 - x_1) \tag{4.67}$$

and

$$\Delta_1 K_1^\bullet \overline{x}_1 S_{1,2}^{-1,-1} = S_{1,2}^{-1,-1} \Delta_1 K_1^\bullet \overline{x}_1 = c_1(\overline{x}_1 - x_1) S_{1,2}^{-1,-1} \,. \tag{4.68}$$

Therefore, we find that

$$\Delta(K_0 K_{1,2}) = c_1 c_2 t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)(t S_1^{-1} S_2^{-2} - S_{1,2}^{-1,-1}) \,, \tag{4.69}$$

and after we combine terms and simplify, we can write down the following corollary.

Corollary 64. *For a two-dimensional highly-symmetric model with a group with two nontrivial generators and $K_0 = K_1^\bullet K_2^*$, we have the functional equation*

$$K\Delta(K_0 Q) = \alpha t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2) \left(t S_1^{-1} S_2^{-1}(1 - b_1 b_2 Q_{1,2}^{0,0}) + S_{1,2}^{-1,-1} Q_{1,2}^{0,0} \right) \,. \tag{4.70}$$

4.2.2 Groups of Order Six

We will go through this case faster because a lot of the work is similar. First, we tabulate all the unknowns Q_i^0 and their orbits under the actions of G. In the following table we write a sequence L for g_L and a reduced sequence for each unknown. For example, $g_1 Q_1^0 = Q_1^0$, so we write $\varnothing$ in the corresponding entry. For another example,

$$g_{2121} Q_1^0 = g_{212} Q_1^0 = g_{121} Q_1^0 = g_{12} Q_1^0 \,, \tag{4.71}$$

since $g_{212} = g_{121}$. The corresponding entry in the table is 12.

Sequence L	Alternative	Q_1^0	Q_2^0	K_1	K_2
1	21212	$\varnothing$	1	1	1
21	1212	2	21	21	21
121	212	12	21	121	212
2121	12	12	1	12	12
12121	2	2	$\varnothing$	2	2

Table 4.2: Reductions of group actions $g \in G$ on Q_i^0 and K_i.

After we apply the group actions, we find six different functions that need to be eliminated. Let

$$u_1 = Q_1^0\,, \quad u_2 = g_2 Q_1^0\,, \quad u_3 = g_{12} Q_1^0\,, \tag{4.72}$$

$$v_1 = Q_2^0\,, \quad v_2 = g_1 Q_2^0\,, \quad v_3 = g_{21} Q_2^0\,. \tag{4.73}$$

Also, there are twelve different K_1's and K_2's that we label sequentially as $A_1, \ldots, A_6$ and $B_1, \ldots, B_6$. The matrix associated with the linear system of equations is

$$M_6 = \begin{bmatrix} A_1 & & & B_1 & & \\ A_2 & & & & B_2 & \\ & A_3 & & & & B_3 \\ & & A_4 & & & B_4 \\ & & A_5 & & B_5 & \\ & A_6 & & B_6 & & \end{bmatrix}\,, \tag{4.74}$$

and it has determinant

$$\det M_6 = A_2 A_4 A_6 B_1 B_3 B_5 - A_1 A_3 A_5 B_2 B_4 B_6\,. \tag{4.75}$$

The even index A's are found by applying g_1 on the odd index A's, and vice versa for the B's. That is,

$$\det M_6 = \Delta_1(A_2 A_4 A_6 B_1 B_3 B_5) = \Delta_1(g_1(K_1) g_{121}(K_1) g_{12}(K_1) K_2 g_{21}(K_2) g_{12}(K_2))\,. \tag{4.76}$$

In the following table we show the relations between A's and B's for the six individual models with one of the two groups of order 6. $S11$, $S12$ and $S13$ have the one group, and $S21$, $S22$ and $S23$, have the other.

Model	Name	A-relations	B-relations
	S11	$A_1 = A_6, A_2 = A_5, A_3 = A_4$	$B_1 = B_4, B_2 = B_3, B_5 = B_6$
	S12	$A_1 = A_4, A_2 = A_3, A_5 = A_6$	$B_1 = B_2, B_3 = B_6, B_4 = B_5$
	S13	None	None
	S21	$A_1 = A_6, A_2 = A_5, A_3 = A_4$	$B_1 = B_2, B_3 = B_6, B_4 = B_5$
	S22	$A_1 = A_4, A_2 = A_3, A_5 = A_6$	$B_1 = B_4, B_2 = B_3, B_5 = B_6$
	S23	None	None

Table 4.3: Relations for $A = K_1$ and $B = K_2$ for groups of order six.

It is easy to check that when a model has nontrivial relation, $\det M_6 = 0$. Also, using a a computer, we can check that the determinant is nonzero in the last two cases where there are no relations. For the models with zero determinant, we seek the same type of relation on β and its orbit as in the case of groups of order four. We have the following linear relation

$$\begin{aligned} 0 =& \gamma\beta - \gamma_1\beta_1 + \gamma_{21}\beta_{21} - \gamma_{121}\beta_{121} + \gamma_{12}\beta_{12} - \gamma_2\beta_2 \\ =& \gamma(A_1u_1 + B_1v_1) - \gamma_2(A_6u_2 + B_6v_1) - \gamma_1A_2u_1 + B_2v_2) \\ &+ \gamma_{12}(A_5u_3 + B_5v_2) + \gamma_{21}(A_3u_2 + B_3v_3) - \gamma_{121}(A_4u_3 + B_4v_3)\,. \end{aligned} \tag{4.77}$$

Expanding out and collecting terms produces the system of equations

$$A_1\gamma = A_2\gamma_1\,, \qquad A_3\gamma_{21} = A_6\gamma_2\,, \qquad A_5\gamma_{12} = A_4\gamma_{121}\,, \tag{4.78}$$

$$B_1\gamma = B_6\gamma_2\,, \qquad B_5\gamma_{12} = B_2\gamma_1\,, \qquad B_3\gamma_{21} = B_4\gamma_{121}\,. \tag{4.79}$$

It does not have a nontrivial symbolic solution, so we need to use the relations here. There is a rational solution in each case, where γ is a free variable. We set γ to be the least common multiple of the denominators. We find that $K_0 = \gamma$ in each of the four cases with relations, as illustrated in the following table.

Model	S-polynomial	Weight $\gamma = K_0$	K-form
	$x_1\overline{x}_2 + \overline{x}_1 + x_2$	$A_3A_5B_6$	$g_{21}(K_1)g_{12}(K_1)g_2(K_2)$
	$\overline{x}_1x_2 + x_1 + \overline{x}_2$	$A_2A_5B_6$	$g_1(K_1)g_{12}(K_1)g_2(K_2)$
	$x_1x_2 + \overline{x}_1 + \overline{x}_2$	$A_2A_3B_5B_6$	$g_1(K_1)g_{21}(K_1)g_{12}(K_2)g_2(K_2)$
	$\overline{x}_1\overline{x}_2 + x_1 + x_2$	A_2B_6	$g_1(K_1)g_2(K_2)$

Table 4.4: Weight K_0 for the Orbit Summation operators.

Here we have picked the respresentations of A's and B's with the least number of transformations to reach the resulting expression. We also tried finding a possible K_0 by brute force for $S13$ and $S23$, but we did not find any among products of distinct A's and B's. It is likely that such a weight does not exist for these cases, and that the Algebraic kernel method does not extend for them.

Theorem 65. *The Algebraic kernel method extends for all interacting lattice walk models with small steps and group G of size six if and only if* $\det M_6 = 0$*. In those cases we have the functional equation*

$$D_G(K_0\beta) = 0\,, \tag{4.80}$$

where $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$ *and* K_0 *depends on the specific step set.*

4.2.3 Groups of Order Eight

The analysis for groups of order 8 is quite similar, so we omit most of it. Here are the results.

Model	S-polynomial
	$x_1 + \overline{x}_1 + x_1\overline{x}_2 + \overline{x}_1x_2$
	$x_1 + \overline{x}_1 + x_1x_2 + \overline{x}_1\overline{x}_2$

Table 4.5: Models with group of size 8.

We name the $A_1 = K_1$ and $B_1 = K_2$ iterates as before; for example, $A_2 = g_1A_1$ and

$A_3 = g_2 g_1 A_1$.

Sequence L	Alternative	Q_1^0	Q_2^0
1	2121212	$\varnothing$	1
21	121212	2	21
121	21212	12	121
2121	1212	212	121
12121	212	212	21
212121	12	12	1
1212121	2	2	$\varnothing$

Table 4.6: Reductions of group actions $g \in G$ on Q_i^0 and K_i.

The relations are the same in both models

$$A_1 = A_6\,, \quad A_2 = A_5\,, \quad A_3 = A_4\,, \quad A_7 = A_8\,, \tag{4.81}$$

$$B_1 = B_4\,, \quad B_2 = B_3\,, \quad B_4 = B_8\,, \quad B_6 = B_7\,. \tag{4.82}$$

In both cases we retrieved the same K_0.

Theorem 66. *The algebraic kernel method extends for all interacting lattice walk models with small steps and group of size eight. The functional equation is*

$$D_G(K_0\beta) = 0\,, \tag{4.83}$$

where $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$ *and*

$$K_0 = A_2 A_3 B_5 B_6 = g_1(K_1) g_{21}(K_1) g_{212}(K_2) g_{1212}(K_2). \tag{4.84}$$

Chapter 5

The Obstinate Kernel Method

We have another tool at our disposal, the Obstinate Kernel method, which was used in the interacting case in [29]. The idea is similar to the weighted algebraic kernel method in that we produce more functional equations from the master equation via group actions $g \in G$. The difference is that here we also set one of $x_1 = \sigma_1$, $x_2 = \sigma_2$ equal to a root of K in order to eliminate the KQ term from the equations. This leaves equations that only include the other variable. It turns out that some combinations of substitutions and group operations do not lead to valid functional equations. This happens when one of the Q_k^0 terms is not a formal power series in t. In all the cases with finite groups of order 4, 6 and 8 we find that there are 2 roots of K for each x_k and that one root is in $O(t)$ and the other in $O(1/t)$. Generally, it works out better if we use the $O(t)$ roots because that at least guarantees that the Q_k^0 terms remain as formal power series in t. However, sometimes (page 11 of [29]) even the $O(1/t)$ term may lead to a valid power series.

It may not be necessary, but it will be sufficient if a combination of a substitution $x_k = \sigma_k$ and a group action g_L together give a power series in t when applied to single variables like x_1 or x_2. For example, we would hope that $g_L(x_j)|_{x_k=\sigma_k}$ is a power series in t for some i, j, L. Again, this is not a necessary condition, but it will be sufficient to show that a lot of the cases involved produce enough different equations to allow us to go through the Obstinate Kernel method. For those cases where we find less equations, it is likely that we will be able to show that the Q_k^0 terms are power series in t via another method like in (two friendly walks).

Definition 67. *A function $f(x_1, \ldots, x_d; t)$ is admissible if it is a power series in t. If σ_k is a root of K, we say f is k-admissible if $f|_{x_k=\sigma_k}$ is admissible. A set of functions is admissible if each element is admissible.*

Proposition 68. *Let $\mathcal{M}$ be a lattice walk model with small nonrepeating (?) steps and group G of order 4. The actions $g \in G$ that preserve the admissibility are given in the following table.*

When $|G| = 6$ we have a richer situation.

Proposition 69. *Let $\mathcal{M}$ be a lattice walk model with small nonrepeating steps and group G of order 4, 6 or 8. Models with the same group have the same admissibility properties with respect to admissible when it comes to x_1, x_2 or the pair (x_1, x_2).*

Proof. We do not know of a conceptual proof of this fact, so we performed the calculations on a computer. They are trivial computationally speaking. For a given model $\mathcal{M}$, we construct its S-polynomial, which can be used to find the admissible roots σ_k for $k = 1, 2$. We can use S to express the generators g_k and their action on the pair (x_1, x_2). Once we have the generators we form the group G. Next, to check the admissibility of $f \in \{x_1, x_2, (x_1, x_2)\}$ all we need to do is find $|\mathrm{Orbit}_G(f)|$ and see if substituting $x_k = \sigma_k$ in the resulting expressions leads to admissible functions. The calculations can be done by hand, but they are tedious, so we have employed a computer to do the legwork. It becomes a trivial matter to check the admissibility properties for the various models and compare the results, summarized in the following tables. □

Type	x_1	x_2	(x_1, x_2)
1-admissibility	1, g_2	all $g \in G$	1, g_2
2-admissibility	1, g_1	all $g \in G$	1, g_1

Table 5.1: k-admissibility of functions when $|G| = 4$.

Type	x_1	x_2	(x_1, x_2)
1-admissibility	1, g_2, g_{21}, g_{121}	1, g_1, g_2, g_{21}	1, g_2, g_{21}
2-admissibility	1, g_1, g_2, g_{12}	1, g_1, g_{12}, g_{121}	1, g_1, g_{12}

Table 5.2: k-admissibility of functions for $S11$, $S12$, $S13$. Here $|G| = 6$.

Type	x_1	x_2	(x_1, x_2)
1-admissibility	1, g_2, g_{21}, g_{121}	1, g_1, g_{12}, g_{121}	1, g_{121}
2-admissibility	1, g_2, g_{21}, g_{121}	1, g_1, g_{12}, g_{121}	1, g_{121}

Table 5.3: k-admissibility of functions for $S21$, $S22$, $S33$. Here $|G| = 6$.

Type	x_1	x_2	(x_1, x_2)
1-admissibility	1, g_2, g_{21}, g_{212}	all $g \in G$	1, g_2, g_{21}, g_{212}
2-admissibility	1, g_1, g_2, g_{12}, g_{121}, g_{2121}	1, g_1, g_{12}, g_{121}	1, g_1, g_{12}, g_{121}

Table 5.4: k-admissibility of functions for $S1$. Here $|G| = 8$.

Type	x_1	x_2	(x_1, x_2)
1-admissibility	1, g_2, g_{21}, g_{212}	all $g \in G$	1, g_2, g_{21}, g_{212}
2-admissibility	1, g_2, g_{21}, g_{121}, g_{212}, g_{2121}	1, g_1, g_{12}, g_{121}	1, g_{121}

Table 5.5: k-admissibility of functions for $S2$. Here $|G| = 8$.

What this classification tells us is that we can easily treat all $|G| = 4$ models with the same method. When $|G| = 6$ we either have to do some more work in the $S2k$ cases to show that the functions are admissible with respect to one more group operation, or just deal with them as they are. The situation is similar in the $|G| = 8$ case. Now suppose that we have a set of valid equations for a substitution $x_k = \sigma_k$. They are a linear system in the variable quantities $Q^{0,0}_{1,2}$, Q^0_1, Q^0_2 and their orbits under the set of valid group actions. We hope for a high level of redundancy in these equations. Then we can eliminate some variables by taking a weighted sum over the valid equations. That is, we attain a weighted Orbit Sum operator that only has a few group operations, perhaps. We also have to keep in mind that we are also making the substitution $x_k = \sigma_k$.

We will perform the first steps in the analysis of the Obstinate Kernel method for each situation that has $|G|/2$ k-admissible elements since that seems to work.

5.1 Group of Order Four

The analysis in this case is the same whether we work with the σ_1 or σ_2 substitutions. We will just work with the σ_1-substitution and group actions 1, g_2, where we let $g_2(F) = F^*$ to keep the notation cleaner. The functional equations corresponding the the group actions are

$$KQ - K_1Q^0_1 - K_2Q^0_2 + K_{12}Q^{0,0}_{1,2} = \alpha\,, \tag{5.1}$$

$$KQ^* - K^*_1Q^{0,*}_1 - K^*_2Q^0_2 - K^*_{12}Q^{0,0}_{1,2} = \alpha\,. \tag{5.2}$$

We will eliminate Q^0_2 because once we make the σ_1-substitution, we would have a tough time trying to expand Q^0_2 into series. We do this by taking K^*_2 times the first equation and subtract K_2 times

the second equation. This leads to the functional equation

$$K_2^* KQ - K_2 KQ^* - (K_2^* K_1 Q_1^0 - K_2 K_1^* Q_1^{0,*}) + (K_2^* K_{12} - K_2 K_{12}^*) Q_{1,2}^{0,0} - \alpha(K_2^* - K_2) = 0 \tag{5.3}$$

where we can set $x_1 = \sigma_1$ to eliminate the K term. We overload notation by not changing any of the symbols. From now on we will know that $x_1 = \sigma_1$ in the equations that follow. We have

$$(K_2^* K_1 Q_1^0 - K_2 K_1^* Q_1^{0,*}) - (K_2^* K_{12} - K_2 K_{12}^*) Q_{1,2}^{0,0} + \alpha(K_2^* - K_2) = 0. \tag{5.4}$$

We can also think of this process as applying a weighted orbit sum over group actions that preserve the admissibility of functions in the master equation. The operator is

$$D_2 = g_2(K_2) - K_2 g_2 = (1 - g_2) K_2^* \tag{5.5}$$

where the weights are given by pre-multiplying an expression by K_2^* before applying the operator $\Delta_2 = 1 - g_2$. The equation can be written as

$$\begin{aligned} 0 &= D_2(KQ - K_1 Q_1^0 - K_2 Q_2^0 + K_{12} Q_{1,2}^{0,0}) \\ 0 &= D_2(K_1 Q_1^0) - D_2(K_{12}) Q_{1,2}^{0,0} + D(\alpha) \end{aligned} \tag{5.6}$$

where the second one follow from substituting $x_1 = \sigma_1$. The situation is similar to the Algebraic Kernel method except that the operators there retain the KQ term and its orbit, while they eliminate the Q_1^0 and Q_2^0 terms and their orbits.

Definition 70. *(?) For a concatenated list $L = L_1 L_2 \ldots$ of positive integer indices, let $g_L = g_{L_1} g_{L_2} \cdots$ whenever this group action makes sense. That is, either the list is finite or there is some form of convergence when L is infinite.*

Definition 71. *Let L be a concatenated list. Definte $\Delta_L = 1 - g_L$. Now let X be some function. The operator $D_L[X]$ is defined by its action on functions F as*

$$D_L[X](F) = \Delta_L(XF) = (1 - g_L)(XF) = XF - g_L(XF)\,. \tag{5.7}$$

We will drop the X from the notation if it is clear from context. In particular, $D_L[1] = \Delta_L$.

For example, $D_2 = D_2[g_2(K_2)]$.

Example 72. If we use $g_L(X)$ instead of X in the above definition, we find that

$$D_L[g_L(X)] = g_L(X) - Xg_L \tag{5.8}$$

and this is essentially an Ore operator. In our analysis we will see these somewhat frequently.

5.1.1 Some Calculations

Before we go onto groups of order 6 we can expand some of the terms in the functional equation. First

$$\begin{aligned} D_2(\alpha) &= \alpha(K_2^* - K_2) = \\ &= \alpha((c_2 - tx_2\overline{\mu}_2 S_2^{-1}) - (c_2 - t\overline{x}_2 S_2^{-1})) \\ &= \alpha t(\overline{x}_2 - x_2\overline{\mu}_2(\sigma_1))S_2^{-1}(\sigma_1)\,. \end{aligned} \tag{5.9}$$

Then

$$\begin{aligned} D_2(K_{12}) &= K_2^* K_{12} - K_2 K_{12}^* = \\ &= (c_2 - tx_2\overline{\mu}_2 S_2^{-1})(c_1c_2 - t\overline{\sigma}_1\overline{x}_2 S_{1,2}^{-1,-1}) \\ &\quad - (c_2 - t\overline{x}_2 S_2^{-1})(c_1c_2 - t\overline{\sigma}_1 x_2\overline{\mu}_2 S_{1,2}^{-1,-1}) \\ &= c_1c_2^2 - c_1c_2tx_2\overline{\mu}_2 S_2^{-1} - c_2t\overline{\sigma}_1\overline{x}_2 S_{1,2}^{-1,-1} + t^2\overline{\sigma}_1\overline{\mu}_2 S_2^{-1}S_{1,2}^{-1,-1} \\ &\quad - c_1c_2^2 + c_1c_2t\overline{x}_2 S_2^{-1} + c_2t\overline{\sigma}_1 x_2\overline{\mu}_2 S_{1,2}^{-1,-1} - t^2\overline{\sigma}_1\overline{\mu}_2 S_2^{-1}S_{1,2}^{-1,-1} \\ &= c_1c_2t(\overline{x}_2 - x_2\overline{\mu}_2)S_2^{-1} - c_2t\overline{\sigma}_1(\overline{x}_2 - x_2\overline{\mu}_2)S_{1,2}^{-1,-1} \\ &= c_2t(\overline{x}_2 - x_2\overline{\mu}_2(\sigma_1))(c_1S_2^{-1}(\sigma_1) - \overline{\sigma}_1 S_{1,2}^{-1,-1}) \end{aligned} \tag{5.10}$$

If the model is symmetric in both directions, we see that $g_2S_1^{-1} = S_1^{-1}$ and therefore, $g_2K_1 = K_1$, and we may bring that factor outside of $D_2(K_1^*F) = K_1D_2(F)$. It is worth investigating if this happens in other cases since this will make it easier to perform coefficient extraction later. We can answer this in the negative.

Proposition 73. *Suppose that a model with group of order 4 has symmetry around the x_i axis and $j \neq i$. Then $D_i[g_iK_j] = K_jD_i[1] = K_j\Delta_i$ in the σ_j-substitution. Also, $D_j[g_j(K_i)] = K_iD_j[1] = K_i\Delta_j$ if and only if the model is also symmetric around the x_j axis. Thus, we can bring out the K term out of both operators if and only if the model is highly symmetric.*

Proof. Assume that the model is symmetric around the x_1 axis, that is, it is vertically symmetric. This implies that $S_1^{-1} = S_1^1$ and $g_1(S_2^k) = S_2^k$ for all k. This immediately tells us that $g_1(K_2) = K_2$

and $D_1[g_1K_2] = K_2\Delta_1$. This remains true when we substitute $x_2 = \sigma_2$. This would also be true for the other operator if the model is symmetric around the x_2 axis. Thus, we only need to show the forward direction of the statement. We will show that if the model is not highly symmetric, the factoring will not happen. It is enough to show that $g_2(K_1) \neq K_1$ even after the $x_1 = \sigma_1$ substitution is made.

The solutions to $K = 0$ in the x_1 variable are

$$x_1 = \frac{(1 - tS_1^0) \pm \sqrt{(1 - tS_1^0)^2 - 4t^2(S_1^1)^2}}{2tS_1^1}. \tag{5.11}$$

If $\sigma_1 = u \pm v\sqrt{w}$, then it turns out that $\overline{\sigma}_1 = u \mp v\sqrt{w}$ due to the symmetry and

$$s = \sigma_1 + \overline{\sigma}_1 = 2u = \frac{1 - tS_1^0}{tS_1^1}. \tag{5.12}$$

Now

$$g_2K_1 = c_2 - t\overline{x}_1 g_2(S_1^{-1}(x_2)) = c_2 - t\overline{x}_1 S_1^{-1}(\overline{x}_2\mu_2(x_1)) \tag{5.13}$$

and $g_2K_1 = K_1$ if and only if

$$S_1^{-1}(\overline{x}_2\mu_2(x_1)) = S_1^{-1}(x_2) \tag{5.14}$$

upon making the substitution $x_1 = \sigma_1$. We will first compute $\mu_2(\sigma_1)$. To that end we have

$$S_2^k(\sigma_1) = S_{1,2}^{0,k} + S_{1,2}^{1,k}(\overline{\sigma}_1 + \sigma_1) = S_{1,2}^{0,k} + S_{1,2}^{1,k}\frac{1 - tS_1^0}{tS_1^1} \tag{5.15}$$

and

$$\mu_2(\sigma_1) = \frac{S_2^{-1}}{S_2^1} = \frac{S_{1,2}^{0,-1} + S_{1,2}^{1,-1}(1 - tS_1^0)/(tS_1^1)}{S_{1,2}^{0,1} + S_{1,2}^{1,1}(1 - tS_1^0)/(tS_1^1)} = \frac{S_{1,2}^{1,-1} + t\left(S_{1,2}^{0,-1}S_1^1 - S_{1,2}^{1,-1}S_1^0\right)}{S_{1,2}^{1,1} + t\left(S_{1,2}^{0,1}S_1^1 - S_{1,2}^{1,1}S_1^0\right)}. \tag{5.16}$$

This is a rational function in x_2. Using software, we can easily compute this expression and further compute $S_1^{-1}(\overline{x}_2\mu_2(x_1))$ to show that we only have equality when the model is highly-symmetric.

□

In a later section (ref) we will investigate the highly-symmetric models in more detail.

5.2 Group of Order Six

We can perform the Obstinate Kernel method on the three models in the first group. They are as follows.

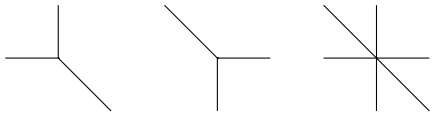


Figure 5.1: Models $S11$, $S12$, $S13$ with group of order 6.

The k-admissibility of (x_1, x_2) is preserved by actions 1, g_j, g_{jk} for $j \neq k$. For consistency, we will work on the 1-admissible case. The actions are 1, g_2, g_{21}. We have the table from the Algebraic Kernel method.

Sequence L	Q_1^0	Q_2^0
$\varnothing$	$\varnothing$	$\varnothing$
2	2	$\varnothing$
21	2	21

Table 5.6: Reductions of group actions $g \in G$ on Q_1^0 and Q_2^0.

In this group we have $g_2(x_1, x_2) = (x_1, x_1\overline{x}_2)$ and $g_{21}(x_1, x_2) = (\overline{x}_2, x_1\overline{x}_2)$, so the terms we wish to eliminate are those involving an x_1 because the substitution $x_1 = \sigma_1$ will make it difficult to expand the Q functions into series. That is, we wish to eliminate $X_2 = g_2 Q_1^0$ and $Y = Q_2^0$. Let

$$\beta = K_{12} Q_{1,2}^{0,0} - \alpha = \alpha(b_1 b_2 Q_{1,2}^{0,0} - 1) \tag{5.17}$$

so that we have equations

$$AX + BY = \beta\,, \quad A_2 X_2 + B_2 Y = \beta_2\,, \quad A_{21} X + B_{21} Y_{21} = \beta_{21}\,. \tag{5.18}$$

We want a linear combination of $\beta, \beta_2, \beta_{21}$ that will eliminate X_2 and Y. We may write

$$a\beta - b\beta_2 + c\beta_{21} = aAX + (aB - bB_2)Y + (cA_{21} - bA_2) + cB_{21}Y_{21} = aAX + cB_{21}Y_{21} \tag{5.19}$$

and this means that

$$aB = bB_2\,, \quad cA_{21} = bA_2\,. \tag{5.20}$$

We do not want to have any denominators, so when we solve for a, b and c, we have a small solution

$$a = A_{21}B_2\,, \quad b = A_{21}B\,, \quad c = A_2B\,, \tag{5.21}$$

and these may or may not have common factors. In other words the elimination procedure yields

$$\begin{aligned} &AA_{21}B_2Q_1^0 + A_2BB_{21}g_{21}Q_2^0 = A_{21}B_2\beta - A_{21}B\beta_2 + A_2B\beta_{21} \\ &= (A_{21}B_2 - A_{21}B + A_2B)\beta\,. \end{aligned} \tag{5.22}$$

Here we may think of a weighted sum operator too. Let

$$D(F) = A_{21}B_2F - A_{21}Bg_2(F) + A_2Bg_{21}(F)\,. \tag{5.23}$$

Then we can rewrite the equation as

$$D(AX + BY) = AA_{21}B_2Q_1^0 + A_2BB_{21}g_{21}Q_2^0 = \beta D(1)\,. \tag{5.24}$$

We may compute $D(1)$ for each case, but the expressions are somewhat complicated. $D(1)$ for each of $S11$, $S12$ and $S13$ will contain a square root like in σ_1. So do the terms $AA_{21}B_2$ and A_2BB_{21}. It is worth mentioning that we can convert the equation to one with no square roots by collecting terms involving the square root on one side and squaring. Then we have a nonlinear equation in X and Y with rational function coefficients. It is not clear how we could go further.

5.3 Group of Order Eight

Since the group G has order 8, we only have two nonisomorphic models, but we will deal with the one with four group actions that preserve the admissibility of (x_1, x_2). The actions are 1, g_2, g_{21}, g_{212}. The step polynomial is



$$:\ S = \overline{x}_1 + x_1 + \overline{x}_1x_2 + x_1\overline{x}_2 \tag{5.25}$$

This is the case covered in [29]. We will go through the same problem but from this new perspective. From the previous chapter we have

Sequence L	Q_1^0	Q_2^0
$\varnothing$	$\varnothing$	$\varnothing$
2	2	$\varnothing$
21	2	21
212	212	21

Table 5.7: Reductions of group actions $g \in G$ on Q_1^0 and Q_2^0 that preserve 1-admissibility.

We will (?) Let $\beta = K_{12}Q_{1,2}^{0,0} - \alpha = \alpha(b_1 b_2 Q_{1,2}^{0,0} - 1)$, $X = Q_1^0$ and $Y = Q_2^0$. Note that β is left unchanged by group actions. We have the four equations

$$AX + BY = \beta\,, \tag{5.26}$$

$$A_2X_2 + B_2Y = \beta\,, \tag{5.27}$$

$$A_{21}X_2 + B_{21}Y_{21} = \beta\,, \tag{5.28}$$

$$A_{212}X_{212} + B_{212}Y_{21} = \beta\,, \tag{5.29}$$

where we wish to eliminate variables X, Y that contain σ_1 in their input. Those are X_2, Y and Y_{21}, and we can eliminate them by first eliminating Y and Y_{21} from the first two and second two equations, and then eliminating X_2 from the resulting two equations. That is, we compute B_2 times the first equation minus B times the second equation, and B_{212} times the third equation minus B_{21} from the last equation. In other words this is the same as if we apply the operators

$$D_2[B_2] = B_2 - Bg_2\,, \tag{5.30}$$

$$-g_{212}D_2[B_2] = g_{212}\,Bg_2 - g_{212}\,B_2 = B_{212}\,g_{21} - B_{21}\,g_{212} \tag{5.31}$$

to both sides of the master functional equation and substitute $x_1 = \sigma_1$. We obtain the equations

$$AB_2X - A_2BX_2 = \beta(B_2 - B)\,, \tag{5.32}$$

$$A_{21}B_{212}X_2 - A_{212}B_{21}X_{212} = \beta(B_{212} - B_{21})\,. \tag{5.33}$$

To eliminate the X_2 terms we take $A_{21}B_{212}$ times the first equation and add A_2B times the second equation. We have

$$A_{21}B_{212}AB_2X - A_2BA_{212}B_{21}X_{212} = \beta(A_{21}B_{212}(B - B_2) + A_2B(B_{212} - B_{21})) \tag{5.34}$$

We could recover the same result by applying the following operator D to both sides of the master functional equation.

$$\begin{aligned} D &= A_{21}B_{212}D_2[B_2] - A_2Bg_{212}D_2[B_2] \\ &= (A_{21}B_{212} - A_2Bg_{212})D_2[B_2] \\ &= (A_{21}B_{212} - g_{212}A_{21}B_{212})D_2[B_2] \\ &= D_{212}[A_{21}B_{212}]D_2[B_2]\,. \end{aligned} \tag{5.35}$$

Starting from the master functional equation, we have The right-hand side

$$D(\beta) = \beta D_{212}[A_{21}B_{212}](B_2 - B) \tag{5.36}$$

and the left-hand side

$$\begin{aligned} D(AX + BY) &= D(AX) \\ &= D_{212}[A_{21}B_{212}]D_2[B_2](AX) \\ &= D_{212}[A_{21}B_{212}](AB_2X - A_2BX_2) \\ &= D_{212}[A_{21}B_{212}](AB_2X) \\ &= D_{212}[AA_{21}B_2B_{212}](X) \end{aligned} \tag{5.37}$$

We note that $A_{212} = A$, so we can actually factor out A from the operator. Finally, we have the following theorem.

Theorem 74. *For the first model with group of order 8, we have*

$$D_{212}[A_{21}B_2B_{212}]\left(Q_1^0\right) = \frac{\beta}{A}D_{212}[A_{21}B_{212}](B_2 - B) \tag{5.38}$$

where $A_L = g_L(K_1)$, $B_L = g_L(K_2)$ and $\beta = \alpha(b_1b_2Q_{1,2}^{0,0} - 1)$, and the equality is valid only once we set $x_1 = \sigma_1$ in this expression.

If we went through the σ_2 substitution instead, we would have obtained an analogous statement. Just switch the A's and the B's, and the 1's and the 2's.

The interesting fact (as found in [29]) is that the coefficients in front of the Q's become Laurent polynomials once we take into consideration the σ_k-substitutions, even though the σ_k's contain square roots. The coefficient in front of β can be reduced to a rational function plus a rational function times a square root after properly conjugating. The calculations are exceedingly complicated and would be too long to be done by hand. The square root part does not depend on a_2 in the first

case, and a_1 in the second case. This is what allows the authors to find relationships between $Q^{0,0}_{1,2}$ and expressions where one or more of a_1 and a_2 equal to 1.

5.4 Highly-symmetric Two-dimensional Models

We will continue further with the OKM for highly-symmetric models. There are only a few such models. If x_k is in S, then so is $\overline{x}_k$. If one of the diagonal terms is in S, then so are the others. To have a group of order 4 we need to have both an x_1 and x_2 variable in S, so we have several nonisomorphic models. One with cartesian steps, one with diagonal steps, one with diagonal step and x_1,$\overline{x}_1$ and finally, the model with all steps. The notation is the same as in previous chapters. We derived the following equations

$$x_1 = \sigma_1 : \quad \Delta_2(K2^* Q^0_1) = \frac{1}{K_1}\left(\Delta_2(K^*_2 K_{12}) Q^{0,0}_{1,2} - \alpha(K^*_2 - K_2)\right) , \tag{5.39}$$

$$x_2 = \sigma_2 : \quad \Delta_1(K1^\bullet Q^0_2) = \frac{1}{K_2}\left(\Delta_1(K^\bullet_1 K_{12}) Q^{0,0}_{1,2} - \alpha(K^\bullet_1 - K_1)\right) . \tag{5.40}$$

It is enough to only consider the analysis for one of the equations for now. Only the third model requires looking at both equations. We already know from the previous chapter that $\mu_2 = S^{-1}_2/S^1_2 = 1$ and

$$\alpha(K^*_2 - K_2) = \alpha t(\overline{x}_2 - x_2) S^{-1}_2(\sigma_1) , \tag{5.41}$$

$$\Delta_2(K^*_2 K_{12}) = c_2 t(\overline{x}_2 - x_2)(c_1 S^{-1}_2(\sigma_1) - \overline{\sigma}_1 S^{-1,-1}_{1,2}) . \tag{5.42}$$

To compute $\Delta_2(K2^* Q^0_1)$ we need to compute $K_2(\sigma_1, x_2)$ and $K^*_2(\sigma_1, x_2)$, which are given by

$$K_2 = c_2 - t\overline{x}_2 S^{-1}_2(\sigma_1) , \quad K^*_2 = c_2 - t x_2 S^{-1}_2(\sigma_1) . \tag{5.43}$$

Therefore, we focus on computing $S^{-1}_2(\sigma)$ first.

$$\begin{aligned} S^{-1}_2(\sigma_1) &= S^{0,1}_{1,2} + S^{1,1}_{1,2}(\overline{\sigma}_1 + \sigma_1) \\ &= S^{0,1}_{1,2} + S^{1,1}_{1,2}\frac{1 - tS^0_1}{tS^1_1} \\ &= \frac{1}{tS^1_1}\left(S^{1,1}_{1,2} + t\left(S^{0,1}_{1,2} S^1_1 - S^{1,1}_{1,2} S^0_1\right)\right) \\ &= \frac{S^{1,1}_{1,2} + tS^{0,1}_{1,2} S^{1,0}_{1,2}}{t\left(S^{1,0}_{1,2} + S^{1,1}_{1,2}(\overline{x}_2 + x_2)\right)} . \end{aligned} \tag{5.44}$$

Therefore, the function $S_2^{-1}(\sigma_1)$ is a rational function of x_2.

Model	$S_{1,2}^{0,1}$	$S_{1,2}^{1,0}$	$S_{1,2}^{1,1}$	$S_2^{-1}(\sigma_1)$
	1	1	0	1
	0	0	1	$\bar{t}/(\overline{x}_2 + x_2)$
	1	0	1	$\bar{t}/(\overline{x}_2 + x_2)$
	0	1	1	$\bar{t}/(\overline{x}_2 + 1 + x_2)$
	1	1	1	$(1 + \bar{t})/(\overline{x}_2 + 1 + x_2)$

Table 5.8: Function $S_2^{-1}(\sigma_1)$ for highly-symmetric models.

Lemma 75. *For highly-symmetric models with $e = S_{1,2}^{1,1}$, $f = S_{1,2}^{1,0}$ and $g = S_{1,2}^{0,1}$, we have*

$$\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(e + tfg)}{S_1^1 K_1}\,, \tag{5.45}$$

$$\frac{\Delta_2(K_2^* K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1(e + tfg) - et\overline{\sigma}_1 S_1^1)}{S_1^1(c_1 - t\overline{\sigma}_1 S_1^1)}\,. \tag{5.46}$$

5.4.1 Cartesian Model – N, S, E, W

In this case $S_2^{-1}(\sigma_1) = 1$, so a lot of the calculations will be simpler, and this is a good place to start further work. Here $K_{12} = c_1 c_2$, so it behaves like α and we have

$$\Delta_2(K2^* Q_1^0) + \frac{\alpha\beta(K_2^* - K_2)}{K_1} = 0\,, \tag{5.47}$$

where

$$\beta = 1 - b_1 b_2 Q_{1,2}^{0,0} \tag{5.48}$$

and

$$\frac{\alpha\beta(K_2^* - K_2)}{K_1} = \frac{\alpha\beta t(\overline{x}_2 - x_2)}{K_1} = \alpha\beta f(c_1, x_2)\,, \tag{5.49}$$

since K_1 does not depend on c_2 or x_1. The equation finally becomes

$$(c_2 - tx_2)Q_1^0(x_2) - (c_2 - t\overline{x}_2)Q_1^0(\overline{x}_2) + \alpha\beta f = 0\,. \tag{5.50}$$

We perform the extraction $[x_2^1]$ to get

$$c_2 Q_{1,2}^{0,1} - tQ_{1,2}^{0,0} + \alpha\beta f_2^1 = 0 \tag{5.51}$$

and the extraction $[b_2^0]$ to get

$$-t\check{Q}_{1,2}^{0,0} + \overline{a}_1 f_2^1\,. \tag{5.52}$$

Using the second equation within the first one and simplifying yields

$$b_2 Q_{1,2}^{0,1} = t(a_2 Q_{1,2}^{0,0} - \beta\check{Q}_{1,2}^{0,0})\,, \tag{5.53}$$

$$b_1 Q_{1,2}^{1,0} = t(a_1 Q_{1,2}^{0,0} - \beta\widehat{Q}_{1,2}^{0,0}) \tag{5.54}$$

Here the second equation is obtained via the same method but with $\bullet$ instead of $*$. Using the same method we can also write

$$b_2 Q_{1,2}^{1,1} = t(a_2 Q_{1,2}^{1,0} - \beta\check{Q}_{1,2}^{1,0})\,. \tag{5.55}$$

Multiplying by b_1 and using the above relations leads to

$$\begin{aligned} b_1 b_2 Q_{1,2}^{1,1} &= t(a_2 b_1 Q_{1,2}^{1,0} - \beta b_1 \check{Q}_{1,2}^{1,0}) \\ &= ta_2 t(a_1 Q_{1,2}^{0,0} - \beta\widehat{Q}_{1,2}^{0,0}) - t\beta t(a_1\check{Q}_{1,2}^{0,0} - \beta\check{\widehat{Q}}_{1,2}^{0,0}) \\ &= t^2(a_1 a_2 Q_{1,2}^{0,0} - a_2\beta\widehat{Q}_{1,2}^{0,0} - a_1\beta\check{Q}_{1,2}^{0,0} + \beta^2\check{\widehat{Q}}_{1,2}^{0,0})\,. \end{aligned} \tag{5.56}$$

Let

$$G = Q_{1,2}^{0,0}, \quad \widehat{G} = G_1, \quad \check{G} = G_2\,. \tag{5.57}$$

From before we have

$$Q_{1,2}^{0,0} = 1 + ta_1 a_2(Q_{1,2}^{1,0} + Q_{1,2}^{0,1})\,. \tag{5.58}$$

Multiplying by $b_1 b_2$ and using the above relations leads to

$$\begin{aligned} b_1 b_2 G &= b_1 b_2 + t a_1 a_2 (b_2 b_1 Q_{1,2}^{1,0} + b_1 b_2 Q_{1,2}^{0,1}) \\ &= b_1 b_2 + t a_1 a_2 (b_2 t (a_1 G - \beta G_1) + b_1 t (a_2 G - \beta G_2) \\ &= b_1 b_2 + t^2 a_1 a_2 (b_2 a_1 G + b_1 a_2 G - b_2 \beta G_1 - b_1 \beta G_2) \,. \end{aligned} \tag{5.59}$$

Using the relation for β and simplifying leads to

$$G = \frac{b_1 b - 2 - t^2 a_1 a_2 (b_2 G_1 + b_1 G_2)}{b_1 b_2 - t^2 a_1 a_2 (a_2 b_1 + a_1 b_2 + b_1 b_2 (b_2 G_1 + b_1 G_2))} \,. \tag{5.60}$$

If we let $F = (b_2 G_1 + b_1 G_2)$ and perform the long division with respect to F, we find that

$$Q_{1,2}^{0,0} = \frac{1}{b_1 b_2} \left(1 - \frac{b_1 b_2 (1 - b_1 b_2) - t^2 a_1 a_2 (a_1 b_2 + a_2 b_1)}{b_1 b_2 + t^2 a_1 a_2 (a_1 b_2 + a_2 b_1 + b_1 b_2 F)} \right) \,. \tag{5.61}$$

Using arguments similar to those in the later section on diagonal steps we can find out more about the various Q's. We have

$$b_2 Q_{1,2}^{m,n+1} = t (a_2 Q_{1,2}^{m,n} - \beta \breve{Q}_{1,2}^{m,n}) \,, \tag{5.62}$$

$$b_1 Q_{1,2}^{m+1,n} = t (a_1 Q_{1,2}^{m,n} - \beta \widehat{Q}_{1,2}^{m,n}) \,. \tag{5.63}$$

Summing the second set of equations multiplied by x_2^n, we find the equations, indexed by m,

$$b_1 Q_1^{m+1} = t (a_1 Q_1^m - \beta \widehat{Q}_1^m) \,. \tag{5.64}$$

Now we sum these equations with respect to x_1 to find

$$b_1 (Q - Q_1^0) = x_1 t (a_1 Q - \beta \widehat{Q}) \,, \tag{5.65}$$

and we can solve this equation for Q_1^0 as follows

$$b_1 Q_1^0 = (b_1 - a_1 x_1 t) Q + x_1 t \beta \widehat{Q} \,. \tag{5.66}$$

We would have found an analogous expression for $b_2 Q_2^0$ if we performed the procedure on the first set of equations by summing with respect to x_1 first and then with respect to x_2. We substitute these

two expressions into the original functional equation, and after some algebra we find that

$$Q = \frac{(1 - b_1 b_2 G)(\alpha + t x_1 \overline{b}_1 K_1 \widehat{Q} + t x_2 \overline{b}_2 K_2 \breve{Q})}{K - (1 - \overline{c}_1 x_1 t) K_1 - (1 - \overline{c}_2 x_2 t) K_2} \tag{5.67}$$

We recognize some of the quantities as modified K_1 and K_2's, so we can finally write down the full solution for Q.

Theorem 76. *The Cartesian model in two dimensions has generating functions satisfying the equations*

$$Q^{0,0}_{1,2} = G = \frac{1}{b_1 b_2}\left(1 - \frac{b_1 b_2 (1 - b_1 b_2) - t^2 a_1 a_2 (a_1 b_2 + a_2 b_1)}{b_1 b_2 + t^2 a_1 a_2 (a_1 b_2 + a_2 b_1 + b_1 b_2 F)}\right), \tag{5.68}$$

where $F = (b_2 \widehat{G} + b_1 \breve{G})$*,*

$$b_1 Q^0_1 = (b_1 - a_1 x_1 t) Q + x_1 t \beta \widehat{Q}\,, \tag{5.69}$$

$$b_2 Q^0_2 = (b_2 - a_2 x_2 t) Q + x_2 t \beta \breve{Q}\,, \tag{5.70}$$

and

$$Q = \frac{(1 - b_1 b_2 Q^{0,0}_{1,2})(\alpha - \overline{b}_1 K_1 \widehat{K}^{\bullet}_1 \widehat{Q} - \overline{b}_2 K_2 \breve{K}^{*}_2 \breve{Q})}{K - \overline{c}_1 K_1 K^{\bullet}_1 - \overline{c}_2 K_2 K^{*}_2}\,. \tag{5.71}$$

This rather complicated expression shows that the generating function$Q$ is built from copies of Q that have some of their marking variables set to 1.

5.4.2 Diagonal Cartesian Model – NE, NW, SE, SW

The next case in terms of difficulty is when we have diagonal steps. This model is very similar to the one covered in [**?**] where they have two interacting walks confined in a semi-infinite strip. The models are different because there we have an extra term in the functional equations. Perhaps the difficulty in the current case is due to $S_2^{-1}(\sigma_1)$ being more complicated, and this is due to the diagonal terms (?). Also, $S_{1,2}^{-1,-1} \neq 1$, so K_{12} is not trivial. Here

$$\frac{\alpha(K^*_2 - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)}{K_1(\overline{x}_2 + x_2)}\,, \tag{5.72}$$

$$\frac{\Delta_2(K^*_2 K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1 - t\overline{\sigma}_1 S^1_1)}{S^1_1(c_1 - t\overline{\sigma}_1 S^1_1)} = \frac{c_2(\overline{x}_2 - x_2)}{\overline{x}_2 + x_2}\,. \tag{5.73}$$

The functional equation becomes

$$K_2^* Q_1 - K_2 Q_1^* - \frac{c_2(\overline{x}_2 - x_2)}{\overline{x}_2 + x_2} Q_{1,2} + \alpha \frac{\overline{x}_2 - x_2}{(\overline{x}_2 + x_2)K_1} = 0\,. \tag{5.74}$$

Surprisingly enough, the only term depending on σ_1, that remains in this expression, is K_1, so this will be of no consequence to the rest of the analysis. We just need to take care of the remaining terms. We will do this by multiplying the equation by $\overline{x}_2 + x_2$. We have

$$R^* = (\overline{x}_2 + x_2)K_2^* = c_2(\overline{x}_2 + x_2) - x_2 = c_2\overline{x}_2 - \overline{a}_2 x_2\,, \tag{5.75}$$

$$R = (\overline{x}_2 + x_2)K_2 = c_2(\overline{x}_2 + x_2) - \overline{x}_2 = c_2 x_2 - \overline{a}_2\overline{x}_2\,. \tag{5.76}$$

There are a couple of relations here

$$R^* - R = v(K_2^* - K_2) = \overline{x}_2 - x_2 \tag{5.77}$$

$$\overline{x}_2 R^* - x_2 R = \overline{x}_2 c_2 v - 1 - x_2 c_2 v + 1 \tag{5.78}$$

$$= c_2 v(\overline{x}_2 - x_2) \tag{5.79}$$

The equation becomes

$$(c_2\overline{x}_2 - \overline{a}_2 x_2)Q_1^0(x_2) + (\overline{a}_2\overline{x}_2 - c_2 x_2)Q_1^0(\overline{x}_2) + (\alpha\overline{K}_1 - c_2 Q_{1,2}^{0,0})(\overline{x}_2 - x_2) = 0 \tag{5.80}$$

and it is ready for extraction. Let $\delta_0 = \overline{a}_1(\overline{x}_2 - x_2)\overline{K}_1$. We rewrite the equation as

$$R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 = 0\,. \tag{5.81}$$

We will use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x}_1 S_1^{-1})Q_1^0 - (c_2 - t\overline{x}_2 S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x}_1\overline{x}_2 S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a}_1\overline{a}_2 \tag{5.82}$$

to further our work here. For diagonal steps it is

$$(1 - tS)Q - (c_1 - tv\overline{x}_1)Q_1^0 - (c_2 - t\overline{x}_2 S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x}_1\overline{x}_2)Q_{1,2}^{0,0} = \overline{a}_1\overline{a}_2 \tag{5.83}$$

extracting the $[x_1^0]$ coefficient yields

$$Q_1^0 = \overline{a}_2 + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1\,. \tag{5.84}$$

We can extract higher coefficients too. Let $k \geqslant 1$. Then the $[x_2^k]$ coefficient is

$$\begin{aligned} 0 &= Q_1^k - tS_1^{-1}Q_1^{k+1} - tS_1^0Q_1^k - tS_1^1Q_1^{k-1} - c_2Q_{1,2}^{k,0} \\ &\quad + t\overline{x}_2(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1}Q_{1,2}^{k,0} + S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) \\ &= Q_1^k - tvQ_1^{k+1} - tvQ_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x}_2(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})\,. \end{aligned} \tag{5.85}$$

We can rewrite the last one in a form more convenient for the next few steps

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2Q_{1,2}^{k,0} - t\overline{x}_2(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})\,. \tag{5.86}$$

We will use induction to prove a nice formula, but for that, we first need to establish another base case because of this splitting of cases for the coefficient extraction. We have

$$Q_1^0 = \overline{a}_2 + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1\,, \tag{5.87}$$

$$Q_1^{0,*} = \overline{a}_2 + c_2Q_{1,2}^{0,0} + a_1tvQ_1^{1,*}\,. \tag{5.88}$$

Combining them with equation (5.81) gives

$$\begin{aligned} 0 &= R^*Q_1^0 - RQ_1^{0,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\ &= R^*(\overline{a}_2 + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1) - R(\overline{a}_2 + c_2Q_{1,2}^{0,0} + a_1tvQ_1^{1,*}) \\ &\quad - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + (R^* - R)(\overline{a}_2 + c_2Q_{1,2}^{0,0}) \\ &\quad - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + \overline{a}_2(\overline{x}_2 - x_2) + \overline{a}_2\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + \overline{a}_2((\overline{x}_2 - x_2) + \delta_0)\,. \end{aligned} \tag{5.89}$$

So we can write the equation

$$R^*Q_1^1 - RQ_1^{1,*} + \overline{a}_2\delta_1 = 0 \tag{5.90}$$

where

$$\delta_1 = \frac{(\overline{x}_2 - x_2) + \delta_0}{a_1tv}\,. \tag{5.91}$$

This equation together with equation (5.81) form the base cases for the following lemma

Lemma 77. *Let m be an integer greater than zero. Then*

$$R^*Q_1^m - RQ_1^{m,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} + \overline{a}_2\delta_m = 0 \tag{5.92}$$

where δ_m is not a function of a_2.

Proof. We show this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$ since we cannot have walks ending on $(1, 0)$. Now we assume that the formula is true for all cases up to and including m. Then we use formula (5.86)

$$\begin{aligned}
&R^*Q_1^m - RQ_1^{m,*} \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(R^*Q_1^{m-1} - RQ_1^{m-1,*}) \\
&+ c_2(R^* - R)Q_{1,2}^{m,0} - t(\overline{x}_2R^* - x_2R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(c_2(\overline{x}_2 - x_2)Q_{1,2}^{m-1,0} - \overline{a}_2\delta_{m-1}) \\
&+ c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x}_2 - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} \\
&- c_2tv(\overline{x}_2 - x_2)Q_{1,2}^{m+1,0}) - \overline{a}_2tv\delta_{m-1}) \,.
\end{aligned} \tag{5.93}$$

Combining this with equation (5.92) leads to

$$0 = tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) - c_2tv(\overline{x}_2 - x_2)Q_{1,2}^{m+1,0} - \overline{a}_2tv\delta_{m-1} + \overline{a}_2\delta_m \,, \tag{5.94}$$

$$0 = R^*Q_1^{m+1} - RQ_1^{m+1,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{m+1,0} + \overline{a}_2\delta_{m+1} \tag{5.95}$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1} \,. \tag{5.96}$$

This function δ_{m+1} is also independent from a_2 and this completes the proof! □

There is an equivalent statement for Q_2^n. We can expand R^* and R to a form we can use

$$\begin{aligned}
0 = &(c_2\overline{x}_2 - \overline{a}_2x_2)Q_1^m(x_2) - (c_2x_2 - \overline{a}_2\overline{x}_2)Q_1^m(\overline{x}_2) \\
&- c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} + \overline{a}_2\delta_m \,.
\end{aligned} \tag{5.97}$$

We start by extracting the $[x_2^1]$ coefficient

$$\begin{aligned} &c_2 Q_{1,2}^{m,2} - \overline{a}_2 Q_{1,2}^{m,0} - c_2 Q_{1,2}^{m,0} + c_2 Q_{1,2}^{m,0} + [x_2^1]\overline{a}_2 \delta_m = 0 \\ &c_2 Q_{1,2}^{m,2} - \overline{a}_2 Q_{1,2}^{m,0} + [x_2^1]\overline{a}_2 \delta_m = 0\,. \end{aligned} \tag{5.98}$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geqslant 2$

$$c_2 Q_{1,2}^{m,n+2} - \overline{a}_2 Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a}_2 \delta_m = 0\,. \tag{5.99}$$

This formula also contains the $n = 0$ case, so we treat them together. We can set $a_2 = 1$ to get

$$-\breve{Q}_{1,2}^{m,n} + [x_2^{n+1}]\delta_m = 0\,. \tag{5.100}$$

Substituting this back into the above equation gives

$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n}\,. \tag{5.101}$$

An equivalent formula can be derived if the process is followed through the use of x_1 instead of x_2 in the analysis.

Corollary 78. *For integers m and n greater than zero, we have the following relations*

$$b_1 Q_{1,2}^{m+2,n} = Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n}\,, \tag{5.102}$$

$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n}\,. \tag{5.103}$$

We can sum up those equations to retrieve the following functional equations.

Corollary 79. *For integers m greater than zero, we have the following relations*

$$Q_1^m - \widehat{Q}_1^m = b_1 Q_1^{m+2}\,, \tag{5.104}$$

$$\breve{Q}_1^m - \breve{\widehat{Q}}_1^m = b_1 \breve{Q}_1^{m+2}\,, \tag{5.105}$$

$$x_1^2(Q_2^n - \widehat{Q}_2^n) = b_1(Q_2^n - Q_{1,2}^{1,n} x_1 - Q_{1,2}^{0,n})\,, \tag{5.106}$$

$$x_1^2(\breve{Q}_2^n - \breve{\widehat{Q}}_2^n) = b_1(\breve{Q}_2^n - \breve{Q}_{1,2}^{1,n} x_1 - \breve{Q}_{1,2}^{0,n})\,, \tag{5.107}$$

$$x_1^2(Q - \widehat{Q}) = b_1(Q - x_1 Q_1^1 - Q_1^0)\,, \tag{5.108}$$

$$x_1^2(\breve{Q} - \breve{\widehat{Q}}) = b_1(\breve{Q} - x_1 \breve{Q}_1^1 - \breve{Q}_1^0)\,. \tag{5.109}$$

Every other equation results from the previous one by either setting $a_1 = 1$ or $a_2 = 1$. We have

an equivalent set of equations for the second coordinate — just replace the subscript 1 with 2 and m with n, to have consistent notation.

Now we can move onto the main result about the diagonal walks, that is, the function for labelled walks ending at $(0,0)$ is connected to the functions for semi-labelled and unlabelled walks ending on $(0,0)$. We can use the master equation to extract the $[x_1^0 x_2^0]$ and $[x_1^1 x_2^1]$ coefficients, which will produce the following two relations

$$Q_{1,2}^{0,0} = 1 + t a_1 a_2 Q_{1,2}^{1,1} \tag{5.110}$$

$$Q_{1,2}^{1,1} = t(Q_{1,2}^{0,0} + Q_{1,2}^{2,0} + Q_{1,2}^{0,2} + Q_{1,2}^{2,2})\,. \tag{5.111}$$

We also have

$$\begin{aligned} b_1 b_2 Q_{1,2}^{2,2} &= b_1(Q_{1,2}^{2,0} - \breve{Q}_{1,2}^{2,0}) \\ &= b_1 Q_{1,2}^{2,0} - b_1 \breve{Q}_{1,2}^{2,0} \\ &= Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} - \breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0}\,, \end{aligned} \tag{5.112}$$

together with

$$b_1 b_2 Q_{1,2}^{2,0} = b_2 Q_{1,2}^{0,0} - b_2 \widehat{Q}_{1,2}^{0,0}\,, \tag{5.113}$$

$$b_1 b_2 Q_{1,2}^{0,2} = b_1 Q_{1,2}^{0,0} - b_1 \breve{Q}_{1,2}^{0,0}\,. \tag{5.114}$$

We also have

$$\begin{aligned} b_1 b_2 Q_{1,2}^{1,1} &= t(b_1 b_2 Q_{1,2}^{0,0} + b_1 b_2 Q_{1,2}^{2,0} + b_1 b_2 Q_{1,2}^{0,2} + b_1 b_2 Q_{1,2}^{2,2}) \\ &= t(b_1 b_2 Q_{1,2}^{0,0} + b_2 Q_{1,2}^{0,0} - b_2 \widehat{Q}_{1,2}^{0,0} + b_1 Q_{1,2}^{0,0} \\ &\quad - b_1 \breve{Q}_{1,2}^{0,0} + Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} - \breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0}) \\ &= t((b_1 b_2 + b_1 + b_2 + 1) Q_{1,2}^{0,0} - (b_2 + 1)\widehat{Q}_{1,2}^{0,0} \\ &\quad - (b_1 + 1)\breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0}) \\ &= t(a_1 a_2 Q_{1,2}^{0,0} - a_2 \widehat{Q}_{1,2}^{0,0} - a_1 \breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0})\,, \end{aligned} \tag{5.115}$$

We can finally write

$$\begin{aligned} b_1 b_2 Q_{1,2}^{0,0} &= b_1 b_2 + t a_1 a_2 b_1 b_2 Q_{1,2}^{1,1} \\ &= b_1 b_2 + t^2 a_1 a_2 (a_1 a_2 Q_{1,2}^{0,0} - a_2 \widehat{Q}_{1,2}^{0,0} - a_1 \breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0})\,. \end{aligned} \tag{5.116}$$

Let $G(b_1, b_2)$ stand for $Q(0, 0, b_1, b_2)$ and G_k has $b_k = 0$. Then we can write it all as

$$b_1 b_2 G = b_1 b_2 + t^2 a_1 a_2 (a_1 a_2 G - a_2 G_1 - a_1 G_2 + G_{1,2}) \,, \tag{5.117}$$

or written as a linear combination,

$$(b_1 b_2 - t^2 a_1^2 a_2^2) G + t^2 a_1 a_2^2 G_1 + t^2 a_1^2 a_2 G_2 - t^2 a_1 a_2 G_{1,2} = b_1 b_2 \,. \tag{5.118}$$

We can divide by $a_1 a_2$ and obtain the equation

$$(c_1 c_2 - a_1 a_2 t^2) G + a_2 t^2 G_1 + a_1 t^2 G_2 - t^2 G_{1,2} = c_1 c_2 \,, \tag{5.119}$$

To conclude, we have the following result.

Theorem 80. *Let $G = Q_{1,2}^{0,0}$ be the generating function for walks ending at the origin in the Diagonal Cartesian model. Let $f(a_1, a_2, t) = c_1 c_2 - a_1 a_2 t^2$ and let $B = B_1 B_2$. Then $B(fG) = c_1 c_2$.*

We can also solve for Q_k^j and other terms, in terms of various $[b_k^l]$ extractions, so overall, it should be possible to find a relationship of the form $B(FQ) = 0$.

5.4.3 The Full Model

We will finally take a look at the model with all the steps.Here we have the most complicated functions. Here

$$\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(1+t)}{S_1^1 K_1} \,, \tag{5.120}$$

$$\frac{\Delta_2(K_2^* K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1(1+t) - t\overline{\sigma}_1 S_1^1)}{S_1^1 (c_1 - t\overline{\sigma}_1 S_1^1)} \tag{5.121}$$

$$= \frac{c_2(\overline{x}_2 - x_2))}{S_1^1} \left(1 - \frac{t}{c_1 - t\overline{\sigma}_1 S_1^1}\right) \,. \tag{5.122}$$

We will need to do some more work to find a representation for that last fraction. For that we need the σ_1 root, which is

$$\begin{aligned} \sigma_1 &= \frac{(1 - tS_1^0) - \sqrt{(1 - tS_1^0)^2 - 4t^2 (S_1^1)^2}}{2tS_1^1} \\ &= \frac{u - \sqrt{u^2 - 4w^2}}{2w} \end{aligned} \tag{5.123}$$

$$\overline{\sigma}_1 = \frac{u + \sqrt{u^2 - 4w^2}}{2w} \,. \tag{5.124}$$

Then we have

$$\begin{aligned}\frac{t}{c_1-\overline{\sigma}_1 w} &= \frac{2t}{2c_1-u-\sqrt{u^2-4w^2)}} \\ &= \frac{2t(2c_1-u+\sqrt{u^2-4w^2})}{(2c_1-u)^2-(u^2-4w^2)} \\ &= \frac{t(2c_1-u+\sqrt{u^2-4w^2})}{2(c_1^2-c_1u+w^2)}\end{aligned} \tag{5.125}$$

Let $v=\overline{x}_2+x_2$. Then the denominator has

$$\begin{aligned}c_1^2-c_1u+w^2 &= c_1^2-c_1(1-tS_1^0)+t^2(S_1^1)^2 \\ &= \overline{a}_1^2b_1-\overline{a}_1b_1vt+(1+v)^2t^2\end{aligned} \tag{5.126}$$

and the numerator has

$$\begin{aligned}u^2-4w^2 &= (1-tv)^2-4t^2(1+v)^2 \\ &= 1-2tv-(3v+2)(v+2)t^2 \\ &= (1-(3v+2)t)(1+(v+2)t)\,.\end{aligned} \tag{5.127}$$

We can find the first few terms of the square root.

$$\sqrt{u^2-4w^2} = 1-vt-2(1+v)^2t^2-2v(1+v)^2t^3+\cdots \tag{5.128}$$

We can give a series representation of the square root via the series

$$\sqrt{1-z} = \sum_{n=0}^{\infty}\frac{(2n)!}{(1-2n)(n!)^2 4^n}z^n\,. \tag{5.129}$$

That is,

$$\begin{aligned}&\sqrt{(1+(v+2)t)(1-(3v+2)t)} \\ &= \left(\sum_{n=0}^{\infty}\frac{(-1)^n(2n)!(v+2)^nt^n}{(1-2n)(n!)^2 4^n}\right)\left(\sum_{n=0}^{\infty}\frac{(2n)!(3v+2)^nt^n}{(1-2n)(n!)^2 4^n}\right) \\ &= \sum_{n=0}^{\infty}\frac{t^n}{4^n}\sum_{k=0}^{n}\frac{(-1)^k(2k)!(2n-2k)!}{(1-2k)(1-2n+2k)k!^2(n-k)!^2}(v+2)^k(3v+2)^{n-k}\,,\end{aligned} \tag{5.130}$$

However, we are not sure if a closed form exists for these coefficients. If we want to extract x_2 coefficients from this expression, we would not have an easy time (?). Now, K_2 and K_2^* have $1+v$

in their denominators, so we can multiply the whole functional equation by $1+v$ to clear out these denominator. We have

$$(1+v)K_2 = c_2(1+v) - \overline{x}_2(1+t) = (c_2 - 1 - t)\overline{x}_2 + c_2 + c_2 x_2\,, \tag{5.131}$$

$$(1+v)K_2^* = c_2(1+v) - x_2(1+t) = c_2\overline{x}_2 + c_2 + (c_2 - 1 - t)x_2\,, \tag{5.132}$$

and

$$(1+v)\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(1+t)}{K_1} = \alpha g(a_1, x_2, t)\,, \tag{5.133}$$

$$(1+v)\frac{\Delta_2(K_2^* K_{12})}{K_1} = c_2(\overline{x}_2 - x_2)\left(1 - t\frac{c_1 - 1 + tv + \sqrt{(1+(v+2)t)(1-(3v+2)t)}}{c_1^2 - c_1 + c_1 tv + (1+v)^2 t^2}\right)\,, \tag{5.134}$$

and let the latter be $c_2(\overline{x}_2 - x_2)f(a_1, x_2, t)$. Then if we let $A = c_2 - (1+t)$, we can write the funtional equation as

$$(c_2\overline{x}_2 + c_2 + Ax_2)Q_1^0(x_2) - (A\overline{x}_2 + c_2 + c_2 x_2)Q_1^0(\overline{x}_2) = c_2(\overline{x}_2 - x_2)f - \alpha g\,. \tag{5.135}$$

As before, we extract the $[x_2]$ coefficient from both sides to find that

$$c_2 Q_{1,2}^{0,2} + c_2 Q_{1,2}^{0,1} - (1+t)Q_{1,2}^{0,0} = -\alpha g_2^1 + c_2(f_2^2 - f_2^0)Q_{1,2}^{0,0}\,, \tag{5.136}$$

and then set $a_2 = 1$ on both sides to find that

$$-(1+t)\breve{Q}_{1,2}^{0,0} = -\overline{a}_1 g_2^1\,. \tag{5.137}$$

When we finally substitute this quantity back into the equation, we can simplify it to

$$b_2(Q_{1,2}^{0,2} + Q_{1,2}^{0,1}) = (b_2(f_2^2 - f_2^0) - a_2(1+t))Q_{1,2}^{0,0} - (1+t)\breve{Q}_{1,2}^{0,0}\,. \tag{5.138}$$

The question is can we find the quantity $f_2^2 - f_2^0$. It appears that

$$-(f_2^2 - f_2^0) = 1 + a_1 t + \frac{a_1^2}{b_1}(a_1^2 - a_1 - 4)t^3 + \frac{4a_1^2}{b_1^2}(a_1^3 - 3a_1^2 + 1)t^4 + \cdots\,. \tag{5.139}$$

Unfortunately we have not been able to go further with this case. One could try making some assumptions for the form of equations like these for $Q_{1,2}^{m,n}$, $\breve{Q}_{1,2}^{m,n}$ and $\widehat{Q}_{1,2}^{m,n}$, but even then it is not clear if we can find a relation between $Q_{1,2}^{0,0}$, $\breve{Q}_{1,2}^{0,0}$, $\widehat{Q}_{1,2}^{0,0}$ and $\breve{\widehat{Q}}_{1,2}^{0,0}$. We would have 6 equations like the one above plus 6 more where $b_i = 0$ for the b_i not present. Then one may use the 4 regular coefficient extractions from the master functional equation for $Q_{1,2}^{m,n}$ and $m, n \leqslant 1$ plus 12 more where we set one or more of b_1, b_2 to zero. This brings us to a grand total of 28 equations in 36 unknown variables, of which we need to eliminate 32. It is not entirely clear if we may do that.

Chapter 6

Generalizations

6.1 Constraint Generalizations

In this final chapter we will finally analyze the kinds of generalizations that the operator perspective allows us to explore. There is already some work that hints at those more general situations. For example, Definition 1.1 of [7] has a functional equation for the non-interacting lattice walks on the quarter plane, which is valid for models with $\mathcal{S}$ containing negative coordinates of size at most -1. Similarly, [11] covers walks in the quarter plane with multiple copies of small steps. Also, we may consider the region of a semi-infinite strip on the plane as in [27], [**?**] and [28]. In [26] the author considers lattice walks in an octant with interactions with the diagonal.

Before we begin the general treatment, we must specify what kinds of generalizations we are ready to make.

- Set $\mathcal{S}$. This is a finite set of steps that we can take from a point in $\mathbb{Z}^n$ to another point in $\mathbb{Z}^n$. In general, this step set may differ from point to point; for example, a honeycomb lattice embedded in $\mathbb{Z}^2$. We will assume that there is one set of steps allowed for every point of the space. The set may involve arbitrarily-many finitely-sized steps with multiplicities at least one. Its polynomial version is denoted by S.

- Set $\mathcal{D}$. This is the region in $\mathbb{Z}^d$ that the lattice walks can occupy. They are not allowed to venture outside. We have already seen half-spaces and orthants. Other interesting regions are semi-infinite strips, finite boxes and wedges.

- Set $\mathcal{D}_0$. This is a sub-region in $\mathcal{D}$ where lattice walk interactions take place — we may track how many times lattice walks visit coordinates from this region. We may count visits to different points from $\mathcal{D}_0$ in different ways. This less-studied paradigm may involve marking

visits to various places: the origin, the axes in two dimensions, various higher-dimensional hyperplanes, a cone, an infinite strip, the sides of a box and so on.

- Set $\mathcal{Q}$. This is a polynomial or series from which we evolve lattice walks. Each monomial may be interpeted as some lattice walk with some iterated counters; for example, $x_1x_2t^2$ may indicate a walk that has started somewhere and ended on $(1,1)$ after two steps. This term could be as trivial as 1, indicating the empty walk ending at the origin, to a series representing a set of infinitely many walks with some pattern. Its polynomial version is denoted by q.

Figure 6.1: General regions in $\mathbb{Z}^2$.

We will be growing lattice walks from an initial condition $\mathcal{Q}$, with steps $\mathcal{S}$, within $\mathcal{D}$ in $\mathbb{Z}^d$, while counting interactions with $\mathcal{D}_0$. Of course, $\mathcal{Q}$ may be a more general term, so we will not restrict it yet. The quadruple of sets should serve as a complete picture of a lattice walk model, so we encapsulate it ina definition.

Definition 81. *Let $\mathcal{M} = (\mathcal{S}, \mathcal{D}, \mathcal{D}_0, \mathcal{Q})$ denote a lattice walk model with step set $\mathcal{S}$, living in the region $\mathcal{D}$, interacting with region $\mathcal{D}_0$, where $\mathcal{Q}$ is an initial condition.*

Of course, we need to keep track of all these quantities inside a generating functions, so we continue using Q for the name.

Definition 82. *Let $\mathcal{M}$ be a lattice walk model and Q its associated generating function counting lattice walks' lengths with t, ending locations with $x_1, \ldots, x_d$, and interactions with $\mathcal{D}_0$ with some multi-index variable a. We write*

$$Q = Q(x_1, \ldots, x_d, a; t) = \sum_{k=0}^{\infty} q_k(x_1, \ldots, x_d, a)t^k \,, \tag{6.1}$$

where q_k are Laurent poynomials in the x and a variables.

The analysis is analogous to the one in Chapter 2 and Chapter 3 — we start with the initial condition $\mathcal{Q}$, that is represented by the function q, and grow lattice walks from it by multiplying by tS. Those walks may venture outside of $\mathcal{D}$ and may interact with $\mathcal{D}_0$. Just like in the previous analysis, we will need operators to help us remove or mark walks.

Definition 83. *Let $\mathcal{M}$ be a given model set in d dimensions. Let E be an operator that eliminates monomials $x^P_{[d]}$ when P is not in $\mathcal{D}$. Similarly, let L be an operator that marks monomials $x^P_{[d]}$ for $P \in \mathcal{D}_0$ in the way specified by the model. Let $K = 1 - tS$ be the kernel of the lattice walk model.*

From what we have seen in previous chapters, it should be clear that the operator L is invertible — just take an operator marking the same locations in $\mathcal{D}_0$ but with inverted labels. That is, if the operator marks a monomial with the label a, then the inverse operator is one marking the same monomial with the label $1/a$. Also, the caution in the definition is there because it may be possible to imagine such convoluted constraints that an operator may no exist. It is not difficult to come up with constraints that lead to unwieldy operators. To make the analysis work, we need the operators to act term-wise on series with respect to the variable t.

Definition 84. *Suppose that Γ is an operator involving the x and a variables but not t. We shall apply Γ on power series $F = \sum_{k=0}^{\infty} f_k t^k$ in the following way:*

$$\Gamma F = \sum_{k=0}^{\infty} \Gamma(f_k) t^k \,. \tag{6.2}$$

We can dive into the main result concerning the general models delineated so far.

Theorem 85. *Let $\mathcal{M}$ be a lattice walk model and Q its associated generating function. Then Q satisfies the following functional equations*

$$(1 - tLES)Q = q \,, \tag{6.3}$$

$$(L^{-1} - tES)Q = L^{-1}(q) = \alpha \,, \tag{6.4}$$

and its initial term is $q_0 = q$.

Proof. The second functional equation is obtained by applying L^{-1} to the first one, so it only remains to show that the first functional equation holds. We do this via iteration. We start with Q and multiply it with tS. This gives tSQ, or an enumeration of all walks extended by a single step. Here the power of t is raised by one to keep track of this extension. Then we apply E to this result to eliminate all lattice walks that venture outside of the region $\mathcal{D}$. Finally, we mark those walks that have arrived at locations in $\mathcal{D}_0$ by applying L to $E(tSQ)$. The resulting expression is almost Q, but one term is missing — namely, $q_0 = q$. That is, we have transformed Q to itself in a nontrivial way and we have the equation

$$Q = q + LE(tSQ) = q + tLE(SQ) \,, \tag{6.5}$$

since the operators do not interact with t. □

Example 86. Also, we can consider a different approach where, instead of removing walks that venture outside of $\mathcal{D}$, we just mark them with some variable z for later deletion. Suppose that we are given some model $\mathcal{M}$ and we want to frame the analysis in this new way. We have a new lattice walk model $\widetilde{\mathcal{M}}$ with $\widetilde{\mathcal{D}} = \mathbb{Z}^d$ and $\widetilde{\mathcal{D}}_0 = \mathcal{D}_0 \cup \mathcal{D}^c$, where we mark monomials in $\mathcal{D}^c$ with z and those in $\mathcal{D}_0$ in the same way as in model $\mathcal{M}$. Here $\widetilde{E}$ is the identity operator since we do not remove any walks. We have the functional equations

$$(1 - t\widetilde{L}S)\widetilde{Q} = q\,, \tag{6.6}$$

$$(\widetilde{L}^{-1} - tS)Q = \widetilde{\alpha}\,. \tag{6.7}$$

It is not clear how one would proceed and if this would be a better way to handle the problem. The upside is that the operators are invertible, but the downside is that we now have more extracted terms in the functional equation, since the lattice walks can now go anywhere.

Example 87. Consider a the lattice walk models in two dimensions from previous chapters. Another way we can look at the quarter plane models is to us the idea from the previous example and also symmetrize the problem by also marking parts on the other side of the origin. Instead of just marking walks on the axes and on lines where one coordinate is -1 we can also mark walks on lines with one coordinate $+1$. This leads to a symmetrical arrangement of marking operations on both sides of the origin. We have the equation

$$(1 - tLS)\widetilde{Q} = (1 - tL_{\text{-}1}L_0L_1S)\widetilde{Q} = 1\,. \tag{6.8}$$

We can skip the marking of the axes with L_0 if we do not wish to investigate wall interactions. In general, we may mark more locations in space so we can combine all of these operators into one called L. Then we have

$$(1 - tLS)\widetilde{Q} = 1\,, \tag{6.9}$$

or in an alternative form,

$$K\widetilde{Q} = L^{-1}(1) + (1 - L^{-1})\widetilde{Q}\,. \tag{6.10}$$

Before the end of this section, we will take a moment to talk about some of the abstract properties of the operators L and E. The operator E has the eigenvalues 1 or 0 for monomials $x^P_{[d]}$ where

$P \in \mathcal{D}$ or $P \in \mathcal{D}^c$, respectively. The latter eigenvalue characterizes the monomials contained in $\ker E$. Similarly, L has eigenvalue 1 for when $P \in \mathcal{D}_0^c$, and the marking monomial a for when $P \in \mathcal{D}_0$, where we keep in mind that a is a multi-index variable and it varies from point to point. Also, L^{-1} has the reciprocal eigenvalues of L.

The task for the rest of the chapter is to find representations for the operators involved. In general, this should will not be an easy task because if we could express arbitrary operators L and E with a small number of coefficient extractions, we would also be able to use this scheme to compress data without loss, and this is impossible in general.

6.1.1 Marking Operators

We start with the observation that the operator E is just a marking operator that applies the label $a = 0$ to all monomials coming from points outside of $\mathcal{D}$. Therefore, we can combine the discussions for operators L and E into one. Whenever invertibility is concerned, the discussion will apply to operators L only. First, we show one form of a marking operator.

Lemma 88. *Let P be a subset of $\mathbb{Z}^d$ and suppose that we have to mark each point p in that set with some label a_p, with some marking operator Γ_P. Then the operator can be written as*

$$\Gamma_P = 1 + \sum_{p \in P} (a_p - 1)[[x^p_{[d]}]] \,. \tag{6.11}$$

Proof. It is enough to confirm this operator's action on single monomials. If p is not in P, then Γ_P returns the monomial itself, since all the term extractions return zero. If p is in P, then the only extraction that returns a nonzero term is the one associated with p. In that case we obtain

$$\Gamma_P C x^p_{[d]} = C x^p_{[d]} + (a_p - 1) C x^p_{[d]} = a_p C x^p_{[d]} \,. \tag{6.12}$$

□

Corollary 89. *Marking operators commute.*

Proof. Since marking operators act on monomials by multiplication with marking variables and multiplication is commutative, then the application of marking operators is also commutative. □

This form of the operator is rather inefficient — just consider one-dimensional models. There, both L and E contain a single term extraction, whereas this representation here would need infinitely many. In other words, a lot of models allow for smaller marking operators that do essentially the same job. The following example illustrates this notion.

Example 90. If we consider lattice walks with small steps on the half-line in one dimension, we have the operator

$$E = 1 - \sum_{n=1}^{\infty} \overline{x}_1^n [\overline{x}_1^n] \,, \tag{6.13}$$

but since we only have small steps, there are no walks that go past $\overline{x}_1$ in the negative direction. Therefore, we can use the operator $E' = 1 - \overline{x}_1[\overline{x}_1]$ to the same effect instead. Although the operators are different in form, they are essentially the same with regards to the problem at hand. We will say the two operators are model-equivalent.

To make this equivalence work we need to consider the whole model. With that in mind, we may think of the lattice walk models as living completely within the sumset $\mathcal{S} + \mathcal{D}$ of points that are combinations of points in $\mathcal{D}$ and steps $\mathcal{S}$. This is how far walks from the model can reach from within $\mathcal{D}$ by taking a single step. We will call this sumset the range of the model.

Definition 91. *Given sets A and B, we let their sumset be*

$$A + B = \{a + b \,|\, a \in A,\, b \in B\} \,. \tag{6.14}$$

Definition 92. *Given a lattice walk model $\mathcal{M}$ with domain $\mathcal{D}$ and step set $\mathcal{S}$, define the range of model $\mathcal{M}$ to be the sumset $\mathcal{D} + \mathcal{S}$.*

With this in mind, we can define a notion of equivalence for operators.

Definition 93. *Let $\mathcal{M}$ be a lattice walk model in $\mathbb{Z}^d$ and let Γ and Γ' be two operators acting on $\mathbb{Z}^d$. Then $\Gamma \equiv_{\mathcal{M}} \Gamma'$ if and only if Γ and Γ' act identically on monomials from the range of $\mathcal{M}$. That is, the operators are model-equivalent if and only if $\Gamma x^p = \Gamma' x^p$ for all $p \in \mathcal{D} + \mathcal{S}$.*

In the preceding example E is equivalent to E', since both operators act the same way on monomials from the range of $\mathcal{M}$, that is, monomials x_1^n for $n \geqslant -1$. Similarly, in the two-dimensional problem we have an equivalence between the operator E and the operator $E' = (1 - \overline{x}_1[\overline{x}_1])(1 - \overline{x}_2[\overline{x}_2])$.

When the boundary of $\mathcal{D}$ is complicated, we may not find nice equivalent operators, but there are some cases where we can say something. In problems featuring half-planes, we have an operator E with infinitely many terms, and we find that there is a model-equivalent operator E' that is only comprised of finitely many terms. We will investigate which operators are equivalent to finite ones.

Definition 94. *Given a model $\mathcal{M}$, a coefficient extraction operator Γ in the variables $x_1, \ldots, x_d$ is finite if it is equivalent to an operator Γ' comprised of finitely many coefficient extraction terms.*

That is,

$$\Gamma \equiv_{\mathcal{M}} \Gamma' = \sum_{\substack{I \subseteq [d] \\ J \subseteq \mathbb{Z}^d}} a_{I,J} x^J [x_I^J] \tag{6.15}$$

where each $|J| < \infty$*. Also,* Γ' *is infinite if it is not finite.*

For most purposes, we will not distinguish between Γ and Γ'. A natural question is to consider which models $\mathcal{M}$ lead to finite operators. This very much depends on the region $\mathcal{D}$ and the nature of the marking of region $\mathcal{D}_0$. We have some cases where it is easy to say.

Lemma 95. *Let* $\mathcal{D}$ *be a* d*-dimensional box in* $\mathbb{Z}^d$ *and* $|\mathcal{S}| < \infty$*. Then the operator* $\Gamma = LE$ *is finite.*

Proof. The operator L is finite since it only needs to mark finitely many points. We can find E explicitly. Let the box have x_k coordinates $[a_k, b_k]$. Suppose that the steps in $\mathcal{S}$ have x_k coordinates of absolute size at most n_k. If $n_k = 0$, the walks do not change their x_k coordinate and the walks do not venture past the boundaries in that coordinate. When $n_k > 0$ we need to remove all monomials x^P with $P_k \in [a_k - n_k, a_k - 1] \cup [b_k + 1, b_k + n_k]$.

Figure 6.2: Finite region enclosed in box B.

We can then write the operator as

$$E_k = 1 - \sum_{j=1}^{n_k} x_k^{a_k - j} [x_k^{a_k - j}] + x_k^{b_k + j} [x_k^{b_k + j}], \tag{6.16}$$

which is equal to $E_k = 1$ when $n = 0$. Applying $E_1, \ldots, E_d$ sequentially ensures that we remove all monomials outside of $\mathcal{D}$. That is, $E = E_1 \cdots E_d$. Since each of the operators is finite, their composition $\Gamma = LE$ is finite as well. □

Theorem 96. *Let* $\mathcal{D}$ *be a finite region in* $\mathbb{Z}^d$ *and* $|\mathcal{S}| < \infty$*. Then the operator* $\Gamma = LE$ *is finite.*

Proof. The region $\mathcal{D}$ lies within a bounding box B, so we may write $E = E'E''$, where $E' = E_1 \cdots E_d$ from the lemma and E'' is the operator removing all monomials outside of $\mathcal{D}$ but inside the interior of B. E'' is finite since there are only finitely many points in that region. Since we are marking only finitely many points $\mathcal{D}_0 \subseteq \mathcal{D}$, L is finite, and finally, Γ is finite. □

Also, we may handle semi-infinite regions if we are marking only finitely many points. The analysis is the same as above, except now one or more of a_k, b_k may be infinite, and in this case we actually lose parts of Γ_{B^c} and the operator is still finite.

Definition 97. *A lattice walk model with finitely many steps is finite if its extraction and marking operators E and L are finite, and its starting condition q has a finite degree. Otherwise, we say the model is infinite.*

Example 98. All the models so far has been finite, but not all models are finite. Consider a one-dimensional model on the whole line where, for each integer i, we mark point x^i with parameter a_i using operator L. Then all equivalent marking operators L' would have to have a term like $(a_i - 1)[[x^i]]$; otherwise, it would not act the same way on monomials in the model's range $\mathbb{Z}$. Therefore, L is not equivalent to a finite operator.

Finite models are nicer because they have functional equations with finitely many terms and there is a chance that they will be amenable to the sort of analysis we have performed for quarter-plane walk models in the past. What follows is a rather easy application of the theory so far.

6.1.2 Examples of Regions

Example 99. Here is an example of an infinite strip. The region $\mathcal{D}$ is comprised of all the points with coordinates (i, j) in $\mathbb{Z}^2$ with $0 \leqslant j \leqslant n$. Maybe we have interactions with the lines $x_2 = 0$ and $x_2 = n$. The operators are

$$E = E_{-1}E_{n+1} = (1 - [[\overline{x}_2]])(1 - [[x_2^{n+1}]]) = 1 - [[\overline{x}_2]] - [[x_2^{n+1}]]\,, \tag{6.17}$$

$$L = L_0 L_n = (1 + b_0[[x_2^0]])(1 + b_n[[x_2^n]]) = 1 + b_0[[x_2^0]] + b_n[[x_2^n]]\,, \tag{6.18}$$

and the functional equation for Q is given by

$$KQ - (c_0 - t\overline{x}_2 S_2^{-1})Q_2^0 - x_2^n(c_n - tx_2 S_2^1)Q_2^n = \overline{a}_0\,. \tag{6.19}$$

If we extract the $[x_2^0]$ term, we find that

$$(1 - a_0 t S_2^0)Q_2^0 - tS_2^{-1}Q_2^1 = 1\,. \tag{6.20}$$

If we extract the $[x_2^n]$ term, we find that

$$(1 - a_n t S_2^0)Q_2^n = tS_2^1 Q_2^{n-1}\,. \tag{6.21}$$

Similarly, we can extract the $[x_2^k]$ terms for $1 \leqslant k \leqslant n-1$. We will have $n+1$ equations in $n+1$ unknown functions $Q_2^0, \ldots, Q_2^n$, and we would be able to solve them to find out what all those functions are. Of course, we only need Q_2^0 and Q_2^n to obtain a solution for Q. In the simplest case, $n = 1$ and the term extraction equations become

$$(1 - a_0 t S_2^0) Q_2^0 = 1 + t S_2^{-1} Q_2^1 \,, \tag{6.22}$$

$$(1 - a_1 t S_2^0) Q_2^1 = t S_2^1 Q_2^0 \,. \tag{6.23}$$

These solutions are

$$Q_2^0 = \frac{1 - a_1 t S_2^0}{(1 - a_0 t S_2^0)(1 - a_1 t S_2^0) - S_2^1 S_2^{-1} t^2} \,, \tag{6.24}$$

$$Q_2^1 = \frac{S_2^1 t}{(1 - a_0 t S_2^0)(1 - a_1 t S_2^0) - S_2^1 S_2^{-1} t^2} \,, \tag{6.25}$$

and they allow us to find Q explicitly from

$$Q = \frac{1}{K}(\overline{a}_0 + (c_0 - t\overline{x}_2 S_2^{-1}) Q_2^0 + x_2 (c_1 - t x_2 S_2^1) Q_2^1) \,. \tag{6.26}$$

Example 100. Another example is a finite box. The region $\mathcal{D}$ is comprised of all the points with coordinates (i, j) in $\mathbb{Z}^2$ with $0 \leqslant i \leqslant m$ and $0 \leqslant j \leqslant n$, for some positive integers m and n. The lattice walks interact with all four sides of the box, each with its own label. As in the previous example, we may write a functional equation directly via the following operators

$$E_{i,j} = 1 - [[x_i^j]] \quad \text{and} \quad L_{i,j} = 1 + b_{i,j}[[x_i^j]] \,. \tag{6.27}$$

For the model itself, we have

$$E = E_{1,\text{-}1} E_{1,m+1} E_{2,\text{-}1} E_{2,n} \,, \tag{6.28}$$

$$L = L_{1,0} L_{1,m} L_{2,0} L_{2,n} \,. \tag{6.29}$$

We may attempt a solution similar to that of the previous example. The function Q should be rational here as well.

Example 101. Suppose that $\mathcal{D}$ is a finite region in $\mathbb{Z}^2$. If we set $x_1 = 0$ for points in $\mathcal{D}$, we obtain the projection of the set onto the x_2 axis, and this tells us how many x_2 coordinates we have in functions in the functional equation. Suppose that this number of points is m_1 and define m_2 analogously. Lastly, let there be m lattice points in $\mathcal{D}$. Next, we examine the number of unknown functions Q_i^j

and $Q_{1,2}^{i,j}$ that we have in the functional equation and its various extractions. The master functional equation may only have m_1 functions Q_2^i and m_2 functions Q_1^i, together with m functions $Q_{1,2}^{i,j}$, because those are the possible nonzero functions Q that we may find through various extractions. Similarly, when we extract terms like $[x_i^j]$ and $[x_1^i x_2^j]$, we only find a subset of the functions listed.

In general, we may even find less of those functions as we perform more extractions. In the end, we have the same number of equations as unknown functions Q and various extractions of Q. This means that the problem reduces to the solution of a linear system of $1 + m_1 + m_2 + m$ equations and that many unknowns Q. Note that, because we have the term KQ in the master functional equation, the $[x_i^j]$ extraction will have the function Q_i^j with a nonzero coefficient. Because there is a unique function Q for the model, this forces the uniqueness of the Q_i^j and $Q_{1,2}^{i,j}$ terms. This means that we have a unique solution to the system of equations. Furthermore, because all the the coefficients are rational functions, so is Q. We will generalize this in the following theorem.

Definition 102. *Given a d-dimensional region $\mathcal{D}$ and a finite set I from $\{1, \ldots, d\}$, let $\mathcal{D}_I$ be defined as follows*

$$\mathcal{D}_I = \{(x_1, \ldots, x_d)_{x_I=0} | (x_1, \ldots, x_d \in \mathcal{D})\}, \tag{6.30}$$

that is, $\mathcal{D}_I$ is the projection of $\mathcal{D}$ onto the space generated by coordinates x_i for i in $[d]\backslash I$. Let $m_I = |\mathcal{D}_I|$.

Note that $I = \varnothing$ corresponds to the set of points in $\mathcal{D}$, and $I = [d]$ corresponds to exactly one point and the master functional equation.

Theorem 103. *If a d-dimensional model $\mathcal{M}$ is set in a finite region $\mathcal{D}$, then the full generating function Q is rational in its variables. To obtain the solution, we solve a system of $N = \sum_{I \subseteq [d]} m_I$ equations and unknown functions Q_I^J, with a unique solution.*

Proof. The master functional equation may at most contain all the possible nonzero functions Q_I^J. Each possible extraction $[x_K^P]$ may contain at most a portion of these variables — all terms of the form $Q_{I,K}^{J,P}$. We just need to count the number of equations and unknowns. The projections $\mathcal{D}_I$ helps us in determining these numbers. The set $\mathcal{D}_I$, with m_I points corresponds to all the functions $Q_{[d]\backslash I}^P$ for $P \in \mathcal{D}_I$. In fact, we have m_I unknowns of the form $Q_{[d]\backslash I}^P$, because the master functional equation contains a term Q, and so we have all the possible extractions among the extraction equations. This means that we have N equations and N unknowns.

Next, since there is a unique generating function solution Q to the master functional equation for the model at hand, the various extractions Q_I^J are also unique, and overall, there must be a unique

solution to the system of equations. Therefore, Q is a rational function in its variables because all the coefficients in the system are rational functions. □

Starting with the master functional equation, we have N unknown functions. When we perform a single extraction $[x_i^i]$ in one of $m_{1,\ldots,i-1,i+1,\ldots,d}$ ways, we no longer see the function Q or any of the functions Q_k^l for k not equal to i. This kind of pattern continues on with further extractions. We can write out a schematic version of the system of equations. For example, in two dimensions we have

$$\begin{bmatrix} Q & Q_1^i & Q_2^i & Q_{1,2}^{i,j} \\ & Q_1^i & & Q_{1,2}^{i,j} \\ & & Q_2^i & Q_{1,2}^{i,j} \\ & & & Q_{1,2}^{i,j} \end{bmatrix}. \tag{6.31}$$

That is, in the master functional equation we have all of the unknown functions. In the next m_2 equations we have variables Q_1^i and $Q_{1,2}^{i,j}$ only, and so on. If we take the last set of equations only involving functions of the form $Q_{1,2}^{i,j}$, we have a system of equations that we can solve for the various functions. Then we may back-substitute these quantities in the rest of the systems of equations. Next, we may solve for the functions Q_1^i and Q_2^i. When we do this, we may finally back-substitute all the quantities in the master functional equation. This procedure is reminiscent of the long-winded approach to finding the transformation from $(1-tLES)Q = 1$ to $(L^{-1}-tES)Q = \alpha$ that we avoided through the use of operator notation. We are hopeful that such an approach exists in this case too.

Example 104. Consider the one-dimensional model set on $\mathcal{D} = \{u, \ldots, v\}$. Here the step set must have steps of size at most $n = v - u$ in either direction, because any larger steps will always step outside of the region, and therefore, we may discard them. We have $S = \sum_{k=-n}^{n} S_k x^k$. The operators E and L are given by

$$E = 1 - \sum_{k=1}^{n} ([[x^{u-k}]] + [[x^{v+k}]]), \tag{6.32}$$

$$L = 1 + \sum_{k=u}^{v} a_k [[x^k]], L^{-1} = 1 - \sum_{k=u}^{v} c_k [[x^k]]. \tag{6.33}$$

We do not need to use all the variables a_k but we may as well do that. The master functional equation is $(L^{-1} - tES)Q = L^{-1}(q)$, as usual. The only extraction equations we obtain here are

those coming from the coefficient extraction with $[x^i]$ with $u \leqslant i \leqslant v$. We have

$$[x^i]E = [x^i]\,, \tag{6.34}$$

$$[x^i]L^{-1} = [x^i] - c_i[x^i] = \overline{a}_i[x^i]\,, \tag{6.35}$$

for those case and the extracted functional equations become

$$[x^i](L^{-1} - tES)Q = [x^i](\overline{a}_i - tS)Q = [x^i]L^{-1}(q) = q_i\,, \tag{6.36}$$

or in other words, $[x^i](1 - a_i tS)Q = [x^i]K(a_i t)Q = a_i q_i$. The explicit equations are

$$Q_i - a_i t \sum_{j=-n}^{n} S_{i-j}Q_j \tag{6.37}$$

We may write all of those equations as the system

$$IX - tHX = (I - tH)X = \alpha\,, \tag{6.38}$$

where I is the $n \times n$ identity matrix, H is the matrix with entries $H_{i,j} = a_i S_{i-j}$ for $u \leqslant i, j \leqslant v$, and α is the vector with entries q_i. Because of the uniqueness of the solution, this matrix $(I - tH)$ must be invertible. We may write the master functional equation as $KQ + F \cdot X = q$ and thus, the function Q is given by

$$Q = \frac{1}{K}(q - F \cdot X)\,. \tag{6.39}$$

Example 105. Consider the one-dimensional model set on $\mathcal{D} = \{0, \ldots, 1\}$. Here $X^T = [Q_0 \;\; Q_1]$ and

$$H = \begin{bmatrix} a_0 & a_0 \\ a_1 & a_1 \end{bmatrix}\,, \tag{6.40}$$

and the determinant of $I - tH$ is $1 - (a_0 + a_1)t$. Therefore,

$$X = \frac{1}{1 - (a_0 + a_1)t} \begin{bmatrix} (1 - a_1 t)q_0 - a_0 t q_1 \\ (1 - a_0 t)q_1 - a_1 t q_0 \end{bmatrix} \tag{6.41}$$

and from the master functional equation

$$KQ - (c_0 - t\overline{x})Q_0 - x(c_1 - tx)Q_1 = q_0 + xq_1\,, \tag{6.42}$$

we find that

$$Q = \frac{1}{(1-(a_0+a_1)t)K}(f_0q_0 + f_1q_1)\,, \tag{6.43}$$

and if we expand the function into series we find that there is a stabilization of the coefficients involving x in front of t^k. We expect that such stabilization will occur for all finite region models.

6.2 Connections With Other Theories

So far we have explored using operators to simplify the derivation of functional equations pertaining to lattice walk models on $\mathbb{Z}^d$, and to simplify the first steps in solutions to these problems. It would be great to know if we can say more about the solutions. It seems that the work so far points to connections with Fredholm and Sturm-Liouville theory. For example, if we interpret coefficient extractions from a complex analysis point of view, by using Cauchy integrals, we may obtain an equation that looks much like a Fredholm integral equation of the first kind. The connection is made stronger by noting that both theories involve a possible Neumann series solution, written differently in each case. Using the Neumann series expansion, we have been able to show that Q may be formed as a Hadamard product of generating functions for lower-dimensional lattice walk models. However, using the Neumann series expansion is unwieldy, so our other hope is to find another kind of series expansion. For example, since the marking operators have eigenvalues and eigenfunctions, we may hope to find some sort of eigenfunction expansion for Q, along the lines of Sturm-Liouville theory. For that we would need a suitable inner product and find the adjoints of the marking operators, with respect to the inner product. Unfortunately, time did not allow to pursue either of these approaches with much effort, and we only have the following writeup with the hopes of inspiring future mathematicians to work along those lines and find out if the approaches are fruitful.

6.2.1 Properties of Marking Operators

The operators we have seen so far are marking operators, like L and E, in combination of multiplication by another function, like ΓS, or a combination of operators, like $L^{-1} - tES$. In this order, we will find out their properties - eigenvalues, eigenvectors, kernels and invertibility. The building block of these operators are coefficient extractions $[x_I^J]$ and term extractions $[[x_I^J]]$.

Combinatorially speaking, we can easily find some of the eigenvalues and eigenfunctions by looking at single monomials. If a monomial is not marked by Γ, then it is an eigenfunction with eigenvalue 1, since Γ does not change it; if it is, then it is an eigenfunction with eigenvalue equal to the label given by Γ. These are all the polynomial eigenfunctions. This does not mean there cannot be more eigenfunctions among those expressible by power/Laurent series and beyond.

Proposition 106. *Let Γ be an operator marking a single monomial x_I^J with label a. Let λ and u_λ be an eigenpair. Then $\lambda = a$ with $u_\lambda = x_I^J$, or $\lambda = 1$ with u_λ a function not containing x_I^J, in the formal power series sense. In particular, every monomial other than x_I^J is an eigenfunction with eigenvalue 1.*

Proof. Let $u_\lambda = u' + u'' x_I^J$ where $u \neq 0$ and u'' does not share variables with x_I^J. Then

$$\Gamma(u) = u' + a u'' x_I^J = \lambda(u' + u'' x_I^J)\,, \tag{6.44}$$

which we may write as

$$(1-\lambda)u' + (a-\lambda)u'' x_I^J = 0\,. \tag{6.45}$$

We have two cases.

- If $u' = 0$, then $u'' \neq 0$ and $\lambda = a$. The eigenfunction is x_I^J.
- If $u' \neq 0$ and there is no term in $u'' x_I^J$ that can compesate for this, then $\lambda = 1$ and $u'' = 0$.

□

Corollary 107. *The kernel of a finite marking operator Γ is generated by the monomials marked with label 0.*

6.2.2 Marking Operators With Multiplication

The operators we see in the theory of lattice walk models often feature application on generating functions that are pre-multiplied by a Laurent polynomial, like S. This leads us to definite such combinations of operators in the following way.

Definition 108. *We call an operator inhomogeneous if it consists of a homogeneous operator Γ and a (possibly infinite) laurent polynomial S, and acts in one of the following ways*

$$\Gamma S(u) = \Gamma(Su) \quad \textit{or} \quad S\,\Gamma u = S \cdot \Gamma u\,. \tag{6.46}$$

This kind of operator is important because it can be found in the functional equations for lattice walk models. To find their eigenpairs, we first show that the definitions are mostly equivalent.

Proposition 109. *The two operators ΓS and $S\,\Gamma$ have the same eigenvalues with eigenfunctions differing by a multiple of S. That is, (λ, u) is an eigenpair for ΓS if and only if (λ, Su) is an eigenpair for $S\,\Gamma$. In particular, if $\lambda = 0$, we have*

$$\ker(\Gamma S) = \overline{S}\ker(\Gamma)\,. \tag{6.47}$$

Proof. Note that

$$\Gamma S u = \lambda u \quad \iff \quad S\,\Gamma(Su) = \lambda(Su)\,. \tag{6.48}$$

Therefore, the operators have the same eigenvalues, and they have the eigenfunctions u and Su respectively. Now, if we pick $\lambda = 0$ to find the kernels of the operators, we find that $u \in \ker(\Gamma S)$ if and only if $Su \in \ker(S\,\Gamma) = \ker(\Gamma)$, and this completes the proof. □

As the next example shows, it may not be easy to find the eigenpairs for such operators.

Example 110. Let Γ be a marking operator and let u be an eigenfunction of ΓS, where

$$S = x\,, \quad u = \sum_k u_k x^k\,. \tag{6.49}$$

Computing $(\Gamma S - \lambda I)u$ leads to the system of equations

$$\lambda u_n = a u_{n-1}\,, \tag{6.50}$$

$$\lambda u_k = u_{k-1}, \quad \text{if } k \neq n\,. \tag{6.51}$$

We have a doubly-infinite recursion, so we can pick an arbitrary initial condition. If $u_n = 0$, the rest of the terms will be zero as well, so we must have $u_n \neq 0$. Set $u_n = 1$. Then, via induction, we can find

$$u_{n+k} = 1/\lambda^k, \quad \text{for } k \geqslant 0\,, \tag{6.52}$$

$$u_{n-k} = \lambda^k/a^k, \quad \text{for } k \geqslant 1\,; \tag{6.53}$$

therefore, we have an eigenfunction for each $\lambda \neq 0$. Computing $u(x)$ formally leads to

$$u(x) = -\frac{\lambda b x^{n+1}}{(\lambda - x)(\lambda - ax)}\,, \tag{6.54}$$

and the situation becomes more complicated for more complicated functions S.

6.2.3 Lattice Walks Operator

In this section we explore the operator $\mathcal{L} = 1 - t\Gamma S$ for a step Laurent polynomial S and some marking operator Γ. Let (u, λ) be an eigenpair for $\mathcal{L}$. Then we calculate

$$\begin{aligned} (\mathcal{L} - \lambda I)u &= u - \lambda u - t\Gamma S u \\ t\Gamma S u &= (1-\lambda)u = \mu u\,. \end{aligned} \tag{6.55}$$

If $\lambda = 1$, we would have to compute the kernel of ΓS, which we did in the previous section. The eigenfunctions of $\mathcal{L}$ for eigenvalue $\lambda = 1$ are the functions $u \in \overline{S} \ker \Gamma$. Let u be given by a finite order Laurent series

$$u = u_n t^n + u_{n+1} t^{n+1} + \cdots\,, \tag{6.56}$$

where $u_n \neq 0$. Then we can rewrite the functional equation as

$$t^{-n}(1 - t\Gamma S)u = \lambda t^{-n} u \tag{6.57}$$

$$(\lambda - 1)t^{-n}u = -t^{1-n}\Gamma S u\,. \tag{6.58}$$

Letting $t \to 0$ in this equation yields

$$(\lambda - 1)u_n = -\lim_{t\to 0} t^{1-n}\Gamma S u = -\Gamma S \lim_{t\to 0}(t^{1-n}u) = 0\,. \tag{6.59}$$

Since $u_n \neq 0$, this implies $\lambda = 1$, and again we see that $u \in \overline{S} \ker \Gamma$. Since S is a Laurent polynomial, the resulting functions will indeed be finite order Laurent series. This analysis shows that the only eigenfunctions of $\mathcal{L}$ among finite order Laurent series in t are those that are members of $\overline{S} \ker \Gamma$ with eigenvalue $\lambda = 1$.

Example 111. Let $\Gamma = LE$ in the one-dimensional problem with small steps on the non-negative line with nontrivial operator E. Then $\ker \Gamma = \ker E$ consists of functions $\alpha(t)\overline{x}$, and the eigenfunctions of $\mathcal{L}$ are $\alpha(t)\overline{x}\overline{S}$ for various $\alpha(t)$. In our case, we only look at finite order Laurent series of

$\alpha(t)$. Note that (?)

$$Q = \frac{t(\overline{x} - \overline{\sigma})}{(1 - tS)(b - a\overline{\sigma}t)} = \frac{1}{xS}\frac{\sigma - x}{b\sigma - at}\frac{tS}{1 - tS} \tag{6.60}$$

where we may write

$$\frac{\sigma - x}{b\sigma - at} = \alpha(t) + \beta(t)x \tag{6.61}$$

where $\alpha(t)$ and $\beta(t)$ are algebraic functions of t and a since $\sigma = \sigma(t)$. This at least suggests the possibility of connecting the eigenfunctions of $\mathcal{L}$ to the solution Q.

If we are to have a theory similar to Sturm-Liouville theory, we still need to find a suitable inner product on multivariate Laurent series that will allow us to write solutions for Q in terms of those eigenfunctions with eigenvalue 1.

6.3 Coefficient Extraction Equations

The work so far is strongly hinting towards a theory of ordinary and partial coefficient extraction equations - OCEs and PCEs. Here we will work out some of the theory. We start with a partial coefficient extraction operator of finite type

Definition 112. *$\mathcal{L}$ is a coefficient operator if*

$$\mathcal{L} = \sum_{\substack{I \subseteq [d] \\ J \subseteq \mathbb{Z}^d}} a_{I,J}[x_I^J] = K + \mathcal{E} \tag{6.62}$$

where K is the kernel of the operator and $\mathcal{E}$ is the pure extraction part

$$\mathcal{E} = \sum_{\substack{I \subseteq [d], I \neq \varnothing \\ J \subseteq \mathbb{Z}^d}} a_{I,J}[x_I^J] \tag{6.63}$$

and all the coefficients are Laurent polynomials in the x variables and perhaps some other variables. The operator is ordinary if $d = 1$ and partial if $d \geqslant 2$. The operator is either of monomial type where the coefficients are monomials or of polynomial type if one or more coefficients is a Laurent polynomial in $x_1, \ldots, x_d$.

Definition 113. *An ordinary/partial coefficient equation (OCE/PCE) is one of the form $\mathcal{L}Q = q$ for some functions Q and q where $\mathcal{L}$ is an ordinary/partial coefficient operator. The equation is homogeneous if $q = 0$ and inhomogeneous if $q \neq 0$.*

The question is whether we can solve these kinds of equations in general, and what their solutions may look like. In the previous section we found that if $\mathcal{L}$ is finite then the function Q is D-finite in the x variables while it may not necessarily be D-finite in the t variable or any other variable (parameter) that appears in the coefficients.

Example 114. The operator Γ in this section is a partial coefficient operator with monomial coefficients.

Example 115. We can transform the functional equation for a finite random walk model into a PCE. To do so, we note that K is the same in both forms. The part requiring some work is the one that has to do with the term $(1-\Gamma)SQ$ which is a pure extraction of SQ. We can expand S into its monomials. Then $(1-\Gamma)S$ fully expands into monomials of the form

$$a_{I,J}x_I^J[x_I^J]x_A^B \tag{6.64}$$

we can pull terms inside $[x_I^J]$ from the right. In the end the S is absorbed within the operator and istead of having monomial coefficients it now has polynomial coefficients. It may be useful to write an operator $\mathcal{L}$ in the form

$$\mathcal{L} = K + \Gamma' S \tag{6.65}$$

for some monomial operator Γ'.

6.3.1 Walks With Unconstrained Steps

We may go through the process again, but this time we have

$$Q = \frac{W}{x_{[d]}^N K} \tag{6.66}$$

where N_k is the maximum negative size of steps in the set $\mathcal{S}$ in the direction x_k or in other words, the maximum N_k for which $S_k^{N_k}$ is nonzero. Then the operators that keep the walks in the quarter plane are

$$E_k = 1 - \sum_{i=1}^{N_k} \overline{x}_k^i[\overline{x}_k^i]\,. \tag{6.67}$$

Their conjugates are

$$E'_k = x_k^{N_k} E_k \overline{x}_k^{N_k} = 1 - \sum_{i=0}^{N_k-1} x_k^i [x_k^i] = 1 - \mathcal{E}_k \,, \tag{6.68}$$

so in the kernel equation we have

$$x_{[d]}^N = x_{[d]}^N E \overline{x}_{[d]}^N W = E'W = W + \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} \mathcal{E}_I(W) \,. \tag{6.69}$$

Here we have various powers x_k^i along with terms with $x_k = 0$ for $k \in I$. Hence, if we seek an operator D that does the same job as the orbit summation operator, we need D to annihilate all the terms with x_k's missing in W on the right-hand side. So then we have

$$D(x_{[d]} KQ) = D(W) = D\left(x_{[d]}^N - \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} \mathcal{E}_k(W) \right) = D(x_{[d]}^N) \,. \tag{6.70}$$

In other words we want D to behave (in some sense) similarly to the derivative

$$\partial_{[d]}^N = \partial_1^{N_1} \cdots \partial_d^{N_d} \,. \tag{6.71}$$

Theorem 116. *A lattice walk model with steps of size at most N in each coordinate, satisfies the functional equation*

$$E'W = x_{[d]}^N \tag{6.72}$$

where $E' = x_{[d]}^N E \overline{x}_{[d]}^N$ and $W = x_{[d]}^N KQ$.

Corollary 117. *The generating function Q for the non-interacting case satisfies the PDE*

$$\partial_{[d]}^N \left(x_{[d]}^N KQ \right) = 1 \tag{6.73}$$

Here $\partial_{[d]}^N$ is an example of an operator D that eliminates all the unnecessary terms but does not eliminate $x_1 \cdots x_d$. It is possible to discuss weighted operators in this setting too, but the situation becomes more complicated. However, we may investigate type (N, M) problems by finding a suitable model whose orbital operator is nice enough. This will show that all type (N, M) satisfy an orbital equation with that particular operator, not just a PDE.

Example 118. Consider the lattice walk model with $S = \sum_{k=1}^d \overline{x}_k^{n_k} + x_k^{n_k}$. Its group G is generated

by the elements of groups G_k with $2n_k$ elements each - each of the elements $g_{k,j,l} = \omega_{n_k}^j x_k^l$ where $\omega_{n_k} = \exp(2\pi i/n_k)$ and $l \in \{-1, 1\}$. These elements commute and allow us to find an orbital operator that annihilates all terms on the left-hand side except $x_{[d]}^N KQ$ and also spares the $x_{[d]}^N$ term on the right-hand side.

We can generalize this somewhat.

Theorem 119. *The full generating function Q for a d-dimensional, finite, non-interacting lattice walk model, satisfies a finite-order inhomogeneous PDE in the variables $x_1, \ldots, x_d$.*

Proof. We will use the kernel form for the functional equation

$$KQ + t(1 - \Gamma)SQ = q. \tag{6.74}$$

Here $\Gamma' = 1 - \Gamma$ is purely a coefficient extractor and it is finite, since L and E are finite. We will prove the theorem by finding the differential equation explicitly. Since the model is finite, we can bound the degree of x_k that we see on either side of the functional equation. In one scheme we let A_k be the largest power of x_k that we need to clear a negative power of x_k in Γ' and in q. Then we let B_k be the largest power of x_k we see on either side. Once we multiply by $x_k^{A_k}$, we see that the degree of x_k in the equation is $A_k + B_k$. We take the functional equation and multiply both sides by $x_{[d]}^A$. That is,

$$x_{[d]}^A(KQ + t\Gamma' SQ) = x_{[d]}^A q \tag{6.75}$$

and we attain a polynomial equation in the variables $x_1, \ldots, x_d$. We take the derivative operator

$$D = \partial_{[d]}^{A+B+\mathbb{1}} = \partial_1^{A_1+B_1+1} \cdots \partial_d^{A_d+B_d+1} \tag{6.76}$$

and apply it to both sides of the equation to retrieve

$$D(x_{[d]}^A(KQ + t\Gamma' SQ) = D(x_{[d]}KQ) = D(x_{[d]}^A q) = 0\,, \tag{6.77}$$

which follows since we are applying a larger number of derivatives than the degrees of the monomials and this clears all of them. The other scheme lets us obtain a potentially lower degree differential equation at the cost of homogeneity, if the degrees of q are large. Let A_k be the power of x_k that we need to clear denominators on the left-hand side of the functional equation and B_k be the largest power of x_k on that side. Then we have D as before, but this time it may have lower A and B

values, since we are not taking q into account. We have

$$D(x^A_{[d]}(KQ + t\Gamma' SQ) = D(x_{[d]}KQ) = D(x^A_{[d]}q)\,. \tag{6.78}$$

□

Example 120. For example, let the powers of x_k be $\{-3, -1, 4\}$ and $\{-2, 0, 6\}$ on the left-hand side and right-hand side respectively. We see that $A_k = 3$ and $B_k = 6$. Then we are taking the $A_k + B_k + 1 = 10$-th derivative of both sides of the functional equation. In the second scheme we take $A_k = 3$ still, but $B_k = 4$ and we are taking the 8-th derivative with respect to x_k this time.

Even in the quarter-plane, we have models with non D-finite generating function (in the t variable), but it turns out that every model so far considered is D-finite in the variables $x_1, \ldots, x_d$. In order to solve the differential equation we need to investigate the kernel of the differential operator.

Corollary 121. *The generating function Q of a finite non-interacting lattice walk model is given by*

$$x^A_{[d]}KQ(x_1, \ldots, x_d; t) = \sum_{i=1}^{d}\sum_{j=0}^{B_i} Aij(x_1, \ldots, x_{i-1}, x_i, \ldots, x_d, t)x_i^j\,, \tag{6.79}$$

where A_{ij} are functions that need to be determined.

This kind of solution would have also appeared if we went through the calculations in the functional equation and extracted coefficients. However, this method is more efficient. If we had used the second scheme in the proof, we would have ended up with a slightly different form but maybe it would have had a slightly lower degree. It would involve an indefinite integral of q and elements in the kernel of the differential operator.

6.3.2 Connection with Fredholm Theory

Suppose that we are given the equation in the form

$$KQ + t\Gamma' SQ = q \tag{6.80}$$

where

$$\Gamma' = \sum_{I,J} a_{I,J}x_I^J[x_I^J] \tag{6.81}$$

where we only have finite-sized J's and $\varnothing \neq I \subseteq [d]$. From complex analysis we know that for a function $f(x)$ given by a power series and for x within the region of convergence

$$[x^n]f(x) = f_n = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z^{n+1}}dz \tag{6.82}$$

and

$$f(x) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z-x}dz \tag{6.83}$$

where in the first case γ winds CCW around the origin within the region of convergence and in the second case γ winds around x within the region of convergence. The idea is to combine these two expressions for many dimensions (at least formally) to represent the operator $\mathcal{L}$ in integral form. We have for $|I| = n$

$$[x_I^J]f(x_1,\ldots,x_d) = \frac{1}{(2\pi i)^n}\int\cdots\int_\gamma \frac{f}{z_I^{J+1}}dz_I \tag{6.84}$$

where f has $x_i = z_i$ for $i \in I$ and the rest are still x's. If we use this representation we will have different numbers of iterations of the integral for different coefficient. We might as well go all-out and write

$$[x_I^J]f(x_1,\ldots,x_d) = \frac{1}{(2\pi i)^d}\int\cdots\int_\gamma \frac{f}{z_I^{J+1}(z-x)_{I^c}}dz_{[d]} \tag{6.85}$$

where now $x_i = z_i$ in f and

$$(z-x)_P = \prod_{p\in P}(z_p - x_p) \tag{6.86}$$

This allows us to write The operator like this

$$(K + t\Gamma' S)Q = \frac{1}{(2\pi i)^d}\int\cdots\int_\gamma \left(\frac{K}{(z-x)_{[d]}} + tS\sum_{I,J}\frac{a_{I,J}}{z_I^{J+1}(z-x)_{I^c}}\right)Q dz_{[d]} \tag{6.87}$$

$$= \frac{1}{(2\pi i)^d}\int\cdots\int_\gamma \frac{1}{(z-x)_{[d]}}\left(K + tS\sum_{I,J}\frac{a_{I,J}(z-x)_I}{z_I^{J+1}}\right)Q dz_{[d]} \tag{6.88}$$

$$= \frac{1}{(2\pi i)^d}\int\cdots\int_\gamma R_0 Q dz_{[d]} \tag{6.89}$$

Now to turn this into a d-dimensional Fredholm problem we will need to parametrize the integral. One way that might work is to set $z_k = r_k \exp(i\theta_k)$ and $x_k = s_k e^{i\phi_k}$. Since $dz_k = r_k i e^{i\theta_k}$, we have

$$(K + t\Gamma' S)Q = \frac{r_{[d]}}{(2\pi)^d} \int_0^{2\pi} \cdots \int_0^{2\pi} e^{i\theta_1} \ldots e^{i\theta_d} R_0 Q d\theta_{[d]} = q(\phi_1, \ldots, \phi_d)\,, \tag{6.90}$$

so if we write

$$R = \exp(i(\theta_1 + \ldots + \theta_d)) R_0(\phi_1, \ldots, \phi_d, \theta_1, \ldots, \theta_d) \tag{6.91}$$

we have the inhomogeneous Fredholm equation of the first kind in d dimensions

$$\int_0^{2\pi} \cdots \int_0^{2\pi} RQ d\theta_1 \ldots d\theta_d = g \tag{6.92}$$

where R is the kernel and $g = (2\pi)^d q / r_{[d]}$ and Q is in terms of θ_k's. Perhaps Fredholm theory can be applied in solving the random walk models in man dimensions. Perhaps this will also lead to a good inner product on the space where Q lives.

Chapter 7

Epilogue

Bibliography

[1] A. A. Monforte and M. Kauers, "Formal laurent series in several variables," *Expositiones Mathematicae*, vol. 31, no. 4, pp. 350–367, 2013. → pages 5

[2] I. P. Goulden and D. M. Jackson, *Combinatorial enumeration*. Courier Corporation, 2004. → pages 5

[3] I. M. Gessel, "A factorization for formal laurent series and lattice path enumeration," *Journal of Combinatorial Theory, Series A*, vol. 28, no. 3, pp. 321–337, 1980. → pages 2

[4] M. Bousquet-Mélou and M. Petkovšek, "Linear recurrences with constant coefficients: the multivariate case," *Discrete Mathematics*, vol. 225, no. 1, pp. 51–75, 2000. → pages 2

[5] P. Duchon, "On the enumeration and generation of generalized dyck words," *Discrete Mathematics*, vol. 225, no. 1, pp. 121–135, 2000. → pages 2

[6] C. Banderier and P. Flajolet, "Basic analytic combinatorics of directed lattice paths," *Theoretical Computer Science*, vol. 281, no. 1, pp. 37–80, 2002. → pages 2

[7] M. Mishna, "Classifying lattice walks restricted to the quarter plane," *Journal of Combinatorial Theory, Series A*, vol. 116, no. 2, pp. 460–477, 2009. → pages 2, 93

[8] M. Bousquet-Mélou and M. Mishna, "Walks with small steps in the quarter plane," *Contemp. Math*, vol. 520, pp. 1–40, 2010. → pages 2, 15, 22, 23, 38, 49, 50, 51, 52

[9] M. Bousquet-Mélou *et al.*, "Walks in the quarter plane: Kreweras algebraic model," *The Annals of Applied Probability*, vol. 15, no. 2, pp. 1451–1491, 2005. → pages 2

[10] A. Bostan and M. Kauers, "The complete generating function for gessel walks is algebraic," *Proceedings of the American Mathematical Society*, vol. 138, no. 9, pp. 3063–3078, 2010. → pages 2

[11] M. Kauers and R. Yatchak, "Walks in the Quarter Plane with Multiple Steps," *CHECK IT!*, November 2014. → pages 2, 38, 93

[12] I. Kurkova and K. Raschel, "Explicit expression for the generating function counting gessel's walks," *Advances in Applied Mathematics*, vol. 47, no. 3, pp. 414–433, 2011. → pages 2

[13] A. Bostan, K. Raschel, and B. Salvy, "Non-d-finite excursions in the quarter plane," *Journal of Combinatorial Theory, Series A*, vol. 121, pp. 45–63, 2014. → pages 2

[14] S. Melczer and M. Mishna, "Singularity analysis via the iterated kernel method," *Combinatorics, Probability and Computing*, vol. 23, no. 05, pp. 861–888, 2014. → pages 2

[15] S. R. Melczer, *Variants of the kernel method for lattice path models*. PhD thesis, SIMON FRASER UNIVERSITY, 2014. → pages 2, 15, 23, 25, 52

[16] I. M. Gessel and D. Zeilberger, "Random walk in a weyl chamber," *Proceedings of the American Mathematical Society*, vol. 115, no. 1, pp. 27–31, 1992. → pages 2

[17] D. J. Grabiner and P. Magyar, "Random walks in weyl chambers and the decomposition of tensor powers," *Journal of Algebraic Combinatorics*, vol. 2, no. 3, pp. 239–260, 1993. → pages 2

[18] A. Bostan, M. Bousquet-Mélou, M. Kauers, and S. Melczer, "On 3-dimensional lattice walks confined to the positive octant," *arXiv preprint arXiv:1409.3669*, 2014. → pages 2, 33, 49, 52

[19] E. J. Van Rensburg and A. Rechnitzer, "Multiple markov chain monte carlo study of adsorbing self-avoiding walks in two and in three dimensions," *Journal of Physics A: Mathematical and General*, vol. 37, no. 27, p. 6875, 2004. → pages 2

[20] R. Brak, P. Dyke, J. Lee, A. Owczarek, T. Prellberg, A. Rechnitzer, and S. Whittington, "A self-interacting partially directed walk subject to a force," *Journal of Physics A: Mathematical and Theoretical*, vol. 42, no. 8, p. 085001, 2009. → pages 2

[21] Y.-b. Chan and A. Rechnitzer, "A monte carlo study of non-trapped self-avoiding walks," *Journal of Physics A: Mathematical and Theoretical*, vol. 45, no. 40, p. 405004, 2012. → pages 2

[22] A. Bacher and M. Bousquet-Mélou, "Weakly directed self-avoiding walks," *Journal of Combinatorial Theory, Series A*, vol. 118, no. 8, pp. 2365–2391, 2011. → pages 2

[23] A. Bedini, A. L. Owczarek, and T. Prellberg, "Lattice polymers with two competing collapse interactions," *Journal of Physics A: Mathematical and Theoretical*, vol. 47, no. 14, p. 145002, 2014. → pages 2

[24] E. Dagrosa, A. Owczarek, and T. Prellberg, "Writhe-induced knotting in a lattice polymer," *Journal of Physics A: Mathematical and Theoretical*, vol. 48, no. 6, p. 065002, 2015. → pages 2

[25] H. Niederhausen, "Random walks in octants, and related structures," *Journal of statistical planning and inference*, vol. 135, no. 1, pp. 165–196, 2005. → pages 2

[26] H. Niederhausen, "A note on the enumeration of diffusion walks in the first octant by their number of contacts with the diagonal," *Journal of Integer Sequences*, vol. 8, no. 2, p. 3, 2005. → pages 2, 93

[27] R. Brak, A. L. Owczarek, A. Rechnitzer, and S. G. Whittington, "A directed walk model of a long chain polymer in a slit with attractive walls," *Journal of Physics A: Mathematical and General*, vol. 38, no. 20, p. 4309, 2005. → pages 2, 93

[28] A. L. Owczarek and T. Prellberg, "Enumeration of area-weighted dyck paths with restricted height," *Aust. J. Comb*, 2012. → pages 2, 93

[29] R. Tabbara, A. Owczarek, and A. Rechnitzer, "An exact solution of two friendly interacting directed walks near a sticky wall," *Journal of Physics A: Mathematical and Theoretical*, vol. 47, no. 1, p. 015202, 2014. → pages 2, 14, 18, 22, 24, 69, 76, 78

[30] I. M. Gessel, "Symmetric functions and p-recursiveness," *Journal of Combinatorial Theory, Series A*, vol. 53, no. 2, pp. 257–285, 1990. → pages 3

[31] F. Chyzak, M. Mishna, and B. Salvy, "Effective scalar products of d-finite symmetric functions," *Journal of Combinatorial Theory, Series A*, vol. 112, no. 1, pp. 1–43, 2005. → pages 3

[32] M. Mishna, *A holonomic systems approach to algebraic combinatorics*. Montréal: Université du Québec à Montréal, 2005. → pages 3

[33] A. Bostan and M. Kauers, "Automatic classification of restricted lattice walks," *DMTCS Proceedings*, no. 01, pp. 201–215, 2009. → pages 3

[34] M. Kauers and D. Zeilberger, "The computational challenge of enumerating high-dimensional rook walks," *Advances in Applied Mathematics*, vol. 47, no. 4, pp. 813–819, 2011. → pages 3

[35] S. Johnson, M. Mishna, and K. Yeats, "Towards a combinatorial understanding of lattice path asymptotics," *arXiv preprint arXiv:1305.7418*, 2013. → pages 3

[36] S. Melczer and M. Mishna, "Asymptotic lattice path enumeration using diagonals," *Probabilistic, Combinatorial and Asymptotic Methods for the Analysis of Algorithms*, p. 313, 2014. → pages 3

[37] S. Melczer and M. Mishna, "Enumerating lattice paths with symmetries through multivariate diagonals," *arXiv preprint arXiv:1402.1230*, 2014. → pages 3